A Fish and Meat Filleting Environment with a Physical Knife-Edge Cutting Law

Scene, tissue mechanics, cutting, bones, robots, evaluation and recorded evidence

Gym Anything for Robotics

Abstract

This environment is a fish preparation bench and a butcher block at real scale. Two Franka FR3 arms with Franka Hands do the knife work: take a flexible fillet knife from its block, head and fillet a gutted rainbow trout, remove a fillet's belly bones, trim, excise and portion a salmon half side, and remove the chain, denude and slice a pork tenderloin. Newton 1.5 builds the rigid world and MuJoCo integrates it at a 1 ms tick. The workpieces are explicit finite-element solids integrated on the GPU by this environment's own solver, together with rigid bones with cuttable joints, thin bones as rods that bend and break, cohesive anatomical interfaces and a skin membrane. Tissue separates only where a knife-edge law, built on Atkins' cutting mechanics, lets the edge advance through it, or where an interface fails by its own cohesive law. Nothing in the world is moved by the interface directly: actions command robot joints and grip force, and the knife moves only while a hand holds it or when it falls. This dossier states every constitutive law as the code implements it, lists every physical constant with its provenance, and reports the recorded experiments. Assisted runs, in which a labelled phantom rig drives the knife, show what the world can produce; they are not robot results. No autonomous robot solution of any cutting task exists or is claimed.

Contents

1 Purpose, the real job and the acceptance boundary 4
1.1 What the environment is for . . . 4
1.2 The real job . . . 4
1.3 How to read this document . . . 4

2 Scene at real scale 5
2.1 Physical geometry . . . 5
2.2 Cameras and visual dressing . . . 5
2.3 Workpieces . . . 6
2.4 Parameter ledger . . . 6

3 Tissue mechanics, integration and the cutting law 15
3.1 Two solvers and one clock . . . 15
3.2 Bulk tissue . . . 16
3.3 Explicit integration, stability mass and substeps . . . 16
3.4 Contact with rigid shapes . . . 17
3.5 The knife-edge cutting law . . . 18
3.6 Crack insertion . . . 19
3.7 The energy ledger . . . 19

4 Bones, rods, interfaces and skin 19
4.1 Rigid bones and their joints . . . 19
4.2 The bilinear cohesive law . . . 20
4.3 Release by the blade and interface fracture energy . . . 20
4.4 Tearing . . . 21
4.5 Thin bones as rods . . . 21
4.6 Skin membrane . . . 22

5 The knife and the robots 22
5.1 The flexible knife . . . 22
5.2 Who integrates the knife . . . 23
5.3 The grip . . . 23
5.4 Motion generator and protective stops . . . 23
5.5 Robot configurations . . . 24
5.6 The bench scale . . . 24

6 Tasks and evaluation 24

7 Experimental design and quality gates 26
7.1 Components before policies . . . 26
7.2 Quality gates . . . 27

7.3 The recorded runs . . . 27
7.4 How this document uses evidence . . . 27
7.5 Statistical scope . . . 27

8 Recorded results **28**
8.1 Machine acceptance of the quality gates . . . 28
8.2 World at real scale and weight (quality gate 1) . . . 37
8.3 The robot doing the work (quality gate 2) . . . 40
8.4 Assisted solvability and negative controls (quality gate 3) . . . 49
8.5 Rest and weight (quality gate 4) . . . 63
8.6 Legal abuse (quality gate 5) . . . 68
8.7 Mechanics against references (quality gate 6) . . . 71
8.8 Runtime (quality gate 7) . . . 82
8.9 Interfaces (quality gate 8) . . . 91

9 Rendered evidence **92**
9.1 Films of the recorded runs . . . 92

10 Measured laws of this solver stack **102**
10.1 Coupling between the two solvers . . . 102
10.2 Tissue solver . . . 103
10.3 Robot and control . . . 103
10.4 Energy, precision and contact . . . 104

11 Limitations **107**

12 Reproduction **109**
12.1 Environment . . . 109
12.2 Running a task . . . 109
12.3 Module map . . . 109

1 Purpose, the real job and the acceptance boundary

1.1 What the environment is for

The deliverable is a digital workplace in which a robot policy can attempt the knife work of a fish preparation bench and a butcher block, and be judged on the product it leaves behind. It is not a solved demonstration. A policy has to take the knife from its block, hold it in a friction grip, bring the edge to the right place on a workpiece it sees only through cameras, and cut along anatomical boundaries it cannot see, while its other arm holds or lifts the product. The environment does not attach the knife to the hand, does not move tissue on command, and does not decide that a cut happened: the tissue parts only where the edge law of Section 3.5 lets the edge advance, or where an interface fails by its own cohesive law (Section 4.2).

Three questions are kept apart throughout. First, does the model implement the mechanics it states, without duplicated forces, unbooked energy or causal shortcuts? Second, do its dimensions and materials follow the best available evidence, and is every value without evidence labelled as such? Third, can a particular policy solve a task? This document addresses the first two. A task that is hard for a scripted robot is not by itself an environment defect, and a successful assisted run is not a policy result.

Every run reported here carries one of three labels, recorded in the run's result file and burned into every film frame. **UNASSISTED**: no command at all, or only legal robot commands. **SCRIPTED robot**: legal robot commands issued by a script that may plan from privileged state (the knife's true pose, settled bone positions); the world is acted on only through the robot. **ASSISTED**: an owner-side phantom rig drives the knife, or a phantom hold or pull acts on a body; every engagement is logged in `world.assist`, and the GAR backend refuses the configuration that enables it (`Config(phantom_rig=True)`).

1.2 The real job

The action group behind this environment gathers 32 occupational task rows: filleting round fish, trimming and portioning fillets, removing pin and belly bones, skinning, and trimming tenderloins and other butcher cuts. The physical core is the same in all of them. A thin, flexible, very sharp blade is drawn through soft, anisotropic, layered tissue along interfaces the cutter cannot see (bones, connective sheets, fat seams), guided by feel, while the other hand holds or lifts the product. Quality is judged on the product: yield, one piece, no bones where there should be none, clean cut faces, no gaping or bruising; the NOAA fillet workmanship method is the reference used for scoring [24].

Three bench presets share the robots, the knife and its law, the cameras and the evaluator framework.

Preset	Setting	Workpiece	Supported tasks
`trout` (A)	fish prep bench	gutted rainbow trout with a rigid head, 17 backbone segments, ribs, pin bones, fin supports and supraneurals	`take-knife`, `head-off`, `fillet-one-side`, `fillet-whole-fish`, `remove-belly-bones`
`salmon-side` (B)	trim station	front half of a salmon side, skin down, with pin bones, a belly fat strip and one melanin spot; bench portion scale	`trim-belly`, `excise-defect`, `portion`
`pork-tenderloin` (C)	butcher block	pork tenderloin: main muscle, the chain along one flank, flake fat, silver skin on part of the upper surface	`remove-chain`, `denude`, `medallions`

The generator-measured masses, bone counts and mesh sizes of each workpiece are recorded results and appear in Section 8 (world at real scale). The task catalogue with every instruction exactly as the policy receives it is Table 4.

1.3 How to read this document

Sections 2–6 describe the environment as built. Their numbers are the code's own values: parameters are read from `filleting/params.py`, named constants are parsed from the source files, and the physical layout and task

catalogue are generated from `filleting/layout.py`, `filleting/task.py` and `filleting/variants.py` when the document is built. Section 7 states how the evidence is organised. Section 8 holds every measured number, read from the recorded result files. Section 11 lists what the world does not do.

2 Scene at real scale

Geometry a robot, the knife or the tissue can touch is physical: it is a collider in MuJoCo and, where tissue or a free knife can reach it, a shape of the tissue solver as well (`scene.py`, `layout.py`, `workpiece.py`). Everything else in the room is visual dressing without colliders (`dressing.py`). World z is up, x runs along the bench's front edge (the robots' right is $+x$) and y points from the front edge into the bench; the origin is on the floor below the middle of the front edge. Table 1 is the layout as built.

2.1 Physical geometry

- **Bench and board.** A stainless bench with its top at catering-table height (authored) and an NSF HDPE board on a damp cloth, top at 0.9137 m. The board is a static body: a board on a damp cloth does not slide.
- **Room floor.** The floor is a collider for MuJoCo and for the tissue solver: a piece that leaves the bench lands on it (`workpiece.py`, the `floor` shape).
- **Robots.** Two FR3 arms on one welded pedestal at the bench's front edge, both facing the bench, each with a Franka Hand. The right hand carries authored knife-handle inserts: per finger a channel of length 60 mm and depth 18 mm, with lips of 4 mm that wrap the handle's top and bottom edges, a liner of friction coefficient 0.8 and a mass of 0.025 kg each (`scene.py`, `INSERT`).
- **Knife.** Victorinox Fibrox 5.3703.18 geometry [21]: blade length 0.173 m, height 25 mm, spine 1.2 mm, bevel 18 degrees per side; handle envelope 0.14 m × 32 mm × 19 mm. The blade is five hinged segments with the blade's flat-wise and edge-wise stiffness (Section 5); its steel mass is computed from its volume in `scene.py`.
- **Knife block.** The block holds the fillet knife and a boning knife handle up. The slot pose is fitted once (`dressing.knife_block_fit`) and the same fit builds the render and the colliders: two block halves around the blade, a slot floor just under the blade's edge line, a ceiling just over its spine and the front chamfer as a collider. The slots are 70 mm apart so that a hand with the inserts fits between the handles. The boning knife's handle is a static collider. In task A0 the fillet knife stands in its slot and is held only by the slot.
- **Vessels.** The perforated fillet tray (GN 1/2) right of the board and a stainless bowl behind it are colliders; the tray is A2's target. In A3 (fillet-whole-fish) the tray stands on the holding-arm side with an empty GN 1/2 pan in front of it for the frame and head, and the ice pan stands right of the board. Presets B and C have a bench portion scale whose weighing platform is a collider in both solvers (Section 5.6).
- **Bench stock within reach.** Everything the render draws on the bench within the arms' reach or a held knife's reach is a static collider sized from the drawn geometry (`Scene.bench_stock`): the GN 1/1 pan of crushed ice with its two stand-in trout (preset A; ice and each trout as a box around the drawn body), the honing steel behind the board, the tweezer tray and the tweezers (preset B), the sanitiser spray bottle and the paper roll at the bench's back corners, each as a box. The collider audit (Section 8) lists every static collider with its dimension source and checks that no prop within reach is visual only.
- **Phantom park.** In assisted runs only, the phantom rig parks the knife above the board behind it (`layout.PHANTOM_KNIFE_PARK`).

2.2 Cameras and visual dressing

The policy observes RGB from an overhead camera 1.0 m above the board, a supervisor camera from the wall side at 1.6 m eye height, and one wrist camera per hand with the D405 envelope. Hero, bench and workpiece cameras are diagnostic and owner-only. Images are rendered by Isaac Sim RTX from recorded physics states in a separate render-only process (`render.py`, `frames.py`, `fish_surfaces.mdl`); the renderer never advances the simulation. Two surfaces are render-only and are moved with the physics rather than simulated: the head's skin surface and the fins. The room, the shelving and the dressing beyond the arms' and a held knife's reach have no colliders.

Table 1: Physical layout of the cell as built, generated from `filleting/layout.py` (metres, degrees; world z up, x along the bench's front edge, y into the bench).

Item	Value
ROOM	size=(5, 4); ceiling=2.8; x_range=(-2.2, 2.8); y_range=(-2.3, 1.7); wall_tile_height=2
TABLE	size=(1.8, 0.75); top_z=0.9; top_thickness=0.04; center=(0, 0.375); undershelf_z=0.25; leg=0.04
BOARD	size=(0.61, 0.457, 0.0127); center=(0, 0.28); cloth_thickness=0.001
BOARD_TOP_Z	0.9137
PEDESTAL	size=(0.9, 0.3, 0.9); center=(0, -0.162)
ROBOT_BASES[left]	pos=(-0.3, -0.1, 0.9); yaw_deg=90
ROBOT_BASES[right]	pos=(0.3, -0.1, 0.9); yaw_deg=90
KNIFE_BLOCK	size=(0.1, 0.16, 0.2); center=(0.33, 0.6); slot_x=(-0.032, 0.038)
FILLET_TRAY	size=(0.325, 0.265, 0.04); center=(0.5, 0.3); wall=0.0008
BOWL	diameter=0.24; depth=0.08; center=(0.52, 0.6)
ICE_PAN	size=(0.53, 0.325, 0.065); center=(-0.62, 0.4); presets=(A)
SCALE	size=(0.3, 0.25, 0.045); center=(-0.55, 0.4); presets=(B, C)
TWEEZER_TRAY	size=(0.2, 0.1, 0.015); center=(0.45, 0.07)
PHANTOM_KNIFE_PARK	handle_com=(-0.12, 0.62, 1.034); yaw_deg=0
CAMERAS[overhead]	pos=(0, 0.28, 1.914); look_at=(0, 0.28, 0.9137); hfov_deg=60; width=1280; height=720
CAMERAS[supervisor]	pos=(0.15, 1.45, 1.6); look_at=(0, 0.2, 0.9137); hfov_deg=55; width=1280; height=720

2.3 Workpieces

Each workpiece is a signed-distance description with tissue regions, reference products (where every gram belongs in a correct job), rigid bones, thin bones, pre-split interfaces and a skin. The tissue solver meshes it by isosurface stuffing, which produces no sliver elements [3]. Seeds vary length (mass scales with the cube of length), bone counts and pose on the board.

- **Trout** (`anatomy/trout.py`): a loft of digitised outlines (FDA reference photographs, MRI cross sections) with myomere cone axes for the connective-sheet normals, a gutted cavity open along the belly slit, a rigid head solid to whose posterior face the flesh is tied, 17 backbone segments (one vertebra each for the first five, four to five each behind) jointed through cuttable intervertebral joints, rib pairs, pin bones, dorsal and anal fin supports, supraneurals and the pelvic girdle, and skin on every outer surface except the cavity and the slit. Pre-split interfaces: the median septum, the rib cage's outer surface and the head face (flesh-bone ties).
- **Trout fillet with its rib strip** (task A4, `trout.build_fillet_rib_strip`): the same trout cut flat along the backbone just above the centra so the upper ribs stay in the belly flap, trimmed and turned skin-down; the ribs stay bonded along the rib-cage interface.
- **Salmon half side** (`anatomy/salmon.py`): white and dark muscle, belly flap, a belly fat strip whose trim line is a reference line through continuous tissue, a seeded melanin spot, pin bones rooted in the flesh with a shaft bond strength derived from a measured pull-out force, and the collarbone.
- **Pork tenderloin** (`anatomy/tenderloin.py`): main muscle with the chain along one flank (a seam interface), flake fat lumps and silver skin as a membrane on part of the upper surface.

Materials by region (`assembly._material_table`): every fish region takes the fish muscle's values with its connective sheets; the salmon's belly fat strip takes them too, through the open factors E_factor 1, R_factor 1 (no measurement of the lipid-rich belly flap's stiffness or cutting toughness was found); the tenderloin's main muscle and chain take pork's values without sheets, its flake fat chilled pork back fat. The skin membrane carries only its density's excess over the flesh it covers, whose elements already fill the volume to the surface. The tissue mesh edge per workpiece is coupon 3.5 mm, trout 3.5 mm, trout_fillet_rib_strip 3.5 mm, salmon 5 mm, tenderloin 3.5 mm (`world.py`, `MESH_H`); the salmon side uses the coarser edge because its finer mesh ran far over the runtime limit. Element counts, masses and generator checks are in Section 8.

2.4 Parameter ledger

Every physical constant the environment uses lives in `filleting/params.py` with its value, unit, a provenance label and the basis for the value. The labels are: *read* (a primary source read), *abstract* (an abstract or quoted table

only), *search* (a search summary only), *literature* (a handbook value), *repo* (a value qualified in another environment of this benchmark), *measured* (measured on this environment's own generator), *derived* (arithmetic shown in the basis), *prior* (recalled, not re-read), *estimate* (an engineering estimate), *authored* (a design choice) and *open* (no digital basis; the basis gives a sweep range). Compound labels such as *read(porcine)+estimate* say that a value was read for another species, tissue or temperature and transferred by estimate.

The ledger has 83 entries; 20 have *open* as their primary class and 21 mention *open* anywhere in their label. The open values that most affect the tasks are the interface fracture energies and strengths (flesh-bone, chain seam, silver skin), blade-tissue friction and chilled fat stiffness. They are placeholders with sweep ranges, and no task judgement has been re-run across those ranges; the coupon's cut has been re-run one value at a time over the bulk tensile strength, the sheets' strength and fracture energy and the blade friction (Section 8). 12 ledger entries are named by no world module (`FISH_HEAD_E, FISH_HEAD_R, SKIN_FLESH_G, SKIN_FLESH_SIGMA, BONE_CRUSH, BONE_SHEAR, PIN_PULL_SALMON, PIN_BREAK_SALMON, PIN_LENGTH_SALMON, TAUO_BLADE_TISSUE, FRANKA_HAND, CODEX_BONE`): they document a value the world does not read from the ledger, either because the mechanism is not modelled or because the world uses its own constant. Table 3 lists the numerical, coupling and assembly constants that live outside the ledger, including numbers written inline in `assembly.py` that carry neither a name nor a provenance label.

Primary class	Count	Entries
read	27	FISH_NU, SKIN_THICKNESS_TROUT, SKIN_THICKNESS_SALMON, SKIN_STRENGTH, BONE_E, BONE_UTS, BONE_CRUSH, PIN_PULL_SALMON, PIN_BREAK_SALMON, PIN_COUNT_SALMON, PIN_LENGTH_SALMON, PIN_ROOT_SHARE, PORK_E, FAT_R, FILLET_BLADE, BONING_BLADE, HANDLE, BSI_HONED, BSI_WORN, REFLEX_TORQUE_NM, REFLEX_TORQUE_CONTACT_NM, JOINT_ACCEL_PANDA, FRANKA_HAND, FR3, TROUT_FILLET_YIELD, IMPS, CODEX_BONE
abstract	2	FISH_DENSITY, JOINT_R
search	2	SKIN_E, TENDERLOIN
literature	3	PORK_DENSITY, STEEL_E, STEEL_G
repo	1	MU_INSERT_HANDLE
measured	1	TROUT
derived	7	FISH_E, TISSUE_VISCO, FISH_R_PERP, FISH_R_PAR, PIN_SHAFT_TAU, PORK_SIGMA_T, SALMON_SIDE
prior	1	JOINT_ACCEL_FR3
estimate	15	FISH_TAU_V, FISH_HEAD_E, FISH_HEAD_R, SHEET_BETA, SKIN_DENSITY, SKIN_R, BONE_DENSITY, BONE_SHEAR, PORK_R, MU_BLADE_BONE, MU_SKIN_HDPE, MU_FLESH_HDPE, MU_PAD_SKIN, MU_PAD_FLESH, MU_TWEEZER_BONE
authored	4	RIG_FORCE_MAX, RIG_TORQUE_MAX, RIG_STIFFNESS, REFLEX_RECOVERY_S
open	20	SHEET_SIGMA, SHEET_G, FISH_SIGMA_T, SKIN_FLESH_G, SKIN_FLESH_SIGMA, FLESH_BONE_G, FLESH_BONE_SIGMA, FAT_E, FAT_SIGMA_T, SALMON_BELLY, SILVER_THICKNESS, SILVER_E, SILVER_STRENGTH, SILVER_R, SILVER_BOND_G, CHAIN_SEAM_G, CHAIN_SEAM_SIGMA, SILVER_BOND_SIGMA, MU_BLADE_TISSUE, TAU0_BLADE_TISSUE
total	83	

Table 2: Parameter ledger, generated from `filleting/params.py` (83 entries). Label is the provenance label as written in the code; the primary class is its first word. *Read by* lists the world modules whose source names the parameter; *none* means no world module reads it from the ledger.

Quantity	Value	Unit	Label	Basis	Read by
Fish muscle density `FISH_DENSITY`	1050	kg/m^3	abstract+ estimate	fresh seafood 1042-1093 kg/m3 at 20 C (Rahman & Driscoll 1994); fatty salmonids at the low end	anatomy/ salmon, anatomy/ trout, assembly, workpiece

Quantity	Value	Unit	Label	Basis	Read by
Fish muscle Young's modulus `FISH_E`	6×10^4	Pa	derived	raw cod G' ~ 22 kPa -> E ~ 66 kPa (Blikra 2019 via arXiv 2112.08280); raw tuna large-strain secant 150-180 kPa (Jumilla-Lorenz & Pares 2025); checked in probe E1 against salmon flat-cylinder breaking force 6.6-20.9 N (Torgersen 2014)	assembly, workpiece
Fish muscle Poisson ratio `FISH_NU`	0.45	–	read (porcine)+ estimate	porcine nu_LT 0.47 (Takaza 2013); fish unmeasured; nearly incompressible	assembly, workpiece
Tissue damping time `FISH_TAU_V`	0.0001	s	estimate	stiffness-proportional (Kelvin-Voigt on the Green strain rate, eta = tau*mu, normalised by the stretch, 3 / max(3, tr C)) damping time. It damps kHz mesh-scale oscillation only and is chosen small enough not to set the explicit step; slow quasi-linear relaxation of muscle (porcine: ~50 % viscous part at 0.5 %/s, Van Loocke 2008) is NOT modelled	assembly, tissue, workpiece
Flesh structural damping branch `TISSUE_VISCO`	g=0.2; tau=0.015	–	derived+ estimate	structural damping of the flesh: one Maxwell branch on the deviatoric Green strain (relative modulus g, relaxation time tau) beside the elastic law. Fish muscle's loss modulus is about one order of magnitude below its storage modulus over 1-100 rad/s (cooked cod/lutefisk, arXiv 2112.08280; raw cod, Blikra et al. 2019): tan delta ~0.1. One branch with g 0.2 has tan delta at most g/(2 sqrt(1+g)) = 0.09, at 1/(tau sqrt(1+g)) = 61 rad/s, near the workpieces' lowest modes (~10 Hz); a broader spectrum (and the slow relaxation, FISH_TAU_V) is not modelled. Used for pork too (estimate). With only the numerical damping (zeta ~0.004 at 13 Hz) a disturbed fish rings for seconds	tissue
Fish toughness across the sheets `FISH_R_PERP`	300	J/m^2	derived	tuna razor shear 4.3-5.5 N across the connective sheets via V = R*w, w = 8.9-30 mm -> 140-620 J/m2 (upper bound: friction and deformation included)	assembly, workpiece
Fish toughness along the sheets `FISH_R_PAR`	100	J/m^2	derived	tuna razor shear 1.8-2.0 N parallel to the sheets, same method -> 60-225 J/m2	assembly, workpiece
Fish head lumped stiffness `FISH_HEAD_E`	1×10^6	Pa	estimate reference, not a model input: the head is a rigid body	head (skull, gill covers, cheek) lumped as a stiff region; no source	*none*
Fish head lumped toughness `FISH_HEAD_R`	4000	J/m^2	estimate reference, not a model input: the head is a rigid body, removed at its joint, not cut through	head lumped toughness; the head is removed at the joint, not cut through	*none*
Myoseptal sheet strength `SHEET_SIGMA`	1×10^4	Pa	**open**	myoseptal sheet strength: no measurement found; sweep range 5-20 kPa, not run	tearing
Myoseptal sheet fracture energy `SHEET_G`	3	J/m^2	**open**	myoseptal sheet fracture energy: no measurement; sweep range 1-10 J/m2, not run	tearing
Sheet failure law shear weight `SHEET_BETA`	1	–	estimate	weight of the shear traction in the sheet failure law sqrt(<t_n>^2 + (beta t_s)^2) ; no measurement	tearing

Quantity	Value	Unit	Label	Basis	Read by
Fish muscle tensile strength off the sheets FISH_SIGMA_T	5×10^4	Pa	**open**	fish muscle tensile strength off the sheets: no accessible measurement; checked against the flat-cylinder puncture (Torgersen 2014: a 12.5 mm flat cylinder at 1 mm/s breaks the salmon fillet surface at 6.6-20.9 N; probe E1)	assembly, workpiece
Trout skin thickness SKIN_THICKNESS_TROUT	0.0003	m	read(other species)+ estimate	striped bass 0.23-0.47 mm; grass carp 0.55 mm	anatomy/ trout
Salmon skin thickness SKIN_THICKNESS_SALMON	0.0005	m	read(other species)+ estimate	as above; salmon skin thicker	anatomy/ salmon
Fish skin in-plane modulus SKIN_E	2×10^7	Pa	search	bass skin ~13 MPa at 0 % strain rising to 32-40 MPa at 20 %	assembly
Fish skin tensile strength SKIN_STRENGTH	3×10^7	Pa	read	striped bass stratum compactum 20-60 MPa, failure strain 0.2-0.5 (Szewciw 2017)	assembly, tearing
Skin density SKIN_DENSITY	1100	kg/m^3	estimate	collagenous dermis	anatomy/ trout, assembly
Skin toughness SKIN_R	2000	J/m^2	estimate	within the dermis scissor-cut range 1-3 kJ/ m2 (search excerpt, mammalian dermis); fish skin is thinner and weaker, so not above it. The 17 kJ/m2 of pork dermis (Comley & Fleck 2010) is a trouser TEAR, not a cut	assembly, tissue
Skin-flesh fracture energy SKIN_FLESH_G	30	J/m^2	**open** reference, not a model input: the skin membrane shares the flesh's surface nodes: no skin-flesh interface (skinning, B4, is refused)	skin-flesh interface fracture energy; no peel test found. Not modelled: the skin membrane shares the flesh's surface nodes (no interface; skinning, B4, is refused for it); kept for the finite-difference check of the interface law	*none*
Skin-flesh strength SKIN_FLESH_SIGMA	2×10^4	Pa	**open** reference, not a model input: as SKIN_FLESH_G	skin-flesh interface strength; as SKIN_ FLESH_G, not modelled	*none*
Flesh-bone fracture energy FLESH_BONE_G	50	J/m^2	**open**	flesh-bone interface energy; no measurement; sweep range 20-200, not run	anatomy/ salmon, anatomy/ trout, assembly
Flesh-bone strength FLESH_BONE_SIGMA	3×10^4	Pa	**open**	flesh-bone interface strength; sweep range 10-80 kPa, not run	anatomy/ salmon, anatomy/ trout, assembly
Intervertebral joint toughness JOINT_R	1500	J/m^2	abstract+ estimate	intervertebral joint tissue; perimysium work of fracture 0.4-1.8 kJ/m2 (cooked beef, Purslow 1985)	anatomy/ trout, workpiece
Fish bone Young's modulus BONE_E	3×10^9	Pa	read	herring intermuscular bone 2.6-3.5 GPa (Fiedler 2019)	assembly
Fish bone tensile strength BONE_UTS	1.1×10^8	Pa	read	herring intermuscular bone 108-111 MPa (Fiedler 2019)	assembly
Fish bone density BONE_DENSITY	2000	kg/m^3	estimate	mineralised fish bone	anatomy/ salmon, anatomy/ trout, assembly

Quantity	Value	Unit	Label	Basis	Read by
Bone compressive strength `BONE_CRUSH`	1.5×10^8	Pa	read(other tissue)+ estimate reference, not a model input: vertebrae are rigid and cannot be cut; thin bones break at BONE_UTS	mammalian cortical compressive strength 154-205 MPa	*none*
Bone shear strength `BONE_SHEAR`	5.5×10^7	Pa	estimate reference, not a model input: no edge-on-bone shear failure is modelled; thin bones break at BONE_UTS	~half the tensile strength; a 1 mm rib needs ~40 N to be severed by an edge	*none*
Pin-bone pull-out force (mean, s.d.) `PIN_PULL_SALMON`	(5.41, 1.71)	N	read reference, not a model input: the source of the derived PIN_SHAFT_TAU (a model input)	post-rigor salmon, 1 mm/s tension (Schroeder et al. 2019)	*none*
Pin-bone breaking force (mean, s.d.) `PIN_BREAK_SALMON`	(3.91, 2.56)	N	read reference, not a model input: pin-boning (B1) is refused	tensile grips 1 mm/s (Schroeder et al. 2019)	*none*
Salmon pin-bone count `PIN_COUNT_SALMON`	29	–	read	micro-CT, 3-5 kg fish (Schroeder et al. 2019)	anatomy/ salmon
Salmon pin-bone length range `PIN_LENGTH_SALMON`	(0.014, 0.05)	m	read reference, not a model input: the salmon's pin bones are generated by anatomy/salmon.py	micro-CT range (Schroeder et al. 2019)	*none*
Pin-bone shaft bond strength `PIN_SHAFT_TAU`	7.8×10^4	Pa	derived	5.41 N / (pi * 0.55 mm * 40 mm)	assembly
Pin-bone root share of extraction `PIN_ROOT_SHARE`	0.5	–	read+derived	cutting the root tendon halves extraction force (SINTEF 2009)	anatomy/ salmon
Pork muscle Young's modulus `PORK_E`	1×10^5	Pa	read+ estimate	raw porcine longissimus in tension (Takaza 2013, abstract read): 10 kPa along and 77 kPa across the fibres at stretch 1.1, i.e. secant moduli 10/0.1 = 100 kPa along and 770 kPa across (across: 'broadly linear'); the isotropic model takes the soft, along-fibre value and so ignores a 7.7x anisotropy that matters most for slicing across the fibres (C3)	assembly
Pork muscle toughness `PORK_R`	400	J/m^2	estimate	Atkins' illustrative beef value 400, bracketed by frozen beef 100-200 (Dobraszczyk 1987) and raw pork 840 +- 800 in notched tension (Aryeetey 2022; not a cutting toughness)	assembly

Quantity	Value	Unit	Label	Basis	Read by
Pork muscle density PORK_DENSITY	1060	kg/m^3	literature	general meat density	anatomy/ tenderloin, assembly
Pork muscle tensile strength PORK_SIGMA_T	1.15×10^5	Pa	derived	raw porcine longissimus across the fibres (Takaza 2013, abstract read): broadly linear, 77 kPa at stretch 1.1, failure at stretch ~1.15, so 77 x 0.15/0.10 ~ 115 kPa at failure (along the fibres failure comes at stretch ~1.65)	assembly
Chilled fat Young's modulus FAT_E	1×10^5	Pa	**open**	chilled back fat: ~1 kPa at 25-37 C (Comley & Fleck 2012) and 'much firmer' chilled (Wijarnprecha 2022); chilled value unmeasured, sweep range 10-500 kPa, not run	assembly
Fat toughness FAT_R	3000	J/m^2	read(warm)+ open	adipose trouser tear 4.1 kJ/m2 at body temperature (Comley & Fleck 2010)	assembly
Fat tensile strength FAT_SIGMA_T	5×10^4	Pa	**open**	chilled fat tensile strength: no measurement	assembly
Salmon belly strip factors on the fish values SALMON_BELLY	E_factor=1; R_ factor=1	–	**open**	salmon belly flap strip (the Trim D strip, region belly_fat): fish muscle richer in lipid, 28.1 % of wet weight against 9.6 % in the white muscle (Aursand et al. 1994, J Sci Food Agric 64:239, abstract), adipocyte-dominated (Aquaculture 2007, abstract); no measurement of its stiffness or cutting toughness at chill temperature was found, so it takes the fish muscle's values (factors on FISH_E and FISH_R); sweep range 0.5-2 each, not run. Chilled pork back fat (FAT_E 100 kPa, FAT_R 3 kJ/m2, 920 kg/m3) would give 10 times the fish toughness on B2's trim line	assembly
Silver skin thickness SILVER_THICKNESS	0.0002	m	**open**	silver skin thickness; no measurement; 0.1-0.5 mm	anatomy/ tenderloin
Silver skin modulus SILVER_E	5×10^7	Pa	**open**	epimysium membrane modulus; no measurement	anatomy/ tenderloin, assembly
Silver skin strength SILVER_STRENGTH	1×10^7	Pa	**open**	no measurement	anatomy/ tenderloin, tearing
Silver skin toughness SILVER_R	2000	J/m^2	**open**	no measurement of epimysium cutting; taken as the collagenous dermis's scissor-cut range 1-3 kJ/m2 (SKIN_R)	anatomy/ tenderloin, assembly, tissue
Silver skin-muscle fracture energy SILVER_BOND_G	150	J/m^2	**open**	hand-peeling is possible but hard; sweep range 50-500, not run	anatomy/ tenderloin
Chain seam fracture energy CHAIN_SEAM_G	30	J/m^2	**open**	no measurement; sweep range 10-100, not run	anatomy/ tenderloin
Chain seam strength CHAIN_SEAM_SIGMA	1×10^4	Pa	**open**	chain seam (loose fat and fascia) peak traction; no measurement; placeholder below the skin-flesh value, sweep range 5-30 kPa, not run	anatomy/ tenderloin
Silver skin-muscle strength SILVER_BOND_SIGMA	5×10^4	Pa	**open**	silver skin to muscle peak traction; no measurement; hand-peeling is possible but hard: placeholder above the flesh-bone value, sweep range 20-100 kPa, not run	anatomy/ tenderloin
Fillet knife blade FILLET_BLADE	length=0.173; height=0.025; spine=0.0012; mass=0.088; bevel_deg=18	m,kg	read	Victorinox Fibrox 5.3703.18 (knivesandtools datasheet)	knife, scene
Boning knife blade BONING_BLADE	length=0.15; height=0.02; spine=0.002; mass=0.102; bevel_deg=18	m,kg	read	Victorinox Fibrox 5.6303.15 (knivesandtools datasheet)	knife, scene

Quantity	Value	Unit	Label	Basis	Read by
Knife handle envelope HANDLE	length=0.14; height=0.032; width=0.019	m	read+ estimate	Victorinox 5.3813.18 handle envelope 19 x 32 mm (manufacturer); 140 mm length (retailer)	scene, world
Blade steel Young's modulus STEEL_E	2×10^{11}	Pa	literature	stainless blade steel	knife, workpiece
Blade steel shear modulus STEEL_G	8×10^{10}	Pa	literature	stainless blade steel	workpiece
Blade sharpness index, honed BSI_HONED	0.22	–	read+ estimate	scalpel sharp BSI 0.216 (McCarthy 2007); transfer to tissue estimated	knife
Blade sharpness index, worn BSI_WORN	0.5	–	read+ estimate	scalpel blunted BSI 0.514 (McCarthy 2007)	knife
Blade-tissue friction MU_BLADE_TISSUE	0.2	–	**open**	no wet-tissue measurement; fitted food mu 0.18-1.0 (Jamdagni & Jia)	scene, workpiece
Blade-tissue adhesive shear TAU0_BLADE_TISSUE	1000	Pa	**open** reference, not a model input: blade-tissue adhesion is not modelled (Coulomb friction, MU_BLADE_TISSUE)	wet adhesive shear; Atkins tau_f 10-20 kPa for sticky cheese/salami at -8 C	*none*
Blade-bone friction MU_BLADE_BONE	0.25	–	estimate	steel on wet bone	workpiece
Skin-board friction MU_SKIN_HDPE	0.15	–	estimate	fish mucus 'ultra-low friction' (Wu et al. 2014, qualitative)	workpiece
Flesh-board friction MU_FLESH_HDPE	0.3	–	estimate	wet flesh on polymer	scene, workpiece
Finger pad-skin friction MU_PAD_SKIN	0.3	–	estimate	rubber on slimy skin	workpiece
Finger pad-flesh friction MU_PAD_FLESH	0.5	–	estimate	rubber on wet flesh	workpiece
Insert liner-handle friction MU_INSERT_HANDLE	0.8	–	repo	shared Franka finger pad friction	scene
Collision reflex thresholds REFLEX_TORQUE_NM	(10, 20)	N m	read	libfranka examples/examples_common.cpp setDefaultBehavior: upper collision torque thresholds (nominal, acceleration) on every joint	world
Collision-reflex torque thresholds, contact profile REFLEX_TORQUE_CONTACT_NM	(100, 100)	N m	read	libfranka examples/force_control.cpp and cartesian_impedance_control.cpp setCollisionBehavior: 100 N m upper torque thresholds on every joint for contact tasks (github.com/frankaemika/libfranka master); the 'contact' collision profile a deployment may choose for cutting and pressing (World Config collision_profile)	world
Phantom rig force cap RIG_FORCE_MAX	150	N	authored	a firm push of a hand on a knife; above the largest cutting load the assisted demonstrations need (55 N, B5 rock-and-draw), so it only caps rams into bone or the board	tissue
Phantom rig torque cap RIG_TORQUE_MAX	15	N m	authored	a wrist holding a knife against a press on its blade (wrist strength of the order of 10-20 N m: recalled, not a sourced measurement); 5 N m let a 33 N press 150 mm down the blade pitch the knife 5.6 deg	tissue
Phantom rig servo stiffness RIG_STIFFNESS	(2×10^5, 600)	N/m, N m/rad	authored	the servo's tracking stiffness (translation, rotation): 0.13 mm behind the path under a 25 N press cut, 0.24 deg under 2.5 N m; critically damped with the knife handle's own mass and inertia (bodies.k_body_ servo)	tissue

Quantity	Value	Unit	Label	Basis	Read by
Protective-stop hold time REFLEX_RECOVERY_S	1	s	authored	after a collision reflex the arm holds and refuses new targets this long (the real FR3 stays stopped until an error recovery)	world
Panda joint acceleration limits JOINT_ACCEL_PANDA	(15, 7.5, 10, 12.5, 15, 20, 20)	rad/s^2	read	libfranka include/franka/rate_limiting.h kMaxJointAcceleration (Panda); the arm's commanded joint targets are rate-limited to these and to the joint speed limits	world
FR3 joint acceleration limits JOINT_ACCEL_FR3	(10, 10, 10, 10, 10, 10, 10)	rad/s^2	prior	Franka's FR3 control-parameter documentation lists 10 rad/s^2 for every joint (recalled; the page could not be re-read, its URL returned 404); the Panda's read values are 7.5-20	world
Tweezer-bone friction MU_TWEEZER_BONE	0.4	–	estimate	textured stainless tips on bone	workpiece
Franka Hand ratings FRANKA_HAND	continuous_n=70; stroke_m=0.08; finger_speed=0.05; pad=(0.018, 0.018)	–	read reference, not a model input: the hand's limits come from the robot catalog's model (robots/models)	Franka Hand manual v1.2 / datasheet	*none*
FR3 arm ratings FR3	payload=3; reach=0.855; tau=(87, 87, 87, 87, 12, 12, 12); k_trans=(10, 3000); k_rot=(1, 300)	–	read	FR3 datasheet v2.4	layout, scene, variants, world
Trout workpiece envelope TROUT	gutted_kg=0.2895; length=0.315; depth=0.0736; width=0.034	kg,m	measured (generator)+ read	31.5 cm gutted farmed trout digitised from FDA RFEIMG-0305, MRI steak cross-sections (Frontiers in Genetics 2021); gutted mass from the generator's own volume x density. A condition-factor estimate (K = 1.2) gives 340 g; FishBase length-weight gives ~268 g gutted at this length	anatomy/ trout, task
Trout skin-on fillet yield TROUT_FILLET_YIELD	(0.69, 0.03)	fraction of gutted	read	Chefs Resources seafood yields: trout skin-on 69 %	anatomy/ trout
Salmon half-side envelope SALMON_SIDE	kg=0.9; length=0.32; width=0.17; loin=0.038	kg,m	derived+ estimate	front half of a 1.5-1.7 kg side (Mowi 2024 yields); thickness unsourced	anatomy/ salmon
Pork tenderloin envelope TENDERLOIN	kg=0.55; length=0.33; diameter=0.065	kg,m	search+ estimate	retail 450-680 g, 30-45 cm, 5-7.5 cm	anatomy/ tenderloin
USDA IMPS pork tolerances IMPS	thickness_tol_le_25mm=0.005; uniformity=0.005; weight_tol_lt_170g=0.007; lean_exposed_denuded=0.9; flake_fat_max=0.003; tail_split_max=0.025; suspension_edge=0.013	m,kg	read	USDA IMPS Series 400 Fresh Pork (2014) sections 3.3, 3.4, 6.3, 6.4, item 415	anatomy/ tenderloin, task

Quantity	Value	Unit	Label	Basis	Read by
Codex fillet bone-defect definition `CODEX_BONE`	per_kg=1; length=0.01; diameter=0.001; ignore_ length=0.005	–	read reference, not a model input: the specification the evaluators' bone checks follow	Codex CXS 190	*none*

Table 3: Numerical, coupling and assembly constants outside the ledger, read by parsing the source (the cited line is the assignment used). Inline numbers have no name in the code.

Meaning	Name	Value	Unit	Source
Rigid (MuJoCo) tick	`rigid_dt`	0.001	s	`world.py:52`
Crack faces inserted into the mesh every this many ticks	`split_every`	10	ticks	`world.py:57`
Tissue mesh edge per workpiece	`MESH_H`	coupon=0.0035; trout=0.0035; trout_fillet_rib_ strip=0.0035; salmon=0.005; tenderloin=0.0035	m	`world.py:35`
MuJoCo solver iterations	`solver_iterations`	60	–	`world.py:60`
MuJoCo friction-to-normal impedance ratio	`solver_impratio`	10	–	`world.py:61`
MuJoCo no-slip post-solver iterations	`solver_noslip_ iterations`	50	–	`world.py:64`
Default force-mode grip per finger	`grip_force`	35	N	`world.py:84`
Finger closing speed	`FRANKA_HAND_FINGER_ SPEED`	0.05	m/s	`world.py:31`
Per-finger normal force that counts as holding the knife	`KNIFE_PINCH_N`	0.5	N	`world.py:36`
Robot contact force that counts as touching the knife	`KNIFE_TOUCH_N`	0.2	N	`world.py:37`
Tissue load below which the knife is not lying on tissue	`KNIFE_TISSUE_N`	0.05	N	`world.py:38`
Ticks without a MuJoCo owner before the knife is free	`KNIFE_RELEASE_TICKS`	3	ticks	`world.py:39`
Ticks with a MuJoCo owner before a free knife is handed back	`KNIFE_GRIP_TICKS`	5	ticks	`world.py:40`
Ticks off the tissue before an unheld knife goes back to MuJoCo	`KNIFE_OFF_TISSUE_TICKS`	100	ticks	`world.py:41`
Largest overlap allowed at hand-back	`KNIFE_GRIP_MAX_PEN`	0.002	m	`world.py:42`
Filter time constant on tissue-to-rigid reactions	`REACTION_TAU`	0.005	s	`workpiece.py:23`
Blade joint linear stiffness	`k_lin`	2×10^5	N/m	`workpiece.py:276`
Swept-surface record interval (CrackRecorder call)	`record_every`	2	ticks	`workpiece.py:140`
Edge-law sample spacing along the blade	`SAMPLE_SPACING`	0.001	m	`knife.py:16`
Half thickness where the bevel meets the flat grind	`EDGE_HALF_SHOULDER`	0.000125	m	`knife.py:15`
Penalty stiffness factor (kappa)	`contact_k_scale`	0.15	–	`tissue.py:1064`
Contact damping ratio (zeta)	`contact_zeta`	0.7	–	`tissue.py:1065`
Node-mass floor, fraction of the median node mass	`mass_floor_frac`	0.2	–	`tissue.py:1077`
Largest tissue substep count per rigid tick	`max_substeps`	400	–	`tissue.py:1080`
Adaptive substep count from the current stiffness bound	`adaptive`	True	–	`tissue.py:1079`
Per-element stiffness softening of slivers	`sliver_cap`	False	–	`tissue.py:1078`
Volume ratio below which tissue under the edge is crushed	`crush_J`	0.5	–	`tissue.py:1410`
Edge tie stiffness per sample (upper bound)	`edge_ke`	5000	N/m	`tissue.py:1409`
Gauss-Seidel passes of the rod length projection	`rod_iters`	8	–	`tissue.py:1932`
Travel over which a blade pays an interface's fracture energy	`IFACE_PAY_M`	0.001	m	`tissue.py:2123`
Contact candidates per particle per tick	`CAND_CAP`	48	–	`tissue.py:38`
Contact candidate margin for the knife's bodies	`CAND_MARGIN_M`	0.03	m	`tissue.py:39`

Meaning	Name	Value	Unit	Source
Contact candidate margin for bones and static shapes	`CAND_SLOW_MARGIN_M`	0.003	m	`tissue.py:40`
Edge-to-support gap below which the remaining layer is severed	`sever_gap`	0.0006	m	`topology.py:69`
Tearing skips elements this many mean edge lengths from the knife edge	`EDGE_ZONE_H`	1.5	–	`tearing.py:182`
Ticks with no tissue load before a MuJoCo-owned knife's blade follows it rigidly	`BLADE_AIR_TICKS`	20	ticks	`workpiece.py:301`
A loose piece below this mass is debris (evaluators)	`DEBRIS_G`	1	g	`task.py:312`
Debris mass allowed, fraction of the workpiece (evaluators)	`DEBRIS_BUDGET`	0.005	–	`task.py:313`
Swept-surface gap tolerance of the face-crossing test	`seal_m`	0.00025	m	`topology.py:76`
Smallest fragment kept separate (crumb rule)	`min_fragment_m3`	3×10^{-8}	m^3	`topology.py:88`
Re-anchoring distance still joined into the swept surface	`bridge_m`	0.0025	m	`topology.py:62`
Flesh within this fraction of the mesh edge h of a bone is tied to it	`TIE_TOL_FRAC`	0.45	–	`assembly.py:27`
Intervertebral joint linear stiffness (upper bound)	`BACKBONE_JOINT_K_LIN`	5×10^4	N/m	`assembly.py:30`
Intervertebral joint rotational stiffness (upper bound)	`BACKBONE_JOINT_K_ANG`	0.02	N m/rad	`assembly.py:29`
Knife-handle insert per finger	`INSERT`	length=0.06; depth=0.018; lip=0.004; lip_ height=0.004; back=0.004; mass=0.025; mu=0.8	m, kg	`scene.py:28`
Blade segment centre-of-mass height / blade height	`SEG_COM_Z_FRAC`	0.6	–	`scene.py:30`
Membrane Poisson ratio, skin and silver skin (inline; not in the ledger)	*inline*	0.45	–	`assembly.py:269`
Fat density (inline; not in the ledger)	*inline*	920	kg/m^3	`assembly.py:42`
Pork and fat Poisson ratio: the fish value is reused	*inline*	0.45	–	`assembly.py:40`
Rod root strength as a fraction of the shaft's tensile capacity	*inline*	0.5	–	`assembly.py:236`

3 Tissue mechanics, integration and the cutting law

The laws below are written as the code implements them; each subsection names the function that does it.

3.1 Two solvers and one clock

Newton 1.5 builds the rigid scene (bench, board, vessels, block, both arms and hands, the knife's handle) and MuJoCo integrates it at a rigid tick $\Delta t_r = 0.001\,\mathrm{s}$. The environment's own GPU solver (`tissue.py`, Warp) advances the workpiece, its rigid bones and the knife's flexible blade over the same interval in n substeps of $\Delta t = \Delta t_r / n$ (nominal 10^{-5} s for the anatomical workpieces and 2×10^{-5} s for the coupon, `world.Config.tissue_dt`). There is one clock. Every rigid collider the workpiece or a free knife can touch is a kinematic shape of the tissue solver: its pose at the start of the tick and its twist over the tick come from the MuJoCo state, so within the substeps it moves exactly as the rigid body moved (`workpiece._pose_shapes`). The tissue's contact, friction and edge reactions on each shape are summed per rigid body as a wrench W_k and applied to MuJoCo during the next tick through a first-order filter (`workpiece.tissue_tick`):

$$\tilde{W}_{k+1} = (1-a)\,\tilde{W}_k + a\,W_k, \qquad a = \frac{\Delta t_r}{\tau_R + \Delta t_r}, \qquad \tau_R = 0.005\,\mathrm{s}. \tag{1}$$

The filter stands in for pad and handle compliance; an unfiltered explicit one-tick coupling to finger links of a few tens of grams is an unstable spring loop. Rendering runs in a separate process from recorded states and never steps either solver.

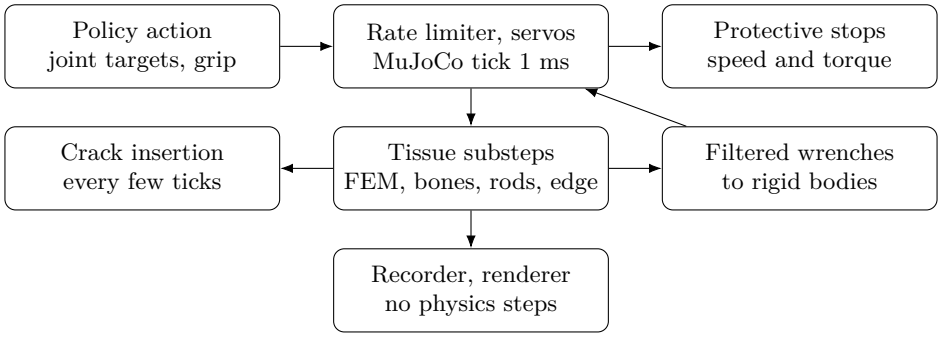


Figure 1: One rigid tick. Rigid shapes enter the tissue solver kinematically; tissue reactions return to MuJoCo filtered by one tick. The renderer reads recorded state only.

3.2 Bulk tissue

Tissue is linear tetrahedra with the stable neo-Hookean energy used by Newton's VBD solver, without its logarithmic term [2] (`tissue.k_tet_forces`, `k_tet_energy`). For an element with rest edge matrix D_m and current edge matrix D_s,

$$F = D_s D_m^{-1}, \quad J = \det F, \quad I_C = \operatorname{tr}(F^T F), \quad \mu = \frac{E}{2(1+\nu)}, \quad \lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}, \quad \lambda' = \lambda + \mu, \quad \alpha = 1 + \frac{\mu}{\lambda'}. \tag{2}$$

The energy density and first Piola–Kirchhoff stress are

$$\Psi(F) = \frac{\mu}{2}(I_C - 3) + \frac{\lambda'}{2}\left[(J-\alpha)^2 - (1-\alpha)^2\right], \qquad P = \frac{\partial \Psi}{\partial F} = \mu F + \lambda'(J-\alpha)\operatorname{cof} F, \quad \operatorname{cof} F = JF^{-T}. \tag{3}$$

At rest $P(I) = \mu I + \lambda'(1-\alpha)I = 0$, so the rest state is stress-free without the logarithmic term, and the energy is defined for inverted elements ($J < 0$). The constant $(1-\alpha)^2$ only shifts Ψ to zero at rest. Viscous stress is Kelvin–Voigt on the rate of the right Cauchy–Green tensor $C = F^T F$,

$$\dot{C} = \dot{F}^T F + F^T \dot{F}, \qquad P_v = \eta' F \dot{C}, \qquad \eta' = \tau_v \mu \frac{3}{\max(3, I_C)}, \qquad \dot{\mathcal{D}}_{\text{visc}} = V_e \frac{\eta'}{2} \dot{C} : \dot{C}, \tag{4}$$

with $\tau_v = 1 \times 10^{-4}$ s for every tissue. It damps kilohertz mesh-scale oscillation only. The flesh's structural damping is a Maxwell branch beside the elastic law (`tissue.k_tet_forces`, g 0.2, tau 0.015): an overstress S_m (second Piola–Kirchhoff) on the deviatoric Green strain $E = \frac{1}{2}(C - I)$,

$$\dot{S}_m + \frac{S_m}{\tau_m} = 2\mu_m \dot{E}_{\text{dev}}, \qquad S_m^{n+1} = a\, S_m^n + 2\mu_m b\, \Delta E_{\text{dev}}, \quad a = e^{-\Delta t/\tau_m}, \; b = \frac{\tau_m}{\Delta t}(1-a), \qquad P_m = F\, S_m^{n+1}, \tag{5}$$

with $\mu_m = g\,\mu$; its stored energy $V_e |S_m|^2/(4\mu_m)$ is counted in Ψ and the step books $V_e\,[S_m^n : \Delta E - \Delta(|S_m|^2/4\mu_m)]$ as viscous dissipation. With $g = 0.2$ and $\tau_m = 15$ ms its loss tangent is at most $g/(2\sqrt{1+g}) = 0.09$, near 10 Hz, against $G''/G' \approx 0.1$ measured for fish muscle; the slow quasi-linear relaxation of muscle over seconds is not modelled. With the numerical term alone (a damping ratio of 0.004 at 13 Hz) a disturbed fish rings for seconds. The factor $3/\max(3, I_C)$ (one at rest) keeps this numerical damping from stiffening with stretch: on the Green strain rate its tangent grows as the largest stretch squared, and beside a cut it sets the explicit step's limit thirty times below its rest value. With the element's material volume V_e, the nodal forces are the columns of $H = -V_e(P + P_v)D_m^{-T}$ on nodes 1–3 and minus their sum on node 0. Fish muscle uses $E = 60$ kPa, $\nu = 0.45$ and density 1050 kg/m^3; pork muscle $E = 100$ kPa with the fish Poisson ratio and density 1060 kg/m^3; chilled fat $E = 100$ kPa (open) with density 920 kg/m^3 written inline (Table 3).

3.3 Explicit integration, stability mass and substeps

Tissue nodes advance by symplectic Euler (`tissue.k_integrate`):

$$v_i^{n+1} = v_i^n + \Delta t \left(\frac{f_i}{m_i} + g\, \frac{m_i^0}{m_i} \right), \qquad x_i^{n+1} = x_i^n + \Delta t\, v_i^{n+1}, \tag{6}$$

where m_i^0 is the node's physical mass and $m_i \geq m_i^0$ its inertia. Gravity acts on the physical mass only; the kinetic energy uses m_i and the gravitational energy m_i^0:

$$K = \sum_i \tfrac{1}{2} m_i |v_i|^2, \qquad U_g = -\sum_i m_i^0\, g \cdot x_i. \tag{7}$$

The physical mass is lumped from the elements plus the point masses of rods and skin, $m_i^0 = \sum_{e \ni i} \rho_e V_e/4 + m_i^{\mathrm{pt}}$; each element keeps the mass it had when it was created, so cutting cannot create or destroy material (`tissue.recompute_mass`). The inertia is raised where a Gershgorin bound on the node's stiffness row would violate the explicit stability limit:

$$K_i = \sum_{e \ni i} (\lambda + 2\mu)(1 + 2\xi_e)\, V_e\, |g_i^e| \sum_j |g_j^e| \;+\; k_i^{\mathrm{ties}} + k_i^{\mathrm{bonds}} + k_i^{\mathrm{skin}}, \qquad \xi_e = \frac{2\tau_v \mu}{\lambda + 2\mu}\,\frac{c_e}{h_e}, \quad c_e = \sqrt{\frac{\lambda + 2\mu}{\rho_e}}, \tag{8}$$

$$m_i = \max\!\left(m_i^0,\; K_i \left(\frac{\Delta t}{1.5}\right)^2,\; m_{\mathrm{floor}} \right), \tag{9}$$

with g^e the element's shape-function gradients, h_e its smallest height and m_{floor} a floor of 0.2 times the median node mass of the uncut mesh on tissue nodes. The resulting $\omega \Delta t \leq 1.5$ for the elements leaves the rest of symplectic Euler's limit of 2 to the contacts, ties and the edge tie that act on the same node. The added mass is inertia, not weight; Section 8 reports it per workpiece. The per-element sliver softening in the same function is switched off (`sliver_cap` = False).

The substep count follows Gershgorin bounds of each node's stiffness and damping, evaluated every tick in the current configuration (`k_stiffness_bound`, `Tissue.step_tick`); $\bar{s}(M)$ is the largest absolute row sum of a symmetric matrix, an upper bound of its largest eigenvalue:

$$\tilde{K}_i = \sum_{e \ni i} V_e \Big[\mu + \lambda' \Big(\bar{s}(\mathrm{cof}\, F^T\, \mathrm{cof}\, F) + |J - \alpha| \sqrt{\bar{s}(F^T F)} \Big) \Big]\, |g_i^e| \sum_j |g_j^e|, \tag{10}$$

$$\tilde{C}_i = \sum_{e \ni i} V_e\, \eta' \Big[|F g_i^e| \sum_j |F g_j^e| + \bar{s}(F^T F)\, |g_i^e| \sum_j |g_j^e| \Big], \tag{11}$$

$$\Delta t_{\mathrm{ok}} = \min_i \frac{2.25\, m_i}{\tilde{C}_i + \sqrt{\tilde{C}_i^2 + 2.25\, \tilde{K}_i m_i}}, \qquad n = \min\!\left(\left\lceil \frac{\Delta t_r}{\min(\Delta t, \Delta t_{\mathrm{ok}})} \right\rceil,\; n_{\max} \right), \tag{12}$$

so that $\tilde{K}_i \Delta t^2 + 2\tilde{C}_i \Delta t \leq 2.25\, m_i$: symplectic Euler with explicit damping is stable for $\omega^2 \Delta t^2 + 2c\, \Delta t < 4$. The volumetric block is $\lambda'(\mathrm{cof}\, F\, g_i)(\mathrm{cof}\, F\, g_j)^T$ and the determinant's curvature block the cross-product matrix of $F(g_i \times g_j)$; the damping block is $V_e \eta' [(Fg_j)(Fg_i)^T + (g_i \cdot g_j) F F^T]$. Without the damping term the bound lets stretched flesh go non-finite. The substep count is capped at $n_{\max} = 400$; a tick that needs more is recorded as a `substep_cap` event. The substep loop runs as a captured CUDA graph.

3.4 Contact with rigid shapes

Each node (with its radius r_i) is tested against the rigid shapes within reach: boxes, capsules and the blade's wedge segments (`tissue.k_contacts`). A blade segment in its local frame (x along the blade, y through the thickness, z from the edge at $z = 0$ to the spine at $z = H$) has half-thickness

$$w(z) = \min\big(z \tan\beta,\; e + (h_y - e)\, z/H \big), \tag{13}$$

with bevel angle β, shoulder half-thickness $e = 0.125\,\mathrm{mm}$ and half spine thickness h_y; points below the edge are outside the wedge, because the edge itself is handled by the cutting law. A node on a crack face is always pushed toward its own side's flank inside the blade slab. The normal force is a penalty scaled to the node's mass with a normal damper of damping ratio ζ,

$$k_n = \kappa\, \frac{\tilde{m}_i}{\Delta t_0^2}, \quad c_n = 2\zeta \sqrt{k_n \tilde{m}_i}, \quad f_n = \max\big(0,\; k_n p - c_n v_n\big), \qquad \tilde{m}_i = \min(m_i, m_{\mathrm{cap}}), \tag{14}$$

with penetration p, normal relative speed v_n, $\kappa = 0.15$, $\zeta = 0.7$, the nominal substep Δt_0 and m_{cap} four times the median node mass. Friction is stick–slip: a tangential spring of stiffness $k_t = k_n/2$ ties the node to an anchor a stored in the shape's frame,

$$f_t = -k_t\,(x_i - a)_\perp, \qquad |f_t| \le \mu f_n, \qquad \Delta\mathcal{D}_{\text{fric}} = \mu f_n\,|\Delta a|, \tag{15}$$

and when $|f_t|$ would exceed μf_n the anchor moves so that the spring sits on the Coulomb limit. The friction coefficients are per pair of shape and node kind (Table 2). On the blade, the nodes of every tetrahedron the blade passes through (nodes on both sides of a segment's mid-plane, the element over the segment; `tissue.k_straddle`, each tick) meet it without friction: a crack follows whole element faces up to $h/2$ from the plane the edge cut, so elements of one piece reach across the 1.2 mm blade, and their push on its flanks is the stretched element's, not the cut surface's. With friction it clamps the blade and lifts the piece with the withdrawn knife. Tissue covered by intact skin does not contact shapes: the skin carries the contact. The reaction on each shape is accumulated as a wrench about its centre of mass, and the power the moving shape delivers, $f \cdot v_{\text{shape}}$, is booked as work put in.

3.5 The knife-edge cutting law

The edge of each blade segment carries samples at a spacing of 1 mm (`knife.py`); a sample stands for an edge length ℓ equal to its segment's length divided by its sample count, and has the local blade thickness t_b, the edge tangent t and the cutting direction n (spine to edge) of its segment (`tissue.k_edge_law`). The law follows the frictionless slicing mechanics of Atkins, Xu and Jeronimidis [1].

Anchoring. A sample that is advancing into material ($v_e \cdot n > 10^{-4}$ m/s) and lies inside an intact element is tied to that material point a (barycentric coordinates in the element); the search runs once per rigid tick. A withdrawing knife never re-anchors in the tissue it passes on the way out.

Tie load. The edge loads the material through a tie of stiffness k_e, limited by the effective mass of the material point for explicit stability:

$$G = k_e\,(e - x_a), \quad G_n = G \cdot n, \quad G_t = G \cdot t, \qquad k_e = \min\Big(k_e^0,\ \frac{0.2}{\Delta t_0^2 \sum_j b_j^2/m_j}\Big), \quad k_e^0 = 5000\,\text{N/m}. \tag{16}$$

If $G_n \le 0$ the material point is ahead of the edge and the edge exerts no force; the anchor is released when the edge has withdrawn more than 0.5 mm.

Toughness. Muscle is tougher across its connective sheets than along them. With s the local sheet normal and $c = |(t \times n) \cdot s|$ the cosine between the cut plane's normal and the sheet normal,

$$R(c) = R_\parallel\,c^2 + R_\perp\,(1 - c^2), \qquad R_\perp = 300\,\text{J/m}^2, \quad R_\parallel = 100\,\text{J/m}^2 \text{ (fish)}; \quad R = 400\,\text{J/m}^2 \text{ (pork)}. \tag{17}$$

Where a membrane (skin, silver skin) lies on a face j of the anchor's element and the anchor is within the membrane's thickness t_m of it (barycentric coordinate $b_j < t_m/H_j$, H_j the element's height over that face), the edge cuts the membrane too and $R \leftarrow R + R_m$, $R_m = 2000\,\text{J/m}^2$ (skin) or $2000\,\text{J/m}^2$ (silver skin), so a membrane crossed by the edge costs $R_m t_m$ per unit edge length. The resistance of a sample is $R\ell$, modified by two factors. While the cumulative slip of a new anchor is below the local blade thickness t_b, the edge is initiating a cut and the resistance is $(1 + \text{BSI})R\ell$, with the blade sharpness index BSI = 0.22 honed or 0.5 worn [4]; the extra work over that distance is $\text{BSI}\,R\ell\,t_b$. Where the element under the edge is compressed below the volume ratio $J_c = 0.5$ the tissue has failed and the edge passes at reduced resistance, $R\ell \leftarrow R\ell\ \text{clamp}\big((J - J_c/2)/(J_c/2),\,0.02,\,1\big)$.

Yield locus and flow. The load is admissible inside a circle of diameter $R\ell$ in the (G_n, G_t) plane:

$$\Phi = G_n^2 + G_t^2 - R\ell\,G_n \le 0. \tag{18}$$

Beyond it the load returns radially to the circle and the anchor slips toward the edge along the load direction $u = G/|G|$ (collinear flow):

$$\lambda = \frac{R\ell\,G_n}{|G|}, \qquad s = \frac{|G| - \lambda}{k_e}, \qquad G \leftarrow \lambda u, \qquad a \leftarrow a + s\,u, \qquad \Delta\mathcal{D}_{\text{cut}} = R\ell\,\max(s\,u \cdot n,\,0). \tag{19}$$

The slip component along n is new crack length, so the dissipation per unit of new crack area is exactly R. The material receives $+G$ at the anchor, distributed by its barycentric weights; the blade segment receives $-G$.

Consequence: pressing and slicing. In steady cutting with slice ratio $\xi = G_t/G_n$ (the ratio of the edge's tangential to normal motion under collinear flow), the load lies on the circle, so

$$G_n = \frac{R\ell}{1+\xi^2}, \qquad G_t = \frac{\xi R\ell}{1+\xi^2}. \tag{20}$$

A press cut ($\xi = 0$) needs $R\ell$; slicing lowers the push force as Atkins' slice ratio predicts. These are the predictions the coupon probe records beside its measurements.

3.6 Crack insertion

Every 2 ticks the rest-space anchor positions of adjacent anchored samples of the same segment define quads of the swept surface, with the blade normal $t \times n$ as their normal; runs of samples are extended by half a sample spacing beyond their ends so a cut reaches a free end face (`topology.CrackRecorder.record`). Every 10 ticks the recorded surface is inserted by one global rule (`topology._surface_split`): the face shared by elements a and b is cut if and only if the segment joining their rest centroids crosses a recorded triangle,

$$\text{face } (a,b) \text{ cut} \iff \exists\, T:\ [\bar{X}_a, \bar{X}_b] \cap T_\delta \neq \emptyset, \qquad \delta = 0.25\,\text{mm}, \tag{21}$$

and a face whose elements host a live crack tip waits until the edge has left them. Node stars are then split into fans of elements connected through uncut faces, so the two sides separate exactly along cut faces. Fragments below $3 \times 10^{-8}\,\text{m}^3$ are re-attached by their signed distance to the recorded surface, and a crumb between a new sheet and an older, already split one stays attached across the new sheet. An anchored sample within 0.6 mm of a support's top face, cutting toward it, extends the recorded surface through the remaining layer to the far surface: a thin layer pressed against the board by the edge is severed. Mass is conserved exactly because every element keeps its own rest volume and mass.

3.7 The energy ledger

Every dissipative term is accumulated in double precision on the GPU: viscous, friction (contacts and body samples), cutting, cohesive damage of ties, dampers (contacts, samples, joints, flesh-bone ties, bonds), interface fracture paid by the blade, and topology changes (`Tissue.ledger`). With work put in by moving shapes and driven bodies W_{in}, the probes report

$$r = W_{\text{in}} - \big(\mathcal{D}_{\text{visc}} + \mathcal{D}_{\text{fric}} + \mathcal{D}_{\text{cut}} + \mathcal{D}_{\text{tie}} + \mathcal{D}_{\text{damp}} + \mathcal{D}_{\text{iface}} + \mathcal{D}_{\text{topo}}\big) - \Delta E, \qquad E = K + U_g + \sum_e V_e \Psi_e + E_{\text{bodies}}, \tag{22}$$

where E_{bodies} is the kinetic and gravitational energy of the tissue solver's rigid bodies. The elastic energy stored in the skin membrane, the cohesive bonds and ties, rod bending and body joints is not part of E; the stored energy probe measures it separately (Section 8, mechanics). A residual is reported as measured, never set to zero by construction.

4 Bones, rods, interfaces and skin

4.1 Rigid bones and their joints

The head and the backbone segments are rigid bodies of the tissue solver, integrated at the tissue substep: a 2 g segment tied to about a hundred flesh nodes would be an unstable explicit loop at MuJoCo's 1 ms tick (`bodies.py`). Each body is advanced by semi-implicit Euler with an explicit gyroscopic term (`bodies.k_body_integrate`):

$$v \leftarrow v + \Big(\frac{F}{m} + g\Big)\Delta t, \quad \omega \leftarrow \omega + I_w^{-1}\big(\tau - \omega \times I_w\omega\big)\Delta t, \quad p \leftarrow p + v\,\Delta t, \quad q \leftarrow \text{normalize}\big(q + \tfrac{\Delta t}{2}\,[\omega, 0]\,q\big). \tag{23}$$

Bones carry capsule shapes that tissue nodes contact like any rigid shape (their reaction goes into the bone), sphere samples along the capsules that contact the external shapes (blade, board, finger pads; their reaction goes to MuJoCo), and flesh-bone ties. Flesh that overlaps a bone at rest is bonded to it and exempt from contact with it. Neighbouring bodies are joined by a spring-damper between two anchor points and a rotational spring-damper on the relative rotation, with separate stiffness about the three axes of the first body's frame (`bodies.k_body_joints`):

$$f = k\,(p_b - p_a) + c\,(v_b - v_a), \qquad \tau = R_a\big(K_\theta\,\theta + C_\theta\,R_a^T(\omega_b - \omega_a)\big), \tag{24}$$

where θ is the rotation vector of $q_b(q_a q_0)^{-1}$ resolved in the first body's frame and q_0 the rest relative rotation. For the intervertebral joints (`assembly.py`)

$$k = \min\Big(K_{\text{lin}},\, 0.05\,\frac{m_{\text{eff}}}{\Delta t^2}\Big), \qquad K_\theta = \min\Big(K_{\text{ang}},\, 0.05\,\frac{I_{\text{eff}}}{\Delta t^2}\Big), \tag{25}$$

with $K_{\text{lin}} = 50000\,\text{N/m}$, $K_{\text{ang}} = 0.02\,\text{N m/rad}$ (an estimate, per joint of 62/13 vertebrae, scaled inversely with the vertebrae a joint stands for: the first five vertebrae have a joint each) and 30 % of critical damping. The vertebrae themselves cannot be cut. A joint parts only when a blade segment has passed through it: every five ticks, a joint whose anchor point p lies in a blade segment's slab on the spine side of the edge,

$$|p_x| \le h_x, \qquad -10^{-4}\,\text{m} \le p_z \le 2h_z, \qquad |p_y| \le h_y + 0.3\,\text{mm} \qquad \text{(segment frame)}, \tag{26}$$

is switched off, and the joint tissue's fracture work $R_J \pi r_j^2$ (joint radius r_j, $R_J = 1500\,\text{J/m}^2$) is charged to that blade segment (`workpiece._joint_cut_rule`, Section 4.3). The knife has to find the joint.

4.2 The bilinear cohesive law

Pre-split anatomical interfaces (median septum, rib-cage lining, the salmon's collarbone, chain seam, silver skin) are pairs of node copies joined by bonds (the salmon's belly fat line is a trimmer's visual line through continuous tissue, not a fascial plane, and is not pre-split); flesh-bone ties join a tissue node to a point on a bone; shaft ties join a rod node to the element containing it. All use one bilinear traction–separation law with irreversible secant damage D (`bodies.k_bonds`, `k_body_ties`, `k_rods_pbd`, `tissue.k_ties`). For a link of stiffness k, peak force $f_{\max}$ and full separation d_f, with opening $d = |x_b - x_a|$ and $d_0 = f_{\max}/k$:

$$D \leftarrow \max\Big(D,\ \text{clamp}\Big(\frac{d_f\,(d - d_0)}{d\,(d_f - d_0)},\,0,\,1\Big)\Big)\ \ (d > d_0), \qquad f = (1 - D)\big[k\,(x_b - x_a) + k_d\,(v_b - v_a)\big]. \tag{27}$$

The secant damage is not the fraction of the fracture energy spent. The code books the spent fraction of the bilinear envelope (`cohesive.cohesive_spent`): with the largest opening reached u,

$$u(D) = \frac{d_0\,d_f}{d_f - D\,(d_f - d_0)}, \qquad s(D) = \text{clamp}\Big(\frac{u - d_0}{d_f - d_0}, 0, 1\Big), \qquad \Delta\mathcal{D}_{\text{tie}} = \tfrac{1}{2} f_{\max} d_f\,\big[s(D_{\text{new}}) - s(D_{\text{old}})\big]. \tag{28}$$

An interface of strength σ and fracture energy G is discretised with the tributary area $A = h^2/2$ of a node pair on a mesh of edge h (`assembly.py`):

$$f_{\max} = \sigma A, \qquad d_f = \frac{2G}{\sigma}, \qquad k = \min\Big(\frac{f_{\max}}{0.2\,d_f},\ 0.1\,\frac{m_{\text{eff}}}{\Delta t^2}\Big), \qquad k_d = 0.2\sqrt{k\,m_{\text{eff}}}, \tag{29}$$

so that tearing a link by pulling dissipates $\frac{1}{2} f_{\max} d_f = GA$, the interface's fracture energy over its area, whatever the stability cap does to k. Flesh-bone ties bond every tissue node within $0.45\,h$ of a bonded bone's surface with the flesh-bone values $\sigma = 30\,\text{kPa}$, $G = 50\,\text{J/m}^2$ (both open). Rod shaft ties use the shaft's lateral area, $f_{\max} = \tau_s\,2\pi r\,\ell_{\text{node}}$ with $\tau_s = 78\,\text{kPa}$ for pin bones (derived from a measured pull-out force [5]) and the flesh-bone strength for other rods, $d_f = 2G/\tau_s$, $k = \min(f_{\max}/(0.2 d_f),\, 0.2\,m/\Delta t^2)$ and 70 % of critical damping.

4.3 Release by the blade and interface fracture energy

A bond or tie is not loaded to failure by the edge first: it is released when a blade segment's slab passes it. A bond whose midpoint lies in the slab (the joint condition above with a lateral margin of 0.5 mm) is set to $D = 1$, and its unspent energy becomes a debt Q_b of that segment's body b (`bodies.k_bond_release`, `k_body_tie_release`):

$$Q_b \leftarrow Q_b + \big(1 - s(D)\big)\,\tfrac{1}{2} f_{\max} d_f. \tag{30}$$

The segment pays its debt from its own motion over a travel $d_p = 1\,\mathrm{mm}$ (`bodies.k_body_integrate`):

$$\Delta v = \min\Big(\frac{Q_b\,\Delta t}{d_p\,m_b},\ \frac{|v|}{2}\Big), \quad v' = v\Big(1 - \frac{\Delta v}{|v|}\Big), \quad \Delta\mathcal{D}_{\mathrm{iface}} = \tfrac{1}{2}m_b\big(|v|^2 - |v'|^2\big), \quad Q_b \leftarrow \max(Q_b - \Delta\mathcal{D}_{\mathrm{iface}}, 0). \tag{31}$$

The resisting force Q_b/d_p never reverses the velocity and only removes kinetic energy; in steady cutting it equals the interface energy released per unit travel, independent of d_p. Cutting along an interface therefore costs what the interface's fracture energy says, and the payment is booked in the ledger.

4.4 Tearing

Flesh separates where it is pulled apart, not only where a blade passes (`tearing.py`). Every split cadence the elastic Cauchy stress of each element, $\sigma = \big(\mu FF^T + \lambda'(J-\alpha)J\,I\big)/J$, is screened against its material's lowest strength (`tearing.k_tet_cauchy`). In fish muscle the traction on the element's own myoseptal sheet plane, whose current normal is $m_c = F^{-T}m/|F^{-T}m|$, is tested against the sheet strength; otherwise, and in every other tissue, the largest principal stress against the tensile strength:

$$\sqrt{\langle t_n\rangle^2 + \beta^2 t_s^2} \ \geq\ \sigma_{\mathrm{sheet}}, \quad t = \sigma m_c,\ t_n = m_c\cdot t,\ t_s = |t - t_n m_c|; \qquad \text{or} \qquad \sigma_1 \ \geq\ \sigma_t, \tag{32}$$

with $\sigma_{\mathrm{sheet}} = 10\,\mathrm{kPa}$ (open), $\beta = 1$ (an estimate), and $\sigma_t = 50\,\mathrm{kPa}$ (fish, open), 115 kPa (pork, derived from raw porcine loin across the fibres), 50 kPa (fat, open); the provenance of each is in the parameter ledger (Section 2.4). A tear is a crack with a fixed plane: the sheet plane (rest normal m), or the plane normal to the principal direction carried back by F^T, through the centre of the element's face best aligned with it. It starts at the most overstressed element of a connected overstressed region that holds no crack, and grows into overstressed elements that touch its torn vertices, that its plane passes through, and where the traction across that plane reaches its strength. In each such element the faces whose two elements' centroids lie on either side of the plane are cut through the knife's own topology code. Where node splitting then separates a node across torn faces, the two copies are joined by an extrinsic cohesive bond of the law above, with $f_{\max} = \sigma A_\perp/3$ and $\frac{1}{2}f_{\max}d_f = GA_\perp/3$ per face ($A_\perp$ the face area projected on the crack plane, $G = 3\,\mathrm{J/m^2}$ for a sheet tear and the anisotropic toughness R of the edge law for a tensile one) and $k = \min(f_{\max}/(0.02d_f),\ 0.1m_{\mathrm{eff}}/\Delta t^2)$: the face carried its strength until it opened, so the bond starts nearly at its peak and only softens. Elements at a point attachment (a flesh-bone tie's node, a pre-inserted bond's node, a rod's tie element) neither start nor grow tears, because the element stress there is set by the mesh; for the same reason neither do elements whose centroid lies within 1.5 mean edge lengths of the knife edge, where the edge law's anchors load single material points and the edge law is the fracture law (`tearing.Tearing._outside_edge_zone`: acting there, tearing's bonds would hold cut halves together); over a tear the skin keeps its nodes joined until its own membrane tension across the edge reaches 30 MPa times its thickness. Measured: a notched fish-muscle bar breaks at a net-section stress 1.24 and 1.05 times the sheet and tensile strengths (pulled across and along the sheets; 0.82 and 0.70 on the gross section) with the ledger closed to 1.1 %.

4.5 Thin bones as rods

Ribs, pin bones, fin supports and supraneurals are chains of nodes with their real mass (bone density $2000\,\mathrm{kg/m^3}$, lumped from each link's volume), advanced once per tissue substep by one thread per rod (`bodies.k_rods_pbd`). Bending acts on the turning vector at each interior node,

$$\kappa_m = t_0 \times t_1, \qquad E_b = \tfrac{1}{2}k_b\,|\kappa_m - \kappa_m^0|^2, \qquad k_b = \frac{E\,I}{\ell_b}, \quad I = \frac{\pi r^4}{4}, \tag{33}$$

with t_0, t_1 the unit link directions, ℓ_b the mean of the two link lengths and $E = 3\,\mathrm{GPa}$. Shaft ties, contact and edge forces and gravity (on the physical mass) give predicted positions; bone is treated as inextensible at these loads, so 8 Gauss–Seidel passes project every link onto its rest length ℓ_0 and the root node onto its vertebral attachment:

$$C = |x_b - x_a| - \ell_0, \quad \lambda = \frac{C}{w_a + w_b}, \quad x_a \leftarrow x_a + w_a\lambda u, \quad x_b \leftarrow x_b - w_b\lambda u, \quad T = \sum_{\text{passes}}\frac{\lambda}{\Delta t^2}, \tag{34}$$

with inverse masses w ($w_a = 0$ for a held root) and T the implied link tension; velocities follow from the position change. A link breaks when $T > \sigma_u\pi r^2$ and a hinge when $k_b|\kappa - \kappa^0| > \sigma_u\pi r^3/4$, with $\sigma_u = 110\,\mathrm{MPa}$ (checked every

five ticks; a snapped hinge also removes its distal link, so the rod becomes two pieces). A root detaches from its segment when its implied force exceeds 0.5 of the shaft's tensile capacity $\sigma_u \pi r_0^2$; otherwise its reaction is applied to the segment. The projection's loss is booked per substep from the rod's own energy and the work of forces from outside it,

$$\mathcal{D}_{\text{proj}} = \big(K + E_b + E_t\big)_{\text{before}} + W_{\text{ext}} - \big(K + E_b + E_t\big)_{\text{after}}, \tag{35}$$

with E_t the shaft ties' elastic energy (the flesh anchors do not move during the update) and W_{ext} the work of contacts, the edge, gravity and the segment's force on a held root along the step: the rod's own springs enter only through their energy, because their start-of-step work exceeds its change by $\frac{1}{2}\, dx^T H dx$ every step.

4.6 Skin membrane

Skin is a membrane of plane-stress stable neo-Hookean triangles on the boundary faces it covers, computed in `tissue.k_tri_forces`. Each triangle is a face of a tetrahedron, so after a cut it takes that tetrahedron's current nodes and follows the cut. With the 3×2 deformation gradient $F = [F_1\, F_2]$ and area ratio $J_a = |F_1 \times F_2|$, normal $n = (F_1 \times F_2)/J_a$:

$$\mu_s = \frac{E_s h_s}{2(1+\nu_s)}, \qquad \lambda_s = \frac{E_s \nu_s h_s}{1-\nu_s^2}, \qquad \lambda_s' = \lambda_s + \mu_s, \qquad \alpha_s = 1 + \frac{\mu_s}{\lambda_s'}, \tag{36}$$

$$P = \mu_s F + \lambda_s'(J_a - \alpha_s)\,[\,F_2 \times n,\ n \times F_1\,] + \tau_s \mu_s \dot{F}, \tag{37}$$

the gradient of $\frac{\mu_s}{2}\,\mathrm{tr}(F^T F) + \frac{\lambda_s'}{2}(J_a - \alpha_s)^2$ per rest area plus a small stiffness-proportional damper ($\tau_s = 10^{-6}$ s by default in `add_membrane`). The nodal forces are $-A_0 P D_m^{-T}$. Fish skin uses $E_s = 20\,\text{MPa}$, thickness 0.3 mm (trout) or 0.5 mm (salmon), $\nu_s = 0.45$ (inline) and density $1100\,\text{kg/m}^3$. The membrane's modulus is named by the workpiece's skin description (`modulus_param`, default `SKIN_E`; `assembly.py`): the tenderloin's silver skin is built by the same call with $E_s = 50\,\text{MPa}$ and thickness 0.2 mm, both open, and with the skin density and the same inline Poisson ratio. Tissue nodes under intact skin do not contact rigid shapes; the skin carries the contact. The membrane has no failure law of its own: it parts where the tetrahedra under it are cut.

5 The knife and the robots

5.1 The flexible knife

The knife exists twice, by design, with one owner at a time (Section 5.2). MuJoCo has a free rigid knife body. The tissue solver has a driven handle body and five blade segment bodies joined in a chain (`workpiece._knife_bodies`). Segment s of length $L_s = L/5$ has its centre of mass at 0.6 of the blade height H above the edge (a spine-heavy flat grind) and a mass from its steel volume. The spine tapers from t at the heel to half of it at the tip, $t_s = t\,(1 - 0.5\,\xi_s)$ with $\xi_s = (s + \frac{1}{2})/5$. Joints sit at the segment boundaries at mid-height, with a linear spring $k = 2 \times 10^5\,\text{N/m}$ (10 % of critical damping) and, in the parent's frame, rotational springs about the blade (torsion), through the thickness (edge-wise) and about the height (flat-wise):

$$K_\theta = \mathrm{diag}\Big(\frac{G_s H t_s^3}{3L_s},\ \frac{E_s\, t_s H^3}{12 L_s},\ \frac{E I_{\text{heel}}}{L_s}\Big(\frac{t_s}{t}\Big)^3\Big), \qquad E I_{\text{heel}} = E_s \sqrt{\frac{H t^3}{12} \cdot \frac{t^3 H}{48}}, \tag{38}$$

the flat-wise heel stiffness being the geometric mean of the rectangle and full-wedge bounds, with $E_s = 200\,\text{GPa}$, $G_s = 80\,\text{GPa}$ and damping ratios of 0.2 (torsion, edge-wise) and 0.02 (flat-wise). One numerical measure is disclosed in the code: the edge-wise spring is stable at the tissue substep only with a rotational inertia $I_{yy} \geq 4 k_{\text{edge}} \Delta t^2 / 0.2$, far above the true value, so that single inertia is raised; it slows edge-wise angular dynamics of a blade that barely bends edge-wise and leaves mass, gravity, flat-wise flex and torsion true. Each segment's contact geometry is the wedge of Section 3 for tissue nodes, plus spheres every 2 mm along the edge (radius 0.15 mm) and the spine (radius $t_s/2$) that contact the board, block and other external shapes. The handle carries a shell of 1.5 mm spheres, active while the knife is free.

5.2 Who integrates the knife

Every tick the world decides the knife's owner (`world.World._update_knife_mode`). With f_{robot} the normal force of robot bodies on MuJoCo's knife, f_{tissue} the load of tissue and bones on the knife's colliders in the tissue solver over the last tick, and π true when both fingers of one hand press the handle with more than 0.5 N each (read from MuJoCo's contacts), MuJoCo is the owner whenever

$$O = \pi \ \vee \ \big[f_{\text{robot}} > 0.2\,\text{N}\big] \ \vee \ \big[f_{\text{tissue}} < 0.05\,\text{N}\big]. \tag{39}$$

A held knife becomes free after O has been false for 3 ticks in a row; a free knife is handed back to MuJoCo after a pinch or a robot touch for 5 ticks in a row, or after 100 ticks in a row off the tissue (a bounce, or lying on the board; with 5 ticks a knife dropped on the fish changes owner 11 times in 0.51 s), and only while MuJoCo's rigid copy overlaps the robot and the static geometry by at most 2 mm. The tissue solver therefore owns the knife only while it lies on tissue with no robot touching it (a knife dropped on the fish).

- **Held (MuJoCo owns).** MuJoCo integrates the knife as one rigid body with the whole knife's mass and inertia, and its rigid blade meets the board, block and robot in MuJoCo's contacts. The tissue solver's handle is driven along MuJoCo's handle over the tick, and the blade chain follows through its root joint. MuJoCo receives only the load the world applies to the knife through the tissue solver (`workpiece._knife_external_load`):

$$F = \bar{F}_h + \frac{P^1 - P^0}{\Delta t_r} - Mg, \qquad \tau_c = \bar{\tau}_h + \frac{L^1 - L^0}{\Delta t_r} + \bar{v}_c \times \bar{P} - \bar{S} \times g, \tag{40}$$

  with $\bar{F}_h, \bar{\tau}_h$ the mean load on the driven handle over the tick, P and L the blade chain's linear and angular momentum about the moving handle point c at the start and end of the tick, M the chain's mass and $S = \sum_s m_s(x_s - c)$; the moment is transferred to MuJoCo's knife centre of mass. The chain's own weight is not applied twice: MuJoCo's knife carries it. While MuJoCo owns the knife and its blade has carried no tissue load for 20 ticks, the tissue solver's blade segments follow MuJoCo's knife at their rest shape as driven bodies and become dynamic again at the first tissue load (`workpiece._blade_in_air`): in that state the segments meet only tissue, so nothing else loads or damps them (otherwise a dropped knife's blade swings through the board).
- **Free (tissue solver owns).** The whole knife, handle and blade, is integrated in the tissue solver against the board, bench, block, floor, tissue, bones and the robots' finger, pad and palm colliders. MuJoCo's copy is placed on it every tick with its true velocity and keeps contacts with the robots only, so the fingers see the knife's real motion; it is kinematic, so a finger cannot push it in MuJoCo.
- **Phantom (ASSISTED only).** A scripted rig drives the handle with the secant twist between its path poses at the start and end of each tick (`workpiece._rig_tick`); every engagement is logged.

A non-finite state in either solver, or any of MuJoCo's instability warnings (acceleration, position, velocity), is a fault that stops the world; it is never a silent continuation.

5.3 The grip

The Franka Hand has a position mode and a force mode (`world.World._rigid_step`). In force mode the fingers close at the commanded speed, 0.05 m/s, until a finger is blocked: behind its command by more than 0.5 mm and nearly stalled ($|\dot{q}| < 5 \times 10^{-3}$), or at the end of its travel. The finger then latches: its position target follows its current position and a constant joint force $-F_g$ acts, by default 35 N per finger (the hand's continuous rating is 70 N for both fingers together). MuJoCo runs with a friction-to-normal impedance ratio of 10 and 50 no-slip post-solver iterations, the settings qualified for held tools in this benchmark's barcode-scanning environment.

5.4 Motion generator and protective stops

Joint targets are not applied as steps. The commanded position q_c of each arm joint moves toward its target q^* at no more than the joint's speed limit v_{lim}, changes speed by at most $a_{\max}$ per second and brakes to arrive, as

libfranka's rate limiter does [23]:

$$e = q^* - q_c, \qquad v_{\text{stop}} = \operatorname{sgn}(e)\min\bigl(v_{\text{lim}}, \sqrt{2a_{\max}|e|}\bigr), \tag{41}$$

$$v_{k+1} = \operatorname{clamp}\bigl(v_{\text{stop}},\, v_k - a_{\max}\Delta t_r,\, v_k + a_{\max}\Delta t_r\bigr), \qquad q_c \leftarrow q_c + v_{k+1}\Delta t_r, \tag{42}$$

snapping to the target on arrival. $a_{\max}$ is (15, 7.5, 10, 12.5, 15, 20, 20) rad/s^2 for the Panda (read) and (10, 10, 10, 10, 10, 10, 10) rad/s^2 for the FR3 (prior: recalled, not re-read). Two protective stops act as on the real arm (`_velocity_reflex`, `_torque_reflex`). A joint faster than $1.05\,v_{\text{lim}}$ stops its arm. The external joint torque

$$\tau_{\text{ext}} = J_c^T \lambda_c + \sum_{b \in \text{robot}} \bigl(J_{p,b}^T F_b + J_{r,b}^T M_b\bigr) \tag{43}$$

sums MuJoCo's contact constraint forces mapped by the contact Jacobians and the tissue solver's wrenches on robot bodies (a finger or pad pressing on the fish); if any joint's $|\tau_{\text{ext}}|$ exceeds the threshold, (10, 20) N m (nominal, and while the targets are still slewing; libfranka's reference thresholds), the arm stops: its targets and motion generator freeze at the current configuration and it refuses new joint targets for 1 s (the fingers excepted). The ten cutting tasks select libfranka's contact-task profile by default ((100, 100) N m, the threshold of its force-control examples; `world.Config.collision_profile`): every robot cutting demonstration and runtime measurement runs with it, and an explicit configuration value is kept.

5.5 Robot configurations

Robot	Rig	Status
`dual-fr3`	`robots/rigs/filleting-fr3.json`	default; every probe in this dossier unless stated
`dual-panda`	`robots/rigs/filleting-panda.json`: Panda joint limits, the same hand and inserts	offered for take-knife only: its holding and rest results are in Section 8; its take-knife run holds the knife but leaves it 65.6° from edge-down and not over the board; its one cutting run, a press cut on the coupon (`robot_press_cut_panda`), reaches the board as the FR3's does and leaves the block whole (one 125.4 g piece); no task cutting run or timing exists, so the cutting tasks are refused on it
`single-fr3`	`robots/rigs/filleting-fr3-single.json`: the right FR3 with the inserts	offered for A0 only (`variants.check_robot`): every cutting task needs a second hand holding the workpiece
`g1`	–	not offered: not integrated (the knife-handle inserts and the grip are modelled on the Franka Hand only, and no G1 run of a filleting task exists)

The three rigs are also accepted as robot-setup objects, rig ids or rig paths (`filleting/robot_setup.py`); any other setup is refused with what the environment needs. Films and RGB observations draw the deployed robot model at the mounts that carry an arm.

5.6 The bench scale

In presets B and C the scale's weighing platform is a static collider in both solvers. Its reading is the normal load the tissue put on the platform over the last tick divided by g (`World.scale_reading_g`); it does not see a hand pressing on it.

6 Tasks and evaluation

The catalogue defines 13 tasks, of which 11 are offered; the resolver refuses the others with the reason printed in Table 4. The evaluator (`task.py`) reads physical state only, on the owner side: fragment connectivity through uncut element faces and intact interface bonds; each element's rest reference label, which says where every gram

of tissue belongs in a correct job (from the anatomy generator, never from the robot's path); thin-bone pieces and what they are bonded to; rigid bones and their joints; the knife's pose in the grasp; bodies against the tray and bowl footprints; and histories recorded every tick (blade proximity to the holding hand, the worst compression of every element, recent motion). It returns the `gar.evaluation.v1` record: a boolean success, a score equal to the fraction of the task's checks passed, and for each check its value, threshold, unit, verdict and note. A solver fault is reported as a fault, not as a task failure. Thresholds are authored or anchored to published practice (NOAA workmanship [24], USDA IMPS [25], Codex CXS 190 [26]); none is fitted to this simulation.

Table 4: Task catalogue, generated from `filleting/task.py` and `filleting/variants.py`. The instruction is the text the policy receives, verbatim. Time limit in simulated seconds.

Id	Name	Start workpiece	Limit	Instruction	Status
A0	take-knife	trout	30	Take the fillet knife from the block and hold it edge-down above the board.	supported
A1	head-off	trout	90	Remove the head with one cut behind the gill cover and pectoral fin, through a joint of the backbone.	supported
A2	fillet-one-side	trout	180	Fillet the upper side in one piece, skin on, leaving the ribs on the frame, and lay it skin-down in the perforated fillet tray.	supported
A3	fillet-whole-fish	trout	360	Fillet both sides with no rib bones left in the fillets, and put the fillets skin-down in the perforated fillet tray and the frame and head in the empty pan.	supported
A4	remove-belly-bones	trout_fillet_rib_strip	120	Remove the rib bones from the belly by sliding the blade under them, taking as little flesh as possible.	supported
B1	pin-bone	salmon	300	Remove all pin bones with the tweezers and drop them in the dish.	**refused**: pin-boning needs the fish-bone tweezers as a physical tool (jaws on a spring hinge gripped by the Franka Hand and pinching 0.3 mm pin bones); in this build the tweezers are a static obstacle in their tray, not a tool a hand can pick up
B2	trim-belly	salmon	180	Trim the belly fat strip and the belly membrane off, following the fat line, leaving a smooth edge.	supported
B3	excise-defect	salmon	120	Cut away the dark spot, taking as little good flesh as possible.	supported
B4	skin	salmon	180	Remove the skin in one piece.	**refused**: skinning needs the skin to separate with under 10 % of its mass in flesh; the tissue mesh separates along 3.5 mm tet faces, so any removed skin carries at least one tet layer of flesh
B5	portion	salmon	180	Cut the loin into 150 g portions with straight cuts and weigh them on the scale.	supported
C1	remove-chain	tenderloin	180	Remove the chain (side muscle) along its seam.	supported
C2	denude	tenderloin	240	Remove the silver skin, taking as little meat as possible.	supported
C3	medallions	tenderloin	180	Cut 25 mm medallions from the thick part.	supported

Table 5 summarises what each offered task is judged on; the checks' docstrings in `task.py` are authoritative.

Table 5: What each offered task is judged on (state, not poses).

Task	Checks
A0 take-knife	right-hand pinch held for 3 s; knife axis within 10 degrees of the insert channel; seated between the lips; slip under 2 mm; edge down above the board; no blade contact with a robot link

Task	Checks
A1 head-off	the head joint cut and no flesh or bond path from head to body; fillet muscle with the head at most 8 g; the fish on the board
A2 fillet-one-side	one piece (at least 97 % of the side's removed muscle); yield at least 0.33 of the gutted mass; no rib piece over 2 mm; no backbone or median fin-support fragment; NOAA workmanship at least 85 on a 25.4 mm grid of the fillet's flesh side (knife scores, tears, bruises with volume ratio under 0.6, bones); skin-down in the tray and settled; no blade contact with the holding hand; knife not dropped
A3 fillet-whole-fish	two different fillets, each A2's checks and at least 0.33 of the gutted mass, skin-down in the tray on the holding-arm side; every rigid bone of the frame and head resting in an empty pan; time at most 360 s
A4 remove-belly-bones	no rib piece over 2 mm left in the fillet; no skin hole over 25 mm^2; flesh removed at most 6 %
B2 trim-belly	at least 90 % of the reference trim removed; loin loss at most 3 %; trimmed edge raggedness (RMS) at most 2 mm
B3 excise-defect	at least 99 % of the spot removed; good flesh removed at most twice the spot; no skin removed
B5 portion	full portions of 150 ± 7 g; blade surfaces within 10 degrees of vertical; no loose piece of 1–5 g; each portion weighed on the scale
C1 remove-chain	at least 95 % of the chain separated; main muscle loss at most 3 %; no crack over 25 mm into the main piece
C2 denude	at least 90 % of the silver skin's area removed; main muscle loss at most 5 % (fat left is reported)
C3 medallions	25 ± 5 mm thick; spread at most 5 mm; blade surfaces within 10 degrees of perpendicular; no loose piece of 1–5 g

Every cutting task's evaluator also scores debris (loose pieces under 1 g: at most 0.005 of the workpiece's mass and ten pieces) and, where pieces are removed, that they rest on a support and not on the knife; every evaluation reports its crack faces by origin (split at build, torn, knife). Every cutting evaluator also checks for thrown tissue (the fastest tissue or bone node over the episode, sampled every 10 ms, at most 1 m/s, and no tissue below the bench top); debris is scored on every A2/A3 path, a medallion counts only up to twice the nominal thickness, and A1 may separate at any backbone joint ahead of the frame. Simplifications of the job, each stated in the evaluator's docstring: the belly membrane (B2) and the IMPS suspension check (C3) are not modelled and not scored; the B and C tasks start from the whole workpiece rather than the designed pre-trimmed states; the pelvic girdle is reported, not scored, in A2 (it lies in the lateral belly wall and normally comes away with a skin-on fillet; the check targets median supports, whose presence means the knife went below the median plane). In every ASSISTED run the `knife_not_dropped` check fails by design, because the knife is on the phantom rig and not in the robot's hand; the check's note says so.

7 Experimental design and quality gates

7.1 Components before policies

A robot policy would conflate perception, planning, grasping, control and mechanics, and a failed cut could expose any of them. The evidence is therefore built in three levels, and a result at one level is never substituted for another. **Component probes** test one mechanism against a physical expectation: a fish-muscle coupon cut by a kinematically driven knife against Atkins' law, a held knife left alone, interface energy placed on a blade, a rest window. **Assisted runs** drive the knife with a labelled phantom rig through a task and let the unmodified evaluator judge the product; they show that the world's own laws can produce it, from a plan authored with privileged state a policy does not see. **Robot runs** use legal robot commands only. A matched **negative control** removes one property of a positive run (the path through the joint, the hold on the fillet, vertical cuts, the medallion planes) and must fail on the check that measures that property; a control that passes would mean the evaluator does not see the property.

7.2 Quality gates

Nine quality gates define the acceptance boundary; an acceptance script checks them against the recorded results and fails on any failed check. The gates, in summary: (1) the world at real scale, every workpiece's physical mass carried by its supports within 2 % at rest; (2) at least one task product made by a robot through the normal action interface with no phantom rig, and the knife pickup from the block completed with the knife held and seated; (3) an assisted run for every supported task, scored by the unmodified evaluator, and at least three tasks with a matched negative control that fails on the property it removes; (4) every preset left alone for 30 s with tissue net drift at most 0.1 mm and knife drift at most 0.05 mm, no protective stop and no fault, published whether or not it passes; (5) legal abuse (random commands, rams into the board and the backbone, the hand swept into the blade, the knife dropped) with every episode finite, tissue mass conserved to 0.1 g and no fragment faster than 1 m/s, and the dropped knife itself at rest; (6) the coupon press cut within 15 % of $R\ell$, mesh and time refinement changing the plateau by at most 5 %, cohesive bonds booking exactly the bilinear law's dissipation and a coupon ledger residual of at most 2 % of the work; (7) operating wall time per simulated second at most 200 through the public interface with an RGB observation, judged on the largest steady action, with the run labelled 'cut' required to make crack faces; (8) the interface checks; (9) evidence custody: every result stamped with the identity of the code that produced it, films decoded frame by frame and reviewed, and every model constant of the ledger read by the world's code. No physical constant or gate is changed to turn a failure into a pass.

7.3 The recorded runs

Every run the acceptance and this document rely on is a named stage. Each stage runs in its own fresh directory with its log kept, and every result is stamped at the start and at the end of its run with the identity of the world that produced it: a SHA-256 over every file that can change what is simulated or how it is scored, separate from the renderer and the scripts. A failure is evidence and is kept.

7.4 How this document uses evidence

Every number in Section 8 is read from a recorded result file when this document is built. Each subsection chooses which recorded fields are shown and may request a ratio, a percentage change or a sum of recorded values, never a number of its own. Plots are drawn only from recorded samples. The rendered evidence comes from the films of the recorded runs: each run that records its state is filmed from its recording, and every frame must decode back from the video. Whether the environment passes its quality gates is the acceptance record's verdict (Section 8).

7.5 Statistical scope

The GPU tissue step uses atomic accumulation, so a rerun of the same command reproduces a run closely but not bit for bit; reruns agree to within a few micrometres of node position (a restore test with no recorded result in this document). A single run of a deterministic scenario is a demonstration of what the world does for that input, not a sample of a success probability. The seeds vary the workpiece, and no result here is a claim about all seeds.

8 Recorded results

Every number in this section is read from a recorded result file when this document is built. Each subsection chooses which recorded fields are shown and may apply plain arithmetic to them (a ratio, a percentage change, a sum), never a number of its own. Plots are drawn from recorded samples.

8.1 Machine acceptance of the quality gates

The acceptance script reads the recorded results and checks every quality gate against its own thresholds. Its record, not prose, says whether the gates pass.

8.1.1 Acceptance record of the nine quality gates

The acceptance script reads the recorded results; every check reads a value from a recorded result or fails.

Observable	Recorded value
All gates pass	false
Generated	2026-10-07T20:59:12Z

Gate	Title	Passed	Passing checks	Checks
1	World at real scale	true	11	11
2	Robot does the work	false	15	23
3	Assisted solvability	false	43	67
4	Rest	true	26	26
5	Legal abuse	false	25	29
6	Mechanics against references	false	25	39
7	Runtime	false	26	29
8	Interfaces	true	5	5
9	Evidence custody	true	109	109

Every check with its recorded value, threshold and verdict. 'Counted' false marks a candidate the gate needs only some of (any one robot product, three matched negative controls): it is listed but not counted in the gate's totals above, and a counted row of the same gate judges the candidates together. The checks of gate 9 (evidence custody) are counted above and not listed.

Gate	Check	Value (or, where none was recorded, the reason)	Threshold	Counted	Passed
1	collider_audit: every collider has a documented dimension source	0 of 124 colliders without a source	a source for every collider	true	true
1	collider_audit: no visual-only object where a hand or the knife works	none (visual only within an arm's reach with the knife)	none in the work volume	true	true
1	mass_budget: trout: physical mass reported	242.8 g physical (+31.2 g stability mass)	> 0 g	true	true
1	mass_budget: trout_fillet_rib_strip: physical mass reported	104.6 g physical (+13.1 g stability mass)	> 0 g	true	true
1	mass_budget: salmon: physical mass reported	954.2 g physical (+34.2 g stability mass)	> 0 g	true	true
1	mass_budget: tenderloin: physical mass reported	534.3 g physical (+30.8 g stability mass)	> 0 g	true	true
1	rest_A2: physical weight carried by the supports at rest	288.5 of 288.5 g (+0.00 %): board 288.5	within 2 %	true	true
1	rest_A4: physical weight carried by the supports at rest	104.6 of 104.6 g (-0.00 %): board 104.6	within 2 %	true	true
1	rest_B2: physical weight carried by the supports at rest	958.6 of 958.7 g (-0.01 %): board 958.6	within 2 %	true	true
1	rest_A0: physical weight carried by the supports at rest	288.4 of 288.5 g (-0.00 %): board 288.4	within 2 %	true	true
1	rest_C1: physical weight carried by the supports at rest	534.3 of 534.3 g (+0.00 %): board 534.3	within 2 %	true	true
2	robot_a0: knife taken from the block, held and seated (A0 evaluation)	success True, score 1.00	success True	true	true
2	robot_a0: no phantom rig (legal commands only)	0 rig entries; labelled 'none: legal robot commands; privileged knife'	0 rig entries	true	true

Gate	Check	Value (or, where none was recorded, the reason)	Threshold	Counted	Passed
2	robot_a1: task product by the robot (A1 evaluation), no phantom rig	success False, score 0.75; failed head_separated, meat_with_ head; 0 ri ...	success True, 0 rig entries	false	false
2	robot_c3: task product by the robot (C3 evaluation), no phantom rig	success False, score 0.56; failed medallions, thickness_25_5mm, thickn ...	success True, 0 rig entries	false	false
2	at least one task product made by the robot	0 (none) of 2	>= 1 product run	true	false
2	robot_press_cut: the press cut divides the block	2 pieces of 10 g or more (62.8, 62.6 g)	>= 2 pieces	true	true
2	robot_press_cut: lifting the knife leaves the pieces on the board	-0.790 N on the blade while it is lifted out	>= -0.1 N	true	false
2	robot_press_cut_slow: the press cut divides the block	2 pieces of 10 g or more (64.9, 60.5 g)	>= 2 pieces	true	true
2	robot_press_cut_slow: lifting the knife leaves the pieces on the board	-0.574 N on the blade while it is lifted out	>= -0.1 N	true	false
2	robot_press_cut_fast: the press cut divides the block	2 pieces of 10 g or more (64.7, 60.6 g)	>= 2 pieces	true	true
2	robot_press_cut_fast: lifting the knife leaves the pieces on the board	-0.772 N on the blade while it is lifted out	>= -0.1 N	true	false
2	robot_press_cut_panda: the press cut divides the block	1 pieces of 10 g or more (125.4 g)	>= 2 pieces	true	false
2	robot_press_cut_panda: lifting the knife leaves the pieces on the board	-0.955 N on the blade while it is lifted out	>= -0.1 N	true	false
2	robot_a1: withdrawing the knife leaves the work on the board (recording)	no piece carried on the withdrawn knife	0 frames	true	true
2	robot_a1: no piece launched faster than 1 m/s or off the bench (recording)	0 pieces of >= 5 nodes faster than 1 m/s (fastest 0.00 m/s); 0 pieces ...	0	true	true
2	robot_c3: withdrawing the knife leaves the work on the board (recording)	no piece carried on the withdrawn knife	0 frames	true	true
2	robot_c3: no piece launched faster than 1 m/s or off the bench (recording)	0 pieces of >= 5 nodes faster than 1 m/s (fastest 0.00 m/s); 0 pieces ...	0	true	true
2	robot_press_cut: withdrawing the knife leaves the work on the board (recording)	8 sampled frames of the knife's withdrawal with a piece raised > 20 mm ...	0 frames	true	false
2	robot_press_cut: no piece launched faster than 1 m/s or off the bench (recording)	0 pieces of >= 5 nodes faster than 1 m/s (fastest 0.00 m/s); 0 pieces ...	0	true	true
2	robot_press_cut_slow: withdrawing the knife leaves the work on the board (recording)	no piece carried on the withdrawn knife	0 frames	true	true
2	robot_press_cut_slow: no piece launched faster than 1 m/s or off the bench (recording)	0 pieces of >= 5 nodes faster than 1 m/s (fastest 0.00 m/s); 0 pieces ...	0	true	true
2	robot_press_cut_fast: withdrawing the knife leaves the work on the board (recording)	8 sampled frames of the knife's withdrawal with a piece raised > 20 mm ...	0 frames	true	false
2	robot_press_cut_fast: no piece launched faster than 1 m/s or off the bench (recording)	0 pieces of >= 5 nodes faster than 1 m/s (fastest 0.00 m/s); 0 pieces ...	0	true	true
2	robot_press_cut_panda: withdrawing the knife leaves the work on the board (recording)	no piece carried on the withdrawn knife	0 frames	true	true
2	robot_press_cut_panda: no piece launched faster than 1 m/s or off the bench (recording)	0 pieces of >= 5 nodes faster than 1 m/s (fastest 0.00 m/s); 0 pieces ...	0	true	true
3	robot_a0: A0 take-knife: run scored by the unmodified evaluator	success True, score 1.00	scored (success recorded)	true	true
3	robot_a0_single: A0 take-knife: run scored by the unmodified evaluator	success True, score 1.00	scored (success recorded)	true	true
3	robot_a0_panda: A0 take-knife: run scored by the unmodified evaluator	success False, score 0.78; failed edge_down_deg, blade_over_ board	scored (success recorded)	true	true
3	A0 take-knife: the task's product made by an assisted or robot run	2 of 3 runs succeed (robot_a0, robot_a0_single: robot_a0, robot_a0_sin ...	success True in at least one run	true	true
3	robot_a1: A1 head-off: run scored by the unmodified evaluator	success False, score 0.75; failed head_separated, meat_with_ head	scored (success recorded)	true	true
3	a1: A1 head-off: run scored by the unmodified evaluator	success False, score 0.62; failed head_separated, meat_with_ head, knif ...	scored (success recorded)	true	true
3	a1: assistance labelled and logged in world.assist	'phantom knife rig + phantom head hold + '; 6 rig entries (drive_body, ...	label and rig log	true	true
3	A1 head-off: the task's product made by an assisted or robot run	0 of 2 runs succeed (none: robot_ a1, a1)	success True in at least one run	true	false
3	a2: A2 fillet-one-side: run scored by the unmodified evaluator	success False, score 0.15; failed one_piece, yield, no_rib_pieces_ over ...	scored (success recorded)	true	true
3	a2: assistance labelled and logged in world.assist	'phantom knife rig + phantom hands on the'; 14 rig entries (drive_body ...	label and rig log	true	true

Gate	Check	Value (or, where none was recorded, the reason)	Threshold	Counted	Passed
3	A2 fillet-one-side: the task's product made by an assisted or robot run	0 of 1 runs succeed (none: a2)	success True in at least one run	true	false
3	A3 fillet-whole-fish: an assisted run scored by the evaluator	*not measurable: no campaign stage produces it (needs an assisted or robot run of A3 (fille …*	scored by the evaluator	true	false
3	a4: A4 remove-belly-bones: run scored by the unmodified evaluator	success False, score 0.33; failed no_rib_pieces_over_2mm, skin_hole_mm …	scored (success recorded)	true	true
3	a4: assistance labelled and logged in world.assist	'phantom knife rig + creator-scripted lef'; 3 rig entries (phantom_kni …	label and rig log	true	true
3	A4 remove-belly-bones: the task's product made by an assisted or robot run	0 of 1 runs succeed (none: a4)	success True in at least one run	true	false
3	b2: B2 trim-belly: run scored by the unmodified evaluator	success False, score 0.43; failed trim_removed, edge_raggedness_mm, de …	scored (success recorded)	true	true
3	b2: assistance labelled and logged in world.assist	'phantom knife rig + creator-scripted lef'; 6 rig entries (phantom_kni …	label and rig log	true	true
3	B2 trim-belly: the task's product made by an assisted or robot run	0 of 1 runs succeed (none: b2)	success True in at least one run	true	false
3	b3: B3 excise-defect: run scored by the unmodified evaluator	success False, score 0.33; failed spot_removed, good_flesh_removed, sk …	scored (success recorded)	true	true
3	b3: assistance labelled and logged in world.assist	'phantom knife rig + creator-scripted lef'; 12 rig entries (phantom_kn …	label and rig log	true	true
3	B3 excise-defect: the task's product made by an assisted or robot run	0 of 1 runs succeed (none: b3)	success True in at least one run	true	false
3	robot_b5_weigh: B5 portion: run scored by the unmodified evaluator	success False, score 0.89; failed weighed	scored (success recorded)	true	true
3	robot_b5_weigh: assistance labelled and logged in world.assist	'cuts: phantom knife rig (earlier run); w'; 1 rig entries (phantom_kni …	label and rig log	true	true
3	b5: B5 portion: run scored by the unmodified evaluator	success False, score 0.89; failed weighed	scored (success recorded)	true	true
3	b5: assistance labelled and logged in world.assist	'phantom knife rig + creator-scripted lef'; 14 rig entries (phantom_kn …	label and rig log	true	true
3	B5 portion: the task's product made by an assisted or robot run	0 of 2 runs succeed (none: robot_b5_weigh, b5)	success True in at least one run	true	false
3	c1: C1 remove-chain: run scored by the unmodified evaluator	success False, score 0.33; failed chain_separated, main_loss, tail_spl …	scored (success recorded)	true	true
3	c1: assistance labelled and logged in world.assist	'phantom knife rig + creator-scripted lef'; 5 rig entries (phantom_kni …	label and rig log	true	true
3	C1 remove-chain: the task's product made by an assisted or robot run	0 of 1 runs succeed (none: c1)	success True in at least one run	true	false
3	c2: C2 denude: run scored by the unmodified evaluator	success False, score 0.67; failed silver_removed, main_loss	scored (success recorded)	true	true
3	c2: assistance labelled and logged in world.assist	'phantom knife rig + creator-scripted lef'; 5 rig entries (phantom_kni …	label and rig log	true	true
3	C2 denude: the task's product made by an assisted or robot run	0 of 1 runs succeed (none: c2)	success True in at least one run	true	false
3	robot_c3: C3 medallions: run scored by the unmodified evaluator	success False, score 0.56; failed medallions, thickness_25_5mm, thickn …	scored (success recorded)	true	true
3	c3: C3 medallions: run scored by the unmodified evaluator	success False, score 0.33; failed medallions, thickness_25_5mm, thickn …	scored (success recorded)	true	true
3	c3: assistance labelled and logged in world.assist	'phantom knife rig + scripted left FR3 ho'; 4 rig entries (phantom_kni …	label and rig log	true	true
3	C3 medallions: the task's product made by an assisted or robot run	0 of 2 runs succeed (none: robot_c3, c3)	success True in at least one run	true	false
3	a1: withdrawing the knife leaves the work on the board (recording)	no piece carried on the withdrawn knife	0 frames	true	true
3	a1: no piece launched faster than 1 m/s or off the bench (recording)	0 pieces of >= 5 nodes faster than 1 m/s (fastest 0.00 m/s); 0 pieces …	0	true	true
3	a2: withdrawing the knife leaves the work on the board (recording)	no piece carried on the withdrawn knife	0 frames	true	true
3	a2: no piece launched faster than 1 m/s or off the bench (recording)	3 pieces of >= 5 nodes faster than 1 m/s (fastest 1.58 m/s); 2 pieces …	0	true	false
3	a4: withdrawing the knife leaves the work on the board (recording)	1 sampled frames of the knife's withdrawal with a piece raised > 20 mm …	0 frames	true	false

Gate	Check	Value (or, where none was recorded, the reason)	Threshold	Counted	Passed
3	a4: no piece launched faster than 1 m/s or off the bench (recording)	0 pieces of >= 5 nodes faster than 1 m/s (fastest 0.00 m/s); 0 pieces ...	0	true	true
3	b2: withdrawing the knife leaves the work on the board (recording)	9 sampled frames of the knife's withdrawal with a piece raised > 20 mm ...	0 frames	true	false
3	b2: no piece launched faster than 1 m/s or off the bench (recording)	2 pieces of >= 5 nodes faster than 1 m/s (fastest 1.31 m/s); 0 pieces ...	0	true	false
3	b3: withdrawing the knife leaves the work on the board (recording)	no piece carried on the withdrawn knife	0 frames	true	true
3	b3: no piece launched faster than 1 m/s or off the bench (recording)	0 pieces of >= 5 nodes faster than 1 m/s (fastest 0.00 m/s); 0 pieces ...	0	true	true
3	b5: withdrawing the knife leaves the work on the board (recording)	no piece carried on the withdrawn knife	0 frames	true	true
3	b5: no piece launched faster than 1 m/s or off the bench (recording)	0 pieces of >= 5 nodes faster than 1 m/s (fastest 0.00 m/s); 0 pieces ...	0	true	true
3	c1: withdrawing the knife leaves the work on the board (recording)	1 sampled frames of the knife's withdrawal with a piece raised > 20 mm ...	0 frames	true	false
3	c1: no piece launched faster than 1 m/s or off the bench (recording)	0 pieces of >= 5 nodes faster than 1 m/s (fastest 0.00 m/s); 0 pieces ...	0	true	true
3	c2: withdrawing the knife leaves the work on the board (recording)	no piece carried on the withdrawn knife	0 frames	true	true
3	c2: no piece launched faster than 1 m/s or off the bench (recording)	0 pieces of >= 5 nodes faster than 1 m/s (fastest 0.00 m/s); 0 pieces ...	0	true	true
3	c3: withdrawing the knife leaves the work on the board (recording)	15 sampled frames of the knife's withdrawal with a piece raised > 20 m ...	0 frames	true	false
3	c3: no piece launched faster than 1 m/s or off the bench (recording)	1 pieces of >= 5 nodes faster than 1 m/s (fastest 1.05 m/s); 0 pieces ...	0	true	false
3	a1_neg_offset: withdrawing the knife leaves the work on the board (recording)	no piece carried on the withdrawn knife	0 frames	true	true
3	a1_neg_offset: no piece launched faster than 1 m/s or off the bench (recording)	0 pieces of >= 5 nodes faster than 1 m/s (fastest 0.00 m/s); 0 pieces ...	0	true	true
3	a4_neg_unheld: withdrawing the knife leaves the work on the board (recording)	29 sampled frames of the knife's withdrawal with a piece raised > 20 m ...	0 frames	true	false
3	a4_neg_unheld: no piece launched faster than 1 m/s or off the bench (recording)	0 pieces of >= 5 nodes faster than 1 m/s (fastest 0.00 m/s); 0 pieces ...	0	true	true
3	b5_neg_tilt: withdrawing the knife leaves the work on the board (recording)	no piece carried on the withdrawn knife	0 frames	true	true
3	b5_neg_tilt: no piece launched faster than 1 m/s or off the bench (recording)	0 pieces of >= 5 nodes faster than 1 m/s (fastest 0.00 m/s); 2 pieces ...	0	true	false
3	c3_neg_offset: withdrawing the knife leaves the work on the board (recording)	20 sampled frames of the knife's withdrawal with a piece raised > 20 m ...	0 frames	true	false
3	c3_neg_offset: no piece launched faster than 1 m/s or off the bench (recording)	0 pieces of >= 5 nodes faster than 1 m/s (fastest 0.00 m/s); 0 pieces ...	0	true	true
3	b2_neg_offset: withdrawing the knife leaves the work on the board (recording)	8 sampled frames of the knife's withdrawal with a piece raised > 20 mm ...	0 frames	true	false
3	b2_neg_offset: no piece launched faster than 1 m/s or off the bench (recording)	2 pieces of >= 5 nodes faster than 1 m/s (fastest 1.18 m/s); 0 pieces ...	0	true	false
3	c1_neg_offset: withdrawing the knife leaves the work on the board (recording)	2 sampled frames of the knife's withdrawal with a piece raised > 20 mm ...	0 frames	true	false
3	c1_neg_offset: no piece launched faster than 1 m/s or off the bench (recording)	0 pieces of >= 5 nodes faster than 1 m/s (fastest 0.00 m/s); 0 pieces ...	0	true	true
3	a1_neg_offset: negative control fails on head_ separated (the cut through a backbone joint ...	control: success False, head_ separated failed; a1: head_ separated fail ...	control fails, matched run passes	false	false
3	a4_neg_unheld: negative control fails on no_ rib_pieces_over_2mm (the hold on the fillet re ...	control: success False, no_rib_ pieces_over_2mm failed; a4: no_ rib_piec ...	control fails, matched run passes	false	false
3	b5_neg_tilt: negative control fails on cut_tilt_ deg (vertical cuts removed), b5 passes it	control: success False, cut_tilt_ deg failed; b5: cut_tilt_deg passed	control fails, matched run passes	false	true
3	c3_neg_offset: negative control fails on thickness_25_5mm (cuts on the medallion planes re ...	control: success False, thickness_ 25_5mm failed; c3: thickness_ 25_5mm ...	control fails, matched run passes	false	false

Gate	Check	Value (or, where none was recorded, the reason)	Threshold	Counted	Passed
3	b2_neg_offset: negative control fails on loin_loss (the cut on the fat line removed), b2 p ...	control: success False, loin_loss failed; b2: loin_loss passed	control fails, matched run passes	false	true
3	c1_neg_offset: negative control fails on main_loss (the cut on the chain seam removed), c1 ...	control: success False, main_loss failed; c1: main_loss failed	control fails, matched run passes	false	false
3	tasks with a matched negative control	2 (B2, B5) of 6 defined	>= 3 tasks	true	false
4	a rest run for every preset	A, B, C	presets A, B, C	true	true
4	rest_A2: left alone for 30 s	30.00 s	>= 30 s	true	true
4	rest_A2: workpiece net drift (tissue, rod nodes, bones)	0.0055 mm max (tissue 0.0055, rods 0.0014, bones 0.0024; tissue median ...	<= 0.1 mm	true	true
4	rest_A2: knife net drift	0.0003 mm	<= 0.05 mm	true	true
4	rest_A2: no protective stop	0	0 stops	true	true
4	rest_A2: no fault	return code 0	no fault	true	true
4	rest_A4: left alone for 30 s	30.00 s	>= 30 s	true	true
4	rest_A4: workpiece net drift (tissue, rod nodes, bones)	0.0003 mm max (tissue 0.0003, rods 0.0001; tissue median 0.0001)	<= 0.1 mm	true	true
4	rest_A4: knife net drift	0.0003 mm	<= 0.05 mm	true	true
4	rest_A4: no protective stop	0	0 stops	true	true
4	rest_A4: no fault	return code 0	no fault	true	true
4	rest_B2: left alone for 30 s	30.00 s	>= 30 s	true	true
4	rest_B2: workpiece net drift (tissue, rod nodes, bones)	0.0090 mm max (tissue 0.0090, rods 0.0022, bones 0.0001; tissue median ...	<= 0.1 mm	true	true
4	rest_B2: knife net drift	0.0003 mm	<= 0.05 mm	true	true
4	rest_B2: no protective stop	0	0 stops	true	true
4	rest_B2: no fault	return code 0	no fault	true	true
4	rest_A0: left alone for 30 s	30.00 s	>= 30 s	true	true
4	rest_A0: workpiece net drift (tissue, rod nodes, bones)	0.0052 mm max (tissue 0.0052, rods 0.0013, bones 0.0022; tissue median ...	<= 0.1 mm	true	true
4	rest_A0: knife net drift	0.0000 mm	<= 0.05 mm	true	true
4	rest_A0: no protective stop	0	0 stops	true	true
4	rest_A0: no fault	return code 0	no fault	true	true
4	rest_C1: left alone for 30 s	30.00 s	>= 30 s	true	true
4	rest_C1: workpiece net drift (tissue, rod nodes, bones)	0.0001 mm max (tissue 0.0001; tissue median 0.0000)	<= 0.1 mm	true	true
4	rest_C1: knife net drift	0.0003 mm	<= 0.05 mm	true	true
4	rest_C1: no protective stop	0	0 stops	true	true
4	rest_C1: no fault	return code 0	no fault	true	true
5	e15: every legal-abuse episode ran	random, ram_board, ram_bone, arm_clash	random, ram_board, ram_bone, arm_clash	true	true
5	e15: random: finite, no fault, full length	finite True, fault None, 3.00 of 3 s	finite for 3 s	true	true
5	e15: random: tissue mass conserved	0.0000 g (every 0.1 s sample and the end)	|error| <= 0.1 g	true	true
5	e15: random: fastest tissue node (0.1 s samples)	0.005 m/s (fastest over every 0.1 s sample)	<= 1 m/s	true	true
5	e15: random: fastest bone body	0.001 m/s (fastest over every 0.1 s sample)	<= 1 m/s	true	true
5	e15: ram_board: finite, no fault, full length	finite True, fault None, 3.00 of 3 s	finite for 3 s	true	true
5	e15: ram_board: tissue mass conserved	0.0000 g (every 0.1 s sample and the end)	|error| <= 0.1 g	true	true
5	e15: ram_board: fastest tissue node (0.1 s samples)	0.331 m/s (fastest over every 0.1 s sample)	<= 1 m/s	true	true
5	e15: ram_board: fastest bone body	0.289 m/s (fastest over every 0.1 s sample)	<= 1 m/s	true	true
5	e15: ram_bone: finite, no fault, full length	finite True, fault None, 3.00 of 3 s	finite for 3 s	true	true
5	e15: ram_bone: tissue mass conserved	0.0000 g (every 0.1 s sample and the end)	|error| <= 0.1 g	true	true
5	e15: ram_bone: fastest tissue node (0.1 s samples)	0.889 m/s (fastest over every 0.1 s sample)	<= 1 m/s	true	true
5	e15: ram_bone: fastest bone body	0.264 m/s (fastest over every 0.1 s sample)	<= 1 m/s	true	true
5	e15: arm_clash: finite, no fault, full length	finite True, fault None, 3.00 of 3 s	finite for 3 s	true	true
5	e15: arm_clash: tissue mass conserved	0.0000 g (every 0.1 s sample and the end)	|error| <= 0.1 g	true	true
5	e15: arm_clash: fastest tissue node (0.1 s samples)	0.006 m/s (fastest over every 0.1 s sample)	<= 1 m/s	true	true
5	e15: arm_clash: fastest bone body	0.001 m/s (fastest over every 0.1 s sample)	<= 1 m/s	true	true
5	e15_drop: knife dropped: finite, no fault	finite True, fault None	finite	true	true
5	e15_drop: knife dropped: tissue mass conserved	0.0000 g (start to end)	|error| <= 0.1 g	true	true
5	e15_drop: knife dropped: fastest tissue node (every 10 ticks)	0.005 m/s	<= 1 m/s	true	true
5	e15_drop: knife dropped: the knife comes to rest (handle and blade)	0.04 mm/s over the last 10 samples (handle and blade segments)	< 5 mm/s	true	true
5	e15_drop_fish: knife dropped on the fish: lands on it, finite, no fault	landed on the fish True, finite True, fault None	landed, finite	true	true
5	e15_drop_fish: knife dropped on the fish: ownership changes	3 ownership changes after the release	<= 2	true	false

Gate	Check	Value (or, where none was recorded, the reason)	Threshold	Counted	Passed
5	e15_drop_fish: knife dropped on the fish: rebound	15.2 mm (blade middle's highest rise after it reached the fish)	< 15 mm	true	false
5	e15_drop_fish: knife dropped on the fish: no splash faster than the knife	fastest tissue node 1.69 m/s against the knife's 1.00 m/s at impact	<= 1x the impact speed	true	false
5	e15_drop_fish: knife dropped on the fish: knife and fish at rest	2.49 s after the release	<= 1.5 s	true	false
5	e15_drop_fish: knife dropped on the fish: tissue mass conserved	0.0000 g	|error| <= 0.1 g	true	true
5	knife_whip: fast wrist swings with the knife held: no fault	fault None, 10.80 s	no fault	true	true
5	knife_whip: deepest loaded contact on the knife	1.49 mm (knife_blade_rigid_4_ 174 / right/shape_17_122, 12.89 N at 5.78 ...	<= 5 mm	true	true
6	e2_press: press-cut plateau against Atkins R*l	24.81 vs 24.00 N (+3.4 %; h 3.5 mm, xi 0.0)	within 15 %	true	true
6	e2_press_fine: press-cut plateau against Atkins R*l	25.03 vs 24.00 N (+4.3 %; h 2.5 mm, xi 0.0)	within 15 %	true	true
6	e2_slice1: slice (xi 1) push plateau against Atkins R*l/(1+xi^2)	14.57 vs 12.00 N (+21.4 %; h 3.5 mm, xi 1.0)	within 15 %	true	false
6	e2_press_fine: mesh refinement change of the plateau (against e2_press)	24.81 N at 3.5 mm -> 25.03 N at 2.5 mm (+0.9 %)	<= 5 %	true	true
6	e2_press_dt: time refinement change of the plateau (against e2_press)	24.81 N at dt 2e-05 s -> 24.81 N at dt 1e-05 s (-0.0 %)	<= 5 %	true	true
6	fd_audit: cohesive bonds book exactly the bilinear law's dissipation	87 of 87 cohesive checks pass (largest booking error 2.2e-07 of G)	every cohesive check passes	true	true
6	fd_audit: force laws are the derivatives of the booked energies (live laws)	286 of 287 checks pass on laws the world runs; failing: float32 positi ...	no failing check on a live law	true	false
6	e2_press: energy ledger residual on the coupon	-0.25 % of 409.9 mJ work	|residual| <= 2 % of the work	true	true
6	e2_press_fine: energy ledger residual on the coupon	-0.51 % of 388.6 mJ work	|residual| <= 2 % of the work	true	true
6	e2_slice1: energy ledger residual on the coupon	-0.17 % of 453.3 mJ work	|residual| <= 2 % of the work	true	true
6	robot_press_cut: the demonstration's energy ledger closes	-1.0 % of 0.85 J work (dissipation 0.87 J, energy change -0.019 J)	|residual| <= 5 % of the work	true	true
6	robot_press_cut_slow: the demonstration's energy ledger closes	-0.7 % of 0.65 J work (dissipation 0.87 J, energy change -0.214 J)	|residual| <= 5 % of the work	true	true
6	robot_press_cut_fast: the demonstration's energy ledger closes	-0.8 % of 0.84 J work (dissipation 0.88 J, energy change -0.029 J)	|residual| <= 5 % of the work	true	true
6	robot_press_cut_panda: the demonstration's energy ledger closes	-1.0 % of 0.66 J work (dissipation 0.88 J, energy change -0.214 J)	|residual| <= 5 % of the work	true	true
6	robot_c3: the demonstration's energy ledger closes	-0.9 % of 7.12 J work (dissipation 7.17 J, energy change +0.020 J)	|residual| <= 5 % of the work	true	true
6	a1: the demonstration's energy ledger closes	-241.6 % of 2.79 J work (dissipation 9.57 J, energy change -0.029 J)	|residual| <= 5 % of the work	true	false
6	a1_neg_offset: the demonstration's energy ledger closes	-61.4 % of 3.65 J work (dissipation 5.93 J, energy change -0.032 J)	|residual| <= 5 % of the work	true	false
6	a2: the demonstration's energy ledger closes	-17.0 % of 10.99 J work (dissipation 13.23 J, energy change -0.372 J)	|residual| <= 5 % of the work	true	false
6	a4: the demonstration's energy ledger closes	-5.6 % of 10.21 J work (dissipation 10.77 J, energy change +0.009 J)	|residual| <= 5 % of the work	true	false
6	a4_neg_unheld: the demonstration's energy ledger closes	+0.5 % of 10.29 J work (dissipation 10.24 J, energy change -0.000 J)	|residual| <= 5 % of the work	true	true
6	b5: the demonstration's energy ledger closes	-12.9 % of 5.45 J work (dissipation 6.16 J, energy change +0.000 J)	|residual| <= 5 % of the work	true	false
6	b5_neg_tilt: the demonstration's energy ledger closes	-9.9 % of 6.08 J work (dissipation 6.68 J, energy change -0.002 J)	|residual| <= 5 % of the work	true	false
6	b2: the demonstration's energy ledger closes	+0.8 % of 7.15 J work (dissipation 7.10 J, energy change -0.005 J)	|residual| <= 5 % of the work	true	true
6	b2_neg_offset: the demonstration's energy ledger closes	+3.9 % of 8.43 J work (dissipation 8.11 J, energy change -0.003 J)	|residual| <= 5 % of the work	true	true
6	b3: the demonstration's energy ledger closes	-13.4 % of 1.22 J work (dissipation 1.38 J, energy change +0.000 J)	|residual| <= 5 % of the work	true	false
6	c1: the demonstration's energy ledger closes	-1.4 % of 14.36 J work (dissipation 14.53 J, energy change +0.032 J)	|residual| <= 5 % of the work	true	true
6	c2: the demonstration's energy ledger closes	-6.1 % of 2.60 J work (dissipation 2.76 J, energy change -0.000 J)	|residual| <= 5 % of the work	true	false
6	c1_neg_offset: the demonstration's energy ledger closes	-1.1 % of 21.73 J work (dissipation 21.96 J, energy change -0.000 J)	|residual| <= 5 % of the work	true	true

Gate	Check	Value (or, where none was recorded, the reason)	Threshold	Counted	Passed
6	c3: the demonstration's energy ledger closes	-0.1 % of 38.22 J work (dissipation 38.25 J, energy change -0.001 J)	|residual| <= 5 % of the work	true	true
6	c3_neg_offset: the demonstration's energy ledger closes	-0.0 % of 33.91 J work (dissipation 33.92 J, energy change +0.000 J)	|residual| <= 5 % of the work	true	true
6	robot_a1: the demonstration's energy ledger closes	+10.8 % of 0.02 J work (dissipation 0.01 J, energy change +0.014 J)	|residual| <= 5 % of the work	true	false
6	flank_friction: ledger residual, blade sliding flat without cutting	*the blade cut 151 faces: not a friction-only run*	|residual| <= 2 % of the work	true	false
6	iface_drag: interface drag books the placed fracture energy	1.000023 of 1 mJ booked; debt left 3.3e-07 mJ; unpaid while still 0.99 ...	|booked - placed| <= 0.001 mJ, paid only ...	true	true
6	rig_cap: assisted rig pressed into the board: force capped, finite, ledger closed	servo force at most 149.3 N (cap 150 N), holding 149.3 N; residual +0. ...	<= cap, |residual| <= 2 %	true	true
6	iface_drag: ledger residual of the interface drag	+0.0157 of 1.061 mJ work (+1.48 %, rest drift removed)	|residual| <= 2 % of the work	true	true
6	tear_across: tearing: notched bar (across the sheets) breaks at its strength	peak net stress 1.24 x the strength, residual +1.03 %, 3 pieces	0.7-1.3 x strength, |residual| <= 2 %	true	true
6	tear_along: tearing: notched bar (along the sheets) breaks at its strength	peak net stress 1.05 x the strength, residual +1.14 %, 9 pieces	0.7-1.3 x strength, |residual| <= 2 %	true	true
6	e1_puncture: flat-cylinder force at 6-8 mm within Torgersen's 6.6-20.9 N	force 6.4-8.3 N at 6-8 mm depth	6.6-20.9 N	true	false
6	e1_puncture: flat-cylinder puncture: the surface breaks at 6-8 mm (G27: tearing is tension ...	largest force at 6-8 mm 8.3 N at 7.9 mm, then at most 0 % lower (peak ...	a 10 % force drop after a peak at 6-8 mm	true	false
7	e16_hold: operating wall/sim through the public API (largest steady action on a quiet host ...	60.1 max, 60.0 mean over 2 quiet-host steady actions of 0.5 s (all 5: ...	<= 200	true	true
7	e16_hold: RGB observation with every action, sim clocks agree	RGB decoded True, clocks agree True (overhead)	every action	true	true
7	e16_hold: startup reported separately	138 s (create 137, reset 1)	reported	true	true
7	e16_hold: host load recorded with every action	load 3.5-4.3; GPU 1.0-2.8 %	every action	true	true
7	e16_cut: operating wall/sim through the public API (largest steady action on a quiet host)	215.0 max, 118.2 mean over 10 quiet-host cutting actions of 0.5 s (all ...	<= 200	true	false
7	e16_cut: the cutting stage cut: crack faces made during the timed actions	1284 crack faces made in 13 of 19 actions	> 0	true	true
7	e16_cut: RGB observation with every action, sim clocks agree	RGB decoded True, clocks agree True (overhead)	every action	true	true
7	e16_cut: startup reported separately	136 s (create 135, reset 1)	reported	true	true
7	e16_cut: host load recorded with every action	load 2.4-9.6; GPU 1.0-20.8 %	every action	true	true
7	e16_cut_2: operating wall/sim through the public API (largest steady action on a quiet hos ...	239.6 max, 133.9 mean over 11 quiet-host cutting actions of 0.5 s (all ...	<= 200	true	false
7	e16_cut_2: the cutting stage cut: crack faces made during the timed actions	1287 crack faces made in 15 of 19 actions	> 0	true	true
7	e16_cut_2: RGB observation with every action, sim clocks agree	RGB decoded True, clocks agree True (overhead)	every action	true	true
7	e16_cut_2: startup reported separately	136 s (create 136, reset 1)	reported	true	true
7	e16_cut_2: host load recorded with every action	load 3.1-10.2; GPU 1.0-1.0 %	every action	true	true
7	e16_cut_B: operating wall/sim through the public API (largest steady action on a quiet hos ...	156.9 max, 118.8 mean over 7 quiet-host cutting actions of 0.5 s (all ...	<= 200	true	true
7	e16_cut_B: the cutting stage cut: crack faces made during the timed actions	594 crack faces made in 11 of 19 actions	> 0	true	true
7	e16_cut_B: RGB observation with every action, sim clocks agree	RGB decoded True, clocks agree True (overhead)	every action	true	true
7	e16_cut_B: startup reported separately	153 s (create 153, reset 0)	reported	true	true
7	e16_cut_B: host load recorded with every action	load 2.5-11.2; GPU 1.0-12.4 %	every action	true	true
7	e16_cut_C: operating wall/sim through the public API (largest steady action on a quiet hos ...	80.5 max, 72.0 mean over 2 quiet-host cutting actions of 0.5 s (all 6: ...	<= 200	true	true
7	e16_cut_C: the cutting stage cut: crack faces made during the timed actions	781 crack faces made in 6 of 19 actions	> 0	true	true
7	e16_cut_C: RGB observation with every action, sim clocks agree	RGB decoded True, clocks agree True (overhead)	every action	true	true
7	e16_cut_C: startup reported separately	177 s (create 177, reset 1)	reported	true	true
7	e16_cut_C: host load recorded with every action	load 3.1-10.6; GPU 1.0-1.0 %	every action	true	true
7	e16_cut_cams: operating wall/sim through the public API (largest steady action on a quiet ...	224.1 max, 132.2 mean over 8 quiet-host cutting actions of 0.5 s (all ...	<= 200	true	false
7	e16_cut_cams: the cutting stage cut: crack faces made during the timed actions	1360 crack faces made in 13 of 19 actions	> 0	true	true

Gate	Check	Value (or, where none was recorded, the reason)	Threshold	Counted	Passed
7	e16_cut_cams: RGB observation with every action, sim clocks agree	RGB decoded True, clocks agree True (overhead)	every action	true	true
7	e16_cut_cams: startup reported separately	139 s (create 137, reset 2)	reported	true	true
7	e16_cut_cams: host load recorded with every action	load 2.7-5.5; GPU 1.0-1.4 %	every action	true	true
8	interfaces: ‘gar run‘ session with owner and policy keys	ready, authenticated, CLI exit 0, policy acts, stopped	all true	true	true
8	interfaces: policy key refused state and checkpoints	checkpoint refused True, state refused True	both 403	true	true
8	interfaces: checkpoint round trip	tick 100 saved, 200 after acting, 100 after restore	tick returns to the saved tick	true	true
8	interfaces: dashboard loaded in a browser, screenshots kept	page 200 True, camera frame shown True, 2 screenshots (owner, policy)	page, frame, 2 screenshots	true	true
8	gar_tests: the GAR filleting tests pass, the physics round trip included	28 passed, 0 failed, 0 errors, 0 skipped (exit 0)	all pass, none skipped	true	true

8.1.2 Every model constant of the ledger is read by the world’s code

A static scan of the world’s source for each constant of params.py; model constants must be read, reference entries carry the reason they are not model inputs (params.REFERENCE_ONLY).

Observable	Recorded value
Constants	83
Model inputs	71
References	12
Open (label)	21
Model constants read by nothing	*none*
References without a reason	*none*

Every constant, its use and the world files that read it.

Constant	Use	Label	Read by	Why a reference
FISH_DENSITY	model	abstract+estimate	filleting/anatomy/ salmon.py, filleting/ anatomy/trout.py, filleting/assembly.py, filleting/workpiece.py	*none*
FISH_E	model	derived	filleting/assembly.py, filleting/workpiece.py	*none*
FISH_NU	model	read(porcine)+estimate	filleting/assembly.py, filleting/workpiece.py	*none*
FISH_TAU_V	model	estimate	filleting/assembly.py, filleting/tissue.py, filleting/workpiece.py	*none*
TISSUE_VISCO	model	derived+estimate	filleting/tissue.py	*none*
FISH_R_PERP	model	derived	filleting/assembly.py, filleting/workpiece.py	*none*
FISH_R_PAR	model	derived	filleting/assembly.py, filleting/workpiece.py	*none*
FISH_HEAD_E	reference	estimate	*none*	the head is a rigid body
FISH_HEAD_R	reference	estimate	*none*	the head is a rigid body, removed at its joint, not cut through
SHEET_SIGMA	model	open	filleting/tearing.py	*none*
SHEET_G	model	open	filleting/tearing.py	*none*
SHEET_BETA	model	estimate	filleting/tearing.py	*none*
FISH_SIGMA_T	model	open	filleting/assembly.py, filleting/workpiece.py	*none*
SKIN_THICKNESS_ TROUT	model	read(other species)+estimate	filleting/anatomy/trout.py	*none*
SKIN_THICKNESS_ SALMON	model	read(other species)+estimate	filleting/anatomy/ salmon.py	*none*
SKIN_E	model	search	filleting/assembly.py	*none*
SKIN_STRENGTH	model	read	filleting/assembly.py, filleting/tearing.py	*none*
SKIN_DENSITY	model	estimate	filleting/anatomy/ trout.py, filleting/ assembly.py	*none*

Constant	Use	Label	Read by	Why a reference
SKIN_R	model	estimate	filleting/assembly.py, filleting/tissue.py	*none*
SKIN_FLESH_G	reference	open	*none*	the skin membrane shares the flesh's surface nodes: no skin-flesh interface (skinning, B4, is refused)
SKIN_FLESH_SIGMA	reference	open	*none*	as SKIN_FLESH_G
FLESH_BONE_G	model	open	filleting/anatomy/salmon.py, filleting/anatomy/trout.py, filleting/assembly.py	*none*
FLESH_BONE_SIGMA	model	open	filleting/anatomy/salmon.py, filleting/anatomy/trout.py, filleting/assembly.py	*none*
JOINT_R	model	abstract+estimate	filleting/anatomy/trout.py, filleting/workpiece.py	*none*
BONE_E	model	read	filleting/assembly.py	*none*
BONE_UTS	model	read	filleting/assembly.py	*none*
BONE_DENSITY	model	estimate	filleting/anatomy/salmon.py, filleting/anatomy/trout.py, filleting/assembly.py	*none*
BONE_CRUSH	reference	read(other tissue)+estimate	*none*	vertebrae are rigid and cannot be cut; thin bones break at BONE_UTS
BONE_SHEAR	reference	estimate	*none*	no edge-on-bone shear failure is modelled; thin bones break at BONE_UTS
PIN_PULL_SALMON	reference	read	*none*	the source of the derived PIN_SHAFT_TAU (a model input)
PIN_BREAK_SALMON	reference	read	*none*	pin-boning (B1) is refused
PIN_COUNT_SALMON	model	read	filleting/anatomy/salmon.py	*none*
PIN_LENGTH_SALMON	reference	read	*none*	the salmon's pin bones are generated by anatomy/salmon.py
PIN_SHAFT_TAU	model	derived	filleting/assembly.py	*none*
PIN_ROOT_SHARE	model	read+derived	filleting/anatomy/salmon.py	*none*
PORK_E	model	read+estimate	filleting/assembly.py	*none*
PORK_R	model	estimate	filleting/assembly.py	*none*
PORK_DENSITY	model	literature	filleting/anatomy/tenderloin.py, filleting/assembly.py	*none*
PORK_SIGMA_T	model	derived	filleting/assembly.py	*none*
FAT_E	model	open	filleting/assembly.py	*none*
FAT_R	model	read(warm)+open	filleting/assembly.py	*none*
FAT_SIGMA_T	model	open	filleting/assembly.py	*none*
SALMON_BELLY	model	open	filleting/assembly.py	*none*
SILVER_THICKNESS	model	open	filleting/anatomy/tenderloin.py	*none*
SILVER_E	model	open	filleting/anatomy/tenderloin.py, filleting/assembly.py	*none*
SILVER_STRENGTH	model	open	filleting/anatomy/tenderloin.py, filleting/tearing.py	*none*
SILVER_R	model	open	filleting/anatomy/tenderloin.py, filleting/assembly.py, filleting/tissue.py	*none*
SILVER_BOND_G	model	open	filleting/anatomy/tenderloin.py	*none*
CHAIN_SEAM_G	model	open	filleting/anatomy/tenderloin.py	*none*
CHAIN_SEAM_SIGMA	model	open	filleting/anatomy/tenderloin.py	*none*
SILVER_BOND_SIGMA	model	open	filleting/anatomy/tenderloin.py	*none*
FILLET_BLADE	model	read	filleting/knife.py, filleting/scene.py	*none*
BONING_BLADE	model	read	filleting/knife.py, filleting/scene.py	*none*
HANDLE	model	read+estimate	filleting/scene.py, filleting/world.py	*none*
STEEL_E	model	literature	filleting/knife.py, filleting/workpiece.py	*none*
STEEL_G	model	literature	filleting/workpiece.py	*none*
BSI_HONED	model	read+estimate	filleting/knife.py	*none*

Constant	Use	Label	Read by	Why a reference
BSI_WORN	model	read+estimate	filleting/knife.py	*none*
MU_BLADE_TISSUE	model	open	filleting/scene.py, filleting/workpiece.py	*none*
TAU0_BLADE_TISSUE	reference	open	*none*	blade-tissue adhesion is not modelled (Coulomb friction, MU_BLADE_TISSUE)
MU_BLADE_BONE	model	estimate	filleting/workpiece.py	*none*
MU_SKIN_HDPE	model	estimate	filleting/workpiece.py	*none*
MU_FLESH_HDPE	model	estimate	filleting/scene.py, filleting/workpiece.py	*none*
MU_PAD_SKIN	model	estimate	filleting/workpiece.py	*none*
MU_PAD_FLESH	model	estimate	filleting/workpiece.py	*none*
MU_INSERT_HANDLE	model	repo	filleting/scene.py	*none*
REFLEX_TORQUE_NM	model	read	filleting/world.py	*none*
REFLEX_TORQUE_CONTACT_NM	model	read	filleting/world.py	*none*
RIG_FORCE_MAX	model	authored	filleting/tissue.py	*none*
RIG_TORQUE_MAX	model	authored	filleting/tissue.py	*none*
RIG_STIFFNESS	model	authored	filleting/tissue.py	*none*
REFLEX_RECOVERY_S	model	authored	filleting/world.py	*none*
JOINT_ACCEL_PANDA	model	read	filleting/world.py	*none*
JOINT_ACCEL_FR3	model	prior	filleting/world.py	*none*
MU_TWEEZER_BONE	model	estimate	filleting/workpiece.py	*none*
FRANKA_HAND	reference	read	*none*	the hand's limits come from the robot catalog's model (robots/models)
FR3	model	read	filleting/layout.py, filleting/scene.py, filleting/variants.py, filleting/world.py	*none*
TROUT	model	measured(generator)+read	filleting/anatomy/trout.py, filleting/task.py	*none*
TROUT_FILLET_YIELD	model	read	filleting/anatomy/trout.py	*none*
SALMON_SIDE	model	derived+estimate	filleting/anatomy/salmon.py	*none*
TENDERLOIN	model	search+estimate	filleting/anatomy/tenderloin.py	*none*
IMPS	model	read	filleting/anatomy/tenderloin.py, filleting/task.py	*none*
CODEX_BONE	reference	read	*none*	the specification the evaluators' bone checks follow

8.2 World at real scale and weight (quality gate 1)

What the workpieces are as generated and meshed, how much inertia the explicit solver adds, and how the settled start states hold.

8.2.1 Mass budget of every workpiece

Builds each workpiece world and reports the physical node mass against the stability inertia the explicit solver adds.

Workpiece	Mesh edge [mm]	Tets	Nodes	Physical mass [g]	Added inertia [g]	Added [%]	Nodes scaled	Largest ratio
trout	3.5	73 273	21 018	242.8	31.2	12.9	7 007	28.6
trout_fillet_rib_strip	3.5	33 501	9 500	104.6	13.1	12.5	3 339	19.5
salmon	5	93 851	21 467	954.2	34.2	3.6	4 730	12.6
tenderloin	3.5	149 862	31 038	534.3	30.8	5.8	5 826	33.3

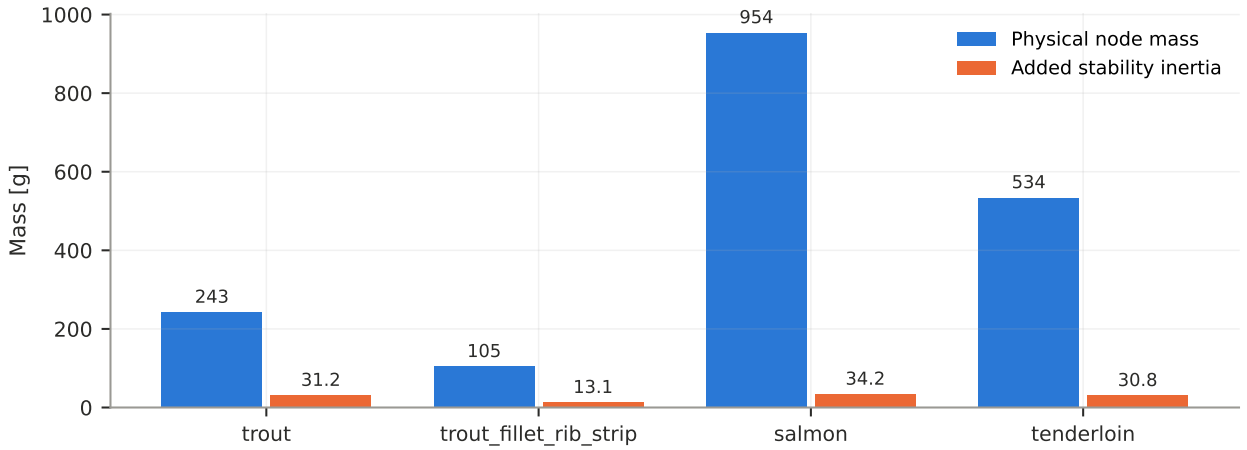


Figure 2: **Mass budget of every workpiece.** Physical node mass and the inertia the explicit solver adds to stiff or tiny nodes, per workpiece, as built. The added mass carries no weight (gravity acts on the physical mass only).

8.2.2 Least muscle loss a denuding can have at this mesh

The silver skin is a membrane on the outer faces of surface tets, so removing it removes at least the tets that carry it; reports that mass against C2's 5 % budget. No stepping.

Observable	Recorded value
Mesh edge [mm]	3.5
Main muscle [g]	469.7
Silver skin area [cm^2]	105
Carrier tets	3 649
Carrier tets' main muscle [g]	6.61
Carrier main muscle / main muscle	0.014
Least main loss for 90 % of the area [g]	5.54
Least main loss for 90 % of the area / main muscle	0.012
C2 budget (fraction)	0.05
Passable at this resolution	true

8.2.3 Least flesh a rib removal (A4) can take at this mesh

A rib is a rod tied to the tets holding its nodes and a cut follows whole tet faces, so taking a rib out takes at least every tet its shaft passes through (the carriers); also the layer a flat stroke under the rib field lifts with the ribs (carriers plus the tets on their cavity side within one edge length); against A4's 6 % budget. No stepping.

Observable	Recorded value
Mesh edge [mm]	3.5
Fillet [g]	104.1
Ribs	36
Carrier tets	1 231
Carrier mass [g]	3.61
Carrier mass / fillet	0.035
Layer under a flat stroke: tets	3 485
Layer mass [g]	9.21
Layer mass / fillet	0.088
A4 budget (fraction)	0.06
Passable at this resolution (carriers)	true

8.2.4 Collider audit: every collider sourced, nothing visual-only within reach

For presets A, B and C, every static collider with its size and dimension source, and every prop within the arms' reach checked for a collider.

Observable	Recorded value
All colliders sourced and no visual-only prop in reach	true
Reach considered [m]	1.255
Colliders without a dimension source	*none*
Props in reach without a collider	*none*

Preset	Colliders	Unsourced	Visual-only in reach	Robot shapes
A	44	0	0	108
B	43	0	0	108
C	37	0	0	108

8.2.5 Thin slice 3 mm deep under a held block

A horizontal slice 3 mm under a fish-muscle block's top, the resolution limit of a layer about one element thick.

Assistance: phantom knife rig + scripted left FR3 hold

Observable	Recorded value
Depth [mm]	3
Mesh edge [mm]	3.19
Slab expected [g]	15.07
Slab freed	*none*
Fragments	8
Bridges left	76
Faces cut	5 956

8.2.6 Thin slice 6 mm deep under a held block

A horizontal slice 6 mm under a fish-muscle block's top, the resolution limit of a layer about one element thick.

Assistance: phantom knife rig + scripted left FR3 hold

Observable	Recorded value
Depth [mm]	6
Mesh edge [mm]	3.19
Slab expected [g]	30.14
Slab freed	*none*
Fragments	4
Bridges left	42
Faces cut	6 468

8.2.7 Thin slice 9 mm deep under a held block

A horizontal slice 9 mm under a fish-muscle block's top, the resolution limit of a layer about one element thick.

Assistance: phantom knife rig + scripted left FR3 hold

Observable	Recorded value
Depth [mm]	9
Mesh edge [mm]	3.19
Slab expected [g]	45.21
Slab freed	*none*
Fragments	1
Bridges left	18
Faces cut	7 420

8.3 The robot doing the work (quality gate 2)

Runs in which only legal robot commands act on the world. A scripted robot may plan from privileged state; that is labelled on every run. No autonomous policy is claimed.

8.3.1 Press cut by the robot-held knife

The right FR3 presses the knife it holds through a workpiece by scripted IK waypoints through the arm servos. 'Work put in' includes the energy the blade carries into and out of the tissue solver when its owner changes (its gravitational energy, -214 mJ when the solver hands it to MuJoCo after the settle, booked back when the press hands it to the solver), so the cutting dissipation can exceed it; the ledger closes to about -1 %.

Assistance: none on the world (legal robot commands); the script plans its waypoints from privileged state

Observable	Recorded value
Simulated time [s]	7.869
Wall seconds per simulated second	165.3
Grip force per finger [N]	35
Knife mass (handle, blade) [kg]	blade: 0.01691; handle: 0.07109; total: 0.088
Fragments	2
Largest fragment masses [g]	62.8, 62.58
Protective stops	0
Ledger: cutting dissipation [mJ]	788.6
Ledger: work put in [mJ]	845.1
Smallest volume ratio at the end	0.956
Fragments (crack faces and intact bonds, the evaluator's count)	2
Fragment masses [g]	62.8, 62.58
Crack faces by origin	total: 1 793; interface_at_build: 0; torn: 685; knife_upper_bound: 1 108
Pieces not resting on the board	1
Blade load during the lift, lowest [N]	-0.79
Block weight [N]	1.23

Script log as recorded (every 20th row of the 10 ms log: phase, edge height, blade load, grip).

t [s]	Phase	Edge mid z [m]	Blade Fz [N]	Blade flex [deg]	Grip angle [deg]
0.01	settle	1.2735	0	0	0
0.21	settle	1.2733	-0.016	0.029	0
0.41	settle	1.2733	0	0.02	0
0.61	orient	1.2729	-0.007	0	0.071
0.81	orient	1.2674	0.003	0.026	0.071
1.01	orient	1.2563	0	0.013	0.07
1.21	orient	1.2439	-0	0.013	0.067
1.41	orient	1.2339	-0.017	0.008	0.071
1.61	orient	1.2303	-0.004	0	0.084
1.81	approach	1.2289	-0.012	0	0.089
2.01	approach	1.2057	-0.01	0	0.088
2.21	approach	1.1524	-0.012	0	0.092
2.41	approach	1.0966	-0.021	0.001	0.122
2.61	approach	1.0362	-0.02	0	0.141
2.81	approach	0.97533	-0.01	0	0.162
3.01	approach	0.94127	-0.016	0	0.139
3.21	press	0.94114	-0.007	0	0.129
3.41	press	0.93716	7.75	0.005	0.122
3.61	press	0.93641	11.7	0.107	0.226
3.81	press	0.93538	17.7	0.476	0.451
4.01	press	0.93439	23.7	0.686	1.03
4.21	press	0.93329	30.1	1.12	1.08
4.41	press	0.93209	37.1	1.57	1.12
4.61	press	0.92951	35.3	1.98	1.15
4.81	press	0.92459	36.2	1.86	1.15
5.01	press	0.92092	36.4	1.96	1.15
5.21	press	0.91695	36.3	2.23	1.15
5.41	press	0.9131	17	2.84	1.15
5.61	press	0.91234	0.571	0.1	1.16
5.81	press	0.9113	0.004	0.288	1.17
6.01	hold	0.91137	0.001	0.278	1.27
6.21	hold	0.91189	-0.097	0.21	1.35
6.41	hold	0.91222	-0.141	0.17	1.4
6.61	lift	0.91385	-0.088	0.014	1.34
6.81	lift	0.92175	-0.687	0.135	1.13
7.01	lift	0.93251	-0.645	0.256	1.13
7.21	lift	0.94287	-0.614	0.114	1.13
7.41	lift	0.9532	-0.618	0.019	1.13
7.61	rest	0.96345	-0.589	0.094	1.12
7.81	rest	0.96625	-0.613	0.105	1.12

8.3.2 Press cut by the robot-held knife (target speed 30 mm/s)

The right FR3 presses the knife it holds through a workpiece by scripted IK waypoints through the arm servos. 'Work put in' includes the energy the blade carries into and out of the tissue solver when its owner changes (its gravitational energy, -214 mJ when the solver hands it to MuJoCo after the settle, booked back when the press hands it to the solver), so the cutting dissipation can exceed it; the ledger closes to about -1 %.

Assistance: none on the world (legal robot commands); the script plans its waypoints from privileged state

Observable	Recorded value
Simulated time [s]	6.934
Wall seconds per simulated second	146.3
Grip force per finger [N]	35
Knife mass (handle, blade) [kg]	blade: 0.01691; handle: 0.07109; total: 0.088
Fragments	2
Largest fragment masses [g]	64.74, 60.64
Protective stops	0
Ledger: cutting dissipation [mJ]	790.3
Ledger: work put in [mJ]	840.1
Smallest volume ratio at the end	0.888
Fragments (crack faces and intact bonds, the evaluator's count)	2
Fragment masses [g]	64.74, 60.64
Crack faces by origin	total: 1 783; interface_at_build: 0; torn: 695; knife_upper_bound: 1 088
Pieces not resting on the board	0
Blade load during the lift, lowest [N]	-0.772
Block weight [N]	1.23

Script log as recorded (every 20th row of the 10 ms log: phase, edge height, blade load, grip).

t [s]	Phase	Edge mid z [m]	Blade Fz [N]	Blade flex [deg]	Grip angle [deg]
0.01	settle	1.2735	0	0	0
0.21	settle	1.2733	-0.016	0.029	0
0.41	settle	1.2733	0	0.02	0
0.61	orient	1.2729	-0.007	0	0.071
0.81	orient	1.2674	0.003	0.026	0.071
1.01	orient	1.2563	0	0.013	0.07
1.21	orient	1.2439	-0	0.013	0.067
1.41	orient	1.2339	-0.017	0.008	0.071
1.61	orient	1.2303	-0.004	0	0.084
1.81	approach	1.2289	-0.012	0	0.089
2.01	approach	1.2057	-0.01	0	0.088
2.21	approach	1.1524	-0.012	0	0.092
2.41	approach	1.0966	-0.021	0.001	0.122
2.61	approach	1.0362	-0.02	0	0.141
2.81	approach	0.97533	-0.01	0	0.162
3.01	approach	0.94127	-0.016	0	0.139
3.21	press	0.94087	-0.013	0	0.128
3.41	press	0.93661	11	0.008	0.142
3.61	press	0.93528	19.1	0.47	0.537
3.81	press	0.93364	29	0.894	1.07
4.01	press	0.93191	38.9	1.57	1.14
4.21	press	0.92639	36.5	1.96	1.15
4.41	press	0.92053	36.5	2.08	1.15
4.61	press	0.91469	36.3	2.41	1.15
4.81	press	0.91234	0.065	0.223	1.16
5.01	hold	0.91134	0.084	0.281	1.23
5.21	hold	0.91165	-0.228	0.225	1.32
5.41	hold	0.91216	-0.349	0.186	1.39
5.61	lift	0.91247	-0.405	0.153	1.43
5.81	lift	0.91856	-0.505	0.058	1.14
6.01	lift	0.9292	-0.434	0.183	1.13
6.21	lift	0.93964	-0.446	0.071	1.13
6.41	lift	0.94992	-0.462	0.007	1.13
6.61	lift	0.9603	-0.429	0.024	1.13
6.81	rest	0.96641	-0.494	0.008	1.13

8.3.3 Press cut by the robot-held knife, dual Panda (target speed 20 mm/s)

The right Panda presses the knife it holds through the fish block by scripted IK waypoints through the arm servos: the Panda's one cutting run. 'Work put in' includes the energy the blade carries into and out of the tissue solver when its owner changes (its gravitational energy, -214 mJ when the solver hands it to MuJoCo after the settle, booked back when the press hands it to the solver), so the cutting dissipation can exceed it; the ledger closes to about -1 %.

Assistance: none on the world (legal robot commands); the script plans its waypoints from privileged state

Observable	Recorded value
Simulated time [s]	7.848
Wall seconds per simulated second	130.1
Grip force per finger [N]	35
Knife mass (handle, blade) [kg]	blade: 0.01691; handle: 0.07109; total: 0.088
Fragments	1
Largest fragment masses [g]	125.4
Protective stops	0
Ledger: cutting dissipation [mJ]	778.1
Ledger: work put in [mJ]	658
Smallest volume ratio at the end	0.956
Fragments (crack faces and intact bonds, the evaluator's count)	1
Fragment masses [g]	125.4
Crack faces by origin	total: 1 844; interface_at_build: 0; torn: 782; knife_upper_bound: 1 062
Pieces not resting on the board	0
Blade load during the lift, lowest [N]	-0.955
Block weight [N]	1.23

Script log as recorded (every 20th row of the 10 ms log: phase, edge height, blade load, grip).

t [s]	Phase	Edge mid z [m]	Blade Fz [N]	Blade flex [deg]	Grip angle [deg]
0.01	settle	1.2735	0	0	0
0.21	settle	1.2731	-0.004	0.038	0
0.41	settle	1.2731	-0.028	0	0
0.61	orient	1.2726	-0.007	0.03	0.078
0.81	orient	1.2669	-0.02	0.03	0.078
1.01	orient	1.2559	-0.006	0.009	0.078
1.21	orient	1.2435	-0.021	0.011	0.05
1.41	orient	1.2335	-0.012	0	0.078
1.61	orient	1.2299	-0.017	0	0.091
1.81	approach	1.2287	-0.001	0	0.096
2.01	approach	1.2047	-0.02	0	0.097
2.21	approach	1.1535	-0	0	0.103
2.41	approach	1.1021	-0.016	0	0.141
2.61	approach	1.0468	-0.008	0.001	0.166
2.81	approach	0.98784	-0.018	0	0.183
3.01	approach	0.94423	-0.015	0	0.155
3.21	press	0.94033	-0.006	0	0.14
3.41	press	0.93718	7.05	0.005	0.132
3.61	press	0.93625	11.8	0.353	0.286
3.81	press	0.93507	18.4	0.698	0.517
4.01	press	0.934	24.8	0.915	1.05
4.21	press	0.93298	30.9	1.06	1.1
4.41	press	0.93181	38.1	1.48	1.15
4.61	press	0.92883	35.4	1.82	1.18
4.81	press	0.924	36.3	1.69	1.18
5.01	press	0.92024	36.5	1.75	1.18
5.21	press	0.91624	36.4	2.01	1.18
5.41	press	0.913	3.44	0.381	1.18
5.61	press	0.91209	-0.007	0.205	1.18
5.81	press	0.911	-0.009	0.194	1.2
6.01	hold	0.91118	-0.009	0.158	1.29
6.21	hold	0.91148	-0.009	0.152	1.33
6.41	hold	0.91164	-0.009	0.149	1.35
6.61	lift	0.91392	-0.202	0.057	1.24
6.81	lift	0.92259	-0.852	0.172	1.18
7.01	lift	0.9338	-0.43	0.117	1.18
7.21	lift	0.9443	-0.012	0	1.18
7.41	lift	0.95466	-0.016	0	1.18
7.61	rest	0.96471	-0.007	0	1.18
7.81	rest	0.9665	-0.007	0	1.17

8.3.4 Press cut by the robot-held knife (target speed 12 mm/s)

The right FR3 presses the knife it holds through a workpiece by scripted IK waypoints through the arm servos. 'Work put in' includes the energy the blade carries into and out of the tissue solver when its owner changes (its gravitational energy, -214 mJ when the solver hands it to MuJoCo after the settle, booked back when the press hands it to the solver), so the cutting dissipation can exceed it; the ledger closes to about -1 %.

Assistance: none on the world (legal robot commands); the script plans its waypoints from privileged state

Observable	Recorded value
Simulated time [s]	9.74
Wall seconds per simulated second	166.5
Grip force per finger [N]	35
Knife mass (handle, blade) [kg]	blade: 0.01691; handle: 0.07109; total: 0.088
Fragments	2
Largest fragment masses [g]	64.89, 60.49
Protective stops	0
Ledger: cutting dissipation [mJ]	772
Ledger: work put in [mJ]	653.8
Smallest volume ratio at the end	0.998
Fragments (crack faces and intact bonds, the evaluator's count)	2
Fragment masses [g]	64.89, 60.49
Crack faces by origin	total: 1 911; interface_at_build: 0; torn: 817; knife_upper_bound: 1 094
Pieces not resting on the board	0
Blade load during the lift, lowest [N]	-0.574
Block weight [N]	1.23

Script log as recorded (every 20th row of the 10 ms log: phase, edge height, blade load, grip).

t [s]	Phase	Edge mid z [m]	Blade Fz [N]	Blade flex [deg]	Grip angle [deg]
0.01	settle	1.2735	0	0	0
0.21	settle	1.2733	-0.016	0.029	0
0.41	settle	1.2733	0	0.02	0
0.61	orient	1.2729	-0.007	0	0.071
0.81	orient	1.2674	0.003	0.026	0.071
1.01	orient	1.2563	0	0.013	0.07
1.21	orient	1.2439	-0	0.013	0.067
1.41	orient	1.2339	-0.017	0.008	0.071
1.61	orient	1.2303	-0.004	0	0.084
1.81	approach	1.2289	-0.012	0	0.089
2.01	approach	1.2057	-0.01	0	0.088
2.21	approach	1.1524	-0.012	0	0.092
2.41	approach	1.0966	-0.021	0.001	0.122
2.61	approach	1.0362	-0.02	0	0.141
2.81	approach	0.97533	-0.01	0	0.162
3.01	approach	0.94127	-0.016	0	0.139
3.21	press	0.94138	-0.008	0	0.129
3.41	press	0.9379	3.46	0.002	0.124
3.61	press	0.93735	5.9	0.082	0.12
3.81	press	0.9365	9.98	0.317	0.118
4.01	press	0.9359	13.1	0.483	0.307
4.21	press	0.9352	16.8	0.732	0.39
4.41	press	0.9348	19	0.885	0.791
4.61	press	0.9341	23.1	1.06	1.03
4.81	press	0.93338	27.3	1.34	1.06
5.01	press	0.93275	31	1.33	1.08
5.21	press	0.93206	35.5	1.55	1.11
5.41	press	0.93137	39.7	1.85	1.14
5.61	press	0.92868	35.7	2.13	1.15
5.81	press	0.92579	36.2	1.9	1.15
6.01	press	0.92363	36.6	1.86	1.15
6.21	press	0.92126	36.8	1.96	1.15
6.41	press	0.9189	36.6	2.05	1.15
6.61	press	0.91646	36.3	2.22	1.15
6.81	press	0.91418	36.4	2.44	1.15
7.01	press	0.91298	5.77	0.854	1.16
7.21	press	0.91243	1.8	0.777	1.16
7.41	press	0.91184	1.07	0.394	1.16
7.61	press	0.91129	0	0.247	1.18
7.81	hold	0.91097	-0.006	0.168	1.22
8.01	hold	0.91163	-0.006	0.166	1.32
8.21	hold	0.91224	-0.006	0.207	1.41
8.41	lift	0.91254	-0.006	0.218	1.45
8.61	lift	0.91836	-0.293	0.021	1.14
8.81	lift	0.92903	0.001	0.207	1.14
9.01	lift	0.93963	-0.014	0	1.14
9.21	lift	0.94984	-0.005	0	1.14
9.41	lift	0.96022	-0.01	0	1.14
9.61	rest	0.9666	-0.008	0	1.13

8.3.5 A0 take the knife from the block (scripted robot, dual FR3)

The right FR3 takes the fillet knife out of the block's slot through legal commands; the script plans from the knife's true pose.

Assistance: none on the world (legal robot commands); the script plans from privileged state

Observable	Recorded value
Simulated time [s]	27.58
Wall time [s]	1.216×10^4
Protective stops	0
Events	0
Assist-log entries (phantom rig or holds; a robot's own scripted hold, legal joint targets, is one)	0

Script log as recorded (at most its first 40 rows). Its times are the world clock, which counts from the world's creation and includes the settle before the episode; the run's simulated time counts the episode's own steps, so the log runs ahead of it by the settle.

t [s]	Step	Knife solver ('held' = MuJoCo)	TCP error [mm]	Stops
4.55	above pre-grasp (joint move)	held	*absent*	0
5.78	pre-grasp	held	0.0693	0
5.78	handle seen from the hand at pre-grasp	held	*absent*	0
13.58	grasp pose	held	0.299	0
14.38	closed at 35 N a finger	held	*absent*	0
16.54	draw 60 mm	held	0.833	0
18.88	draw 190 mm	held	0.625	0
20.04	lift clear 60 mm (orientation within 2.9 deg)	held	0.592	0
20.04	knife out of the block	held	*absent*	0
25.88	over the board, edge down	held	0.736	0
29.08	held 3.2 s	held	*absent*	0

Evaluator output (unmodified task evaluator, schema gar.evaluation.v1): success true, score 1.000.

Check	Value	Threshold	Unit	Passed	Note
held_by_right_hand_s	15.28	3	s of continuous right-hand pinch	**yes**	
not_touched_by_left_hand	0	0	bool	**yes**	
axis_in_channel_deg	0.06294	10	deg	**yes**	
seated_lateral_offset_mm	0.7051	4	mm	**yes**	
slip_mm_over_3s	0.01875	2	mm	**yes**	
edge_down_deg	0.1017	30	deg from vertical	**yes**	
blade_over_board	1	1	bool	**yes**	
blade_never_touched_robot	0	0	ticks with contact	**yes**	
time	27.58	30	s	**yes**	

8.3.6 A0 take the knife (scripted robot, single FR3)

The same pickup with the single-FR3 robot setup.

Assistance: none on the world (legal robot commands); the script plans from privileged state

Observable	Recorded value
Simulated time [s]	27.18
Wall time [s]	1.176×10^4
Protective stops	0
Events	0
Assist-log entries (phantom rig or holds; a robot's own scripted hold, legal joint targets, is one)	0

Script log as recorded (at most its first 40 rows). Its times are the world clock, which counts from the world's creation and includes the settle before the episode; the run's simulated time counts the episode's own steps, so the log runs ahead of it by the settle.

t [s]	Step	Knife solver ('held' = MuJoCo)	TCP error [mm]	Stops
4.55	above pre-grasp (joint move)	held	*absent*	0
5.78	pre-grasp	held	0.0693	0
5.78	handle seen from the hand at pre-grasp	held	*absent*	0
13.58	grasp pose	held	0.299	0
14.38	closed at 35 N a finger	held	*absent*	0
16.54	draw 60 mm	held	0.834	0
18.88	draw 190 mm	held	0.625	0
20.04	lift clear 60 mm (orientation within 2.9 deg)	held	0.593	0
20.04	knife out of the block	held	*absent*	0
25.48	over the board, edge down	held	0.59	0
28.68	held 3.2 s	held	*absent*	0

Evaluator output (unmodified task evaluator, schema gar.evaluation.v1): success true, score 1.000.

Check	Value	Threshold	Unit	Passed	Note
held_by_right_hand_s	14.88	3	s of continuous right-hand pinch	**yes**	
not_touched_by_left_hand	0	0	bool	**yes**	
axis_in_channel_deg	0.06686	10	deg	**yes**	
seated_lateral_offset_mm	0.83	4	mm	**yes**	
slip_mm_over_3s	0.2114	2	mm	**yes**	
edge_down_deg	0.08772	30	deg from vertical	**yes**	
blade_over_board	1	1	bool	**yes**	
blade_never_touched_robot	0	0	ticks with contact	**yes**	
time	27.18	30	s	**yes**	

8.3.7 A0 take the knife (scripted robot, dual Panda)

The same pickup with the dual-Panda robot setup.

Assistance: none on the world (legal robot commands); the script plans from privileged state

Observable	Recorded value
Simulated time [s]	18.66
Wall time [s]	8200
Protective stops	0
Events	0
Assist-log entries (phantom rig or holds; a robot's own scripted hold, legal joint targets, is one)	0

Script log as recorded (at most its first 40 rows). Its times are the world clock, which counts from the world's creation and includes the settle before the episode; the run's simulated time counts the episode's own steps, so the log runs ahead of it by the settle.

t [s]	Step	Knife solver ('held' = MuJoCo)	TCP error [mm]	Stops
4.55	above pre-grasp (joint move)	held	*absent*	0
5.79	pre-grasp	held	0.00308	0
5.79	handle seen from the hand at pre-grasp	held	*absent*	0
13.6	grasp pose	held	0.00299	0
14.4	closed at 35 N a finger	held	*absent*	0
16.55	draw 60 mm	held	0.429	0
18.88	draw 190 mm	held	1.24	0
19.33	lift clear 60 mm (orientation within 2.9 deg)	*absent*	*absent*	*absent*
19.83	lift clear 60 mm (orientation within 5.2 deg)	*absent*	*absent*	*absent*
20.16	lift clear 30 mm (orientation within 5.2 deg)	held	2.46	0
20.16	knife out of the block	held	*absent*	0
20.16	carry	*absent*	*absent*	*absent*

Evaluator output (unmodified task evaluator, schema gar.evaluation.v1): success false, score 0.778.

Check	Value	Threshold	Unit	Passed	Note
held_by_right_hand_s	6.36	3	s of continuous right-hand pinch	**yes**	
not_touched_by_left_hand	0	0	bool	**yes**	
axis_in_channel_deg	0.07733	10	deg	**yes**	
seated_lateral_offset_mm	0.09123	4	mm	**yes**	
slip_mm_over_3s	0.7149	2	mm	**yes**	
edge_down_deg	65.57	30	deg from vertical	no	
blade_over_board	0	1	bool	no	
blade_never_touched_robot	0	0	ticks with contact	**yes**	
time	18.66	30	s	**yes**	

8.3.8 C3 medallions cut by the robot

The right FR3 slices tenderloin medallions with the knife it holds while the left FR3 holds the loin; legal commands, privileged plan.

Assistance: none on the world (legal robot commands); the script plans from privileged state

Observable	Recorded value
Simulated time [s]	59.98
Wall time [s]	1.502×10^4
Protective stops	0
Events	33
Assist-log entries (phantom rig or holds; a robot's own scripted hold, legal joint targets, is one)	1

Script log as recorded (at most its first 40 rows). Its times are the world clock, which counts from the world's creation and includes the settle before the episode; the run's simulated time counts the episode's own steps, so the log runs ahead of it by the settle.

t [s]	Step	Knife solver ('held' = MuJoCo)	TCP error [mm]	Stops
1.8	hold: reach: planned	*absent*	*absent*	*absent*
4.59	hold: descended	held	0.54	0
6.44	hold: pressing	held	*absent*	0
6.44	cut 0: planned	*absent*	*absent*	*absent*
24.82	cut 0: bottom of the stroke	held	*absent*	0
26.6	cut 0: lifted	held	*absent*	0
26.6	cut 1: planned	*absent*	*absent*	*absent*
42.05	cut 1: bottom of the stroke	held	*absent*	0
43.83	cut 1: lifted	held	*absent*	0
43.83	cut 2: planned	*absent*	*absent*	*absent*
59.21	cut 2: bottom of the stroke	held	*absent*	0
60.98	cut 2: lifted	held	*absent*	0

Evaluator output (unmodified task evaluator, schema gar.evaluation.v1): success false, score 0.556.

Check	Value	Threshold	Unit	Passed	Note
medallions	0	1	count	no	
thickness_25_5mm	*none*	5	mm worst	no	
thickness_spread_mm	*none*	5	mm	no	
cut_tilt_deg	28.48	10	deg from perpendicular	no	nothing made yet: a quality check of a product passes only when the product exists
no_tag_ends	0	0	pieces of 1-5 g	**yes**	
time	59.98	180	s	**yes**	
debris	0	2.671	g in pieces under 1 g (at most 10 pieces)	**yes**	0 pieces
no_tissue_thrown	0.212	1	m/s fastest piece (31493 nodes, at 24.12 s; the fastest single node 9.68 m/s, a flesh node at 24.08 s); 0 nodes below the bench top	**yes**	
products_resting	1	1	fraction of the removed pieces' mass on a support, off the knife	**yes**	

8.3.9 A1 head-off by the robot

The robot removes the trout's head through legal commands.

Assistance: none on the world (legal robot commands); the script plans from privileged state

Observable	Recorded value
Simulated time [s]	4.84
Wall time [s]	3249
Protective stops	0
Events	0
Assist-log entries (phantom rig or holds; a robot's own scripted hold, legal joint targets, is one)	1

Script log as recorded (at most its first 40 rows). Its times are the world clock, which counts from the world's creation and includes the settle before the episode; the run's simulated time counts the episode's own steps, so the log runs ahead of it by the settle.

t [s]	Step	Knife solver ('held' = MuJoCo)	TCP error [mm]	Stops
1.8	tail hold: reach: planned	*absent*	*absent*	*absent*
3.99	tail hold: descended	held	0.678	0
6.04	tail hold: pressing	held	*absent*	0
6.04	head-off: no plan	*absent*	*absent*	*absent*

Evaluator output (unmodified task evaluator, schema gar.evaluation.v1): success false, score 0.750.

Check	Value	Threshold	Unit	Passed	Note
head_separated	0	1	bool	no	the head's flesh is joined to the body
meat_with_head	0	8	g of fillet muscle behind the collar line	no	nothing made yet: a quality check of a product passes only when the product exists
fish_on_board	1	0.95	fraction of body nodes over the board	**yes**	
knife_not_dropped	0.001044	15	mm (handle offset in grasp)	**yes**	
no_blade_contact_with_holding_hand	0	0	ticks with contact	**yes**	
debris	0	1.206	g in pieces under 1 g (at most 10 pieces)	**yes**	0 pieces
no_tissue_thrown	0.004193	1	m/s fastest piece (19761 nodes, at 4.52 s; the fastest single node 0.0224 m/s, a flesh node at 3.6 s); 0 nodes below the bench top	**yes**	
time	4.84	90	s	**yes**	

8.3.10 B5 portions weighed by the robot

The robot weighs the portions the assisted B5 run cut, continuing from that run's final state.

Assistance: The portions were cut by the phantom knife in the assisted B5 run; the robot only weighs them

Observable	Recorded value
Simulated time [s]	137
Wall time [s]	3955
Protective stops	1
Events	882
Assist-log entries (phantom rig or holds; a robot's own scripted hold, legal joint targets, is one)	1

Script log as recorded (at most its first 40 rows). Its times are the world clock, which counts from the world's creation and includes the settle before the episode; the run's simulated time counts the episode's own steps, so the log runs ahead of it by the settle.

t [s]	Step	Knife solver ('held' = MuJoCo)	TCP error [mm]	Stops
136.2	portion 1 (147.8 g) to the scale: above the piece	phantom	0.368	0
138.5	portion 1 (147.8 g) to the scale: descend	*absent*	*absent*	*absent*

Evaluator output (unmodified task evaluator, schema gar.evaluation.v1): success false, score 0.889.

Check	Value	Threshold	Unit	Passed	Note
full_portions	1	1	count	**yes**	
portions_150_7g	2.195	7	g worst deviation	**yes**	
cut_tilt_deg	3.286	10	deg from vertical (area-weighted)	**yes**	
no_tag_ends	0	0	pieces of 1-5 g	**yes**	
weighed	0	1	fraction of full portions	no	
time	137	180	s	**yes**	
debris	0.08062	4.763	g in pieces under 1 g (at most 10 pieces)	**yes**	2 pieces
no_tissue_thrown	0.8011	1	m/s fastest piece (6 nodes, at 136.94 s; the fastest single node 4.75 m/s, a flesh node at 61.35 s); 0 nodes below the bench top	**yes**	
products_resting	1	1	fraction of the removed pieces' mass on a support, off the knife	**yes**	

8.4 Assisted solvability and negative controls (quality gate 3)

The knife is driven by a labelled phantom rig (and, where stated, a phantom hold). These runs test whether the world's own cutting law can produce each product, not whether a policy can: only B5 makes one (a 147.8 g portion); every other task's product is not made, and its evaluation below says which checks fail. Negative controls remove one property and must fail on it; where a control fails no less than its positive run, the entry says the pair does not discriminate. An assisted run's simulated time counts the world clock from its creation, including the 1.5 s settle; the evaluator's time counts the episode from its start, so it is 1.5 s less.

8.4.1 A1 head-off (assisted)

A sawing cut by the phantom rig, square to the backbone through the joint between the fourth and fifth vertebrae, 14.8 mm behind the occiput, through bulk flesh; the evaluator is unmodified.

Assistance: phantom knife rig and a phantom hold of the head

Observable	Recorded value
Simulated time [s]	13.55
Wall time [s]	8525
Saw frequency [Hz]	1.5
Saw amplitude [mm]	40
Descent [m/s]	0.01
Path offset into the head [mm]	0
Blade entry into the flesh [mm]	0
Uncut faces bridging head and body	495
Intact interface bonds across the cut	19
Intact flesh-bone ties across the cut	0
Pull test: head moved [mm]	40
Pull test: body nodes	20 758
Pull test: body motion, median [mm]	30.2
Pull test: body motion, 95th percentile [mm]	37.9
Pull test: body motion, max [mm]	42.1
Ledger: work put in [mJ]	2792
Ledger: cutting dissipation [mJ]	381.5
Ledger: interface fracture paid [mJ]	49.66
Ledger: friction [mJ]	1574
Smallest volume ratio at the end	0.610
Placement: head carried to the bowl (phantom hand), in the bowl	*absent*
Placement: head fragment's tissue over the bowl (fraction)	*absent*
Placement: head speed 1.5 s after release [m/s]	*absent*

Evaluator output (unmodified task evaluator, schema gar.evaluation.v1): success false, score 0.625.

Check	Value	Threshold	Unit	Passed	Note
head_separated	0	1	bool	no	the head's flesh is joined to the body
meat_with_head	0	8	g of fillet muscle behind the collar line	no	nothing made yet: a quality check of a product passes only when the product exists
fish_on_board	1	0.95	fraction of body nodes over the board	**yes**	
knife_not_dropped	534.3	15	mm (handle offset in grasp)	no	ASSISTED run: the knife is on the phantom rig, not in the robot's hand (not a policy result)
no_blade_contact_with_holding_hand	0	0	ticks with contact	**yes**	
debris	0.04502	1.206	g in pieces under 1 g (at most 10 pieces)	**yes**	1 pieces
no_tissue_thrown	0.9155	1	m/s fastest piece (13 nodes, at 10.82 s; the fastest single node 10.5 m/s, a flesh node at 7.81 s); 0 nodes below the bench top	**yes**	
time	12.05	90	s	**yes**	

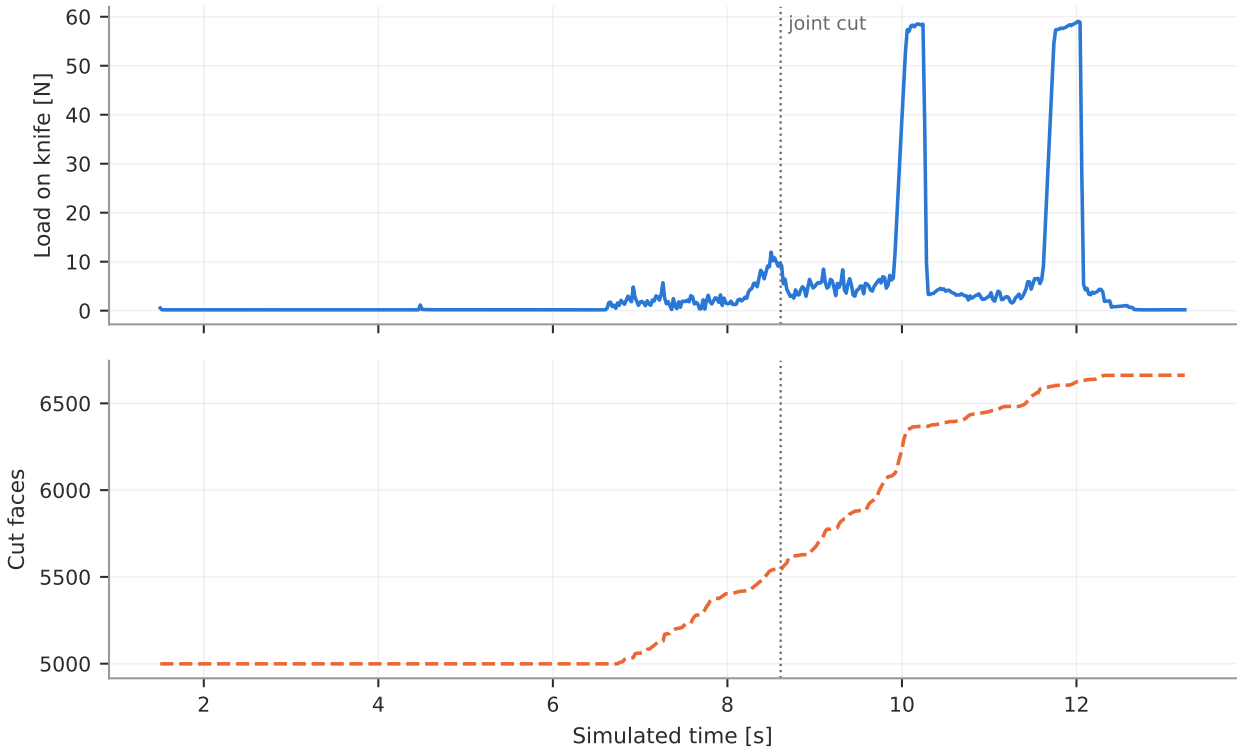


Figure 3: **A1 head-off (assisted).** Head-off: magnitude of the tissue's load on the knife and the number of cut mesh faces, from the recorded log; a dotted line marks the recorded event of the planned backbone joint being cut, drawn only when one is recorded.

8.4.2 A1 negative control: blade path 6 mm toward the head, onto a vertebra

NEGATIVE CONTROL: the same sawing cut shifted 6 mm toward the head, off the V4/V5 joint and onto a vertebra, so the backbone cannot part there; the evaluator is unmodified. It must fail on head_separated; A1 itself fails on it too, so the pair does not discriminate.

Assistance: as A1, the path offset 6 mm toward the head (off the joint, onto a rigid vertebra)

Observable	Recorded value
Simulated time [s]	13.55
Wall time [s]	6173
Saw frequency [Hz]	1.5
Saw amplitude [mm]	40
Descent [m/s]	0.01
Path offset into the head [mm]	6
Blade entry into the flesh [mm]	0
Uncut faces bridging head and body	501
Intact interface bonds across the cut	16
Intact flesh-bone ties across the cut	1
Pull test: head moved [mm]	40
Pull test: body nodes	20 920
Pull test: body motion, median [mm]	39.7
Pull test: body motion, 95th percentile [mm]	40.4
Pull test: body motion, max [mm]	40.6
Ledger: work put in [mJ]	3654
Ledger: cutting dissipation [mJ]	426.2
Ledger: interface fracture paid [mJ]	55.46
Ledger: friction [mJ]	2352
Smallest volume ratio at the end	0.610

Evaluator output (unmodified task evaluator, schema gar.evaluation.v1): success false, score 0.625.

Check	Value	Threshold	Unit	Passed	Note
head_separated	0	1	bool	no	the head's flesh is joined to the body
meat_with_head	0	8	g of fillet muscle behind the collar line	no	nothing made yet: a quality check of a product passes only when the product exists
fish_on_board	1	0.95	fraction of body nodes over the board	**yes**	
knife_not_dropped	529.8	15	mm (handle offset in grasp)	no	ASSISTED run: the knife is on the phantom rig, not in the robot's hand (not a policy result)
no_blade_contact_with_holding_hand	0	0	ticks with contact	**yes**	
debris	0.2566	1.206	g in pieces under 1 g (at most 10 pieces)	**yes**	3 pieces
no_tissue_thrown	0.8602	1	m/s fastest piece (21 nodes, at 11.18 s; the fastest single node 14.4 m/s, a flesh node at 9.56 s); 0 nodes below the bench top	**yes**	
time	12.05	90	s	**yes**	

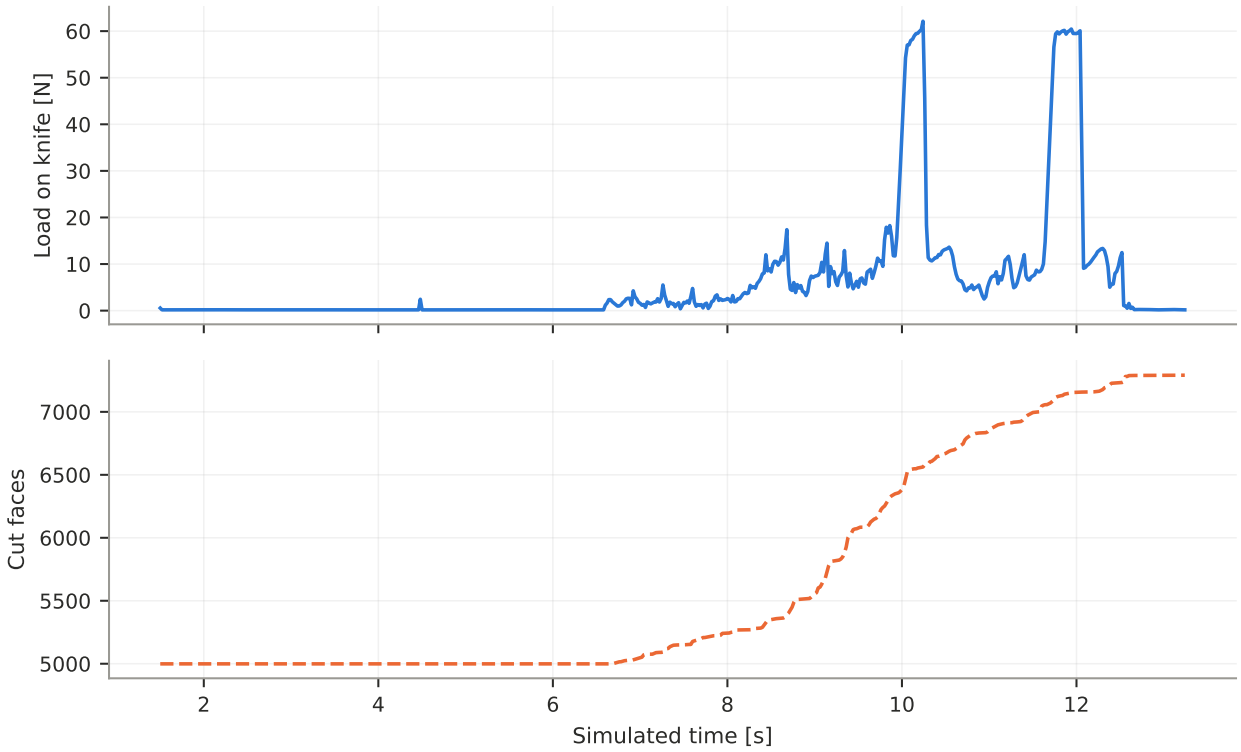


Figure 4: **A1 negative control: blade path 6 mm toward the head, onto a vertebra.** Head-off: magnitude of the tissue's load on the knife and the number of cut mesh faces, from the recorded log; a dotted line marks the recorded event of the planned backbone joint being cut, drawn only when one is recorded. **Not drawn:** event marker 'joint cut': no matching event recorded.

8.4.3 A2 fillet one side (assisted)

The phantom rig fillets one side, following the planned path; the evaluator is unmodified.

Assistance: Phantom knife rig; phantom hands hold the head (carried away after the head-off) and two vertebral segments (those of V16 and V46: segment_07 and segment_13) to the end, released after the evaluation

Observable	Recorded value
Simulated time [s]	72.61
Wall time [s]	3.286×10^{4}
Events	775
Assist-log entries (phantom rig or holds; a robot's own scripted hold, legal joint targets, is one)	14
Ledger: work put in [mJ]	1.099×10^{4}
Ledger: cutting dissipation [mJ]	2538
Smallest volume ratio at the end	0.711
After the evaluation: holds released, held segments moved [mm]	segment_07: 0.05; segment_13: 0.65
After the release: largest kinetic energy [J]	5.891×10^{-5}
Elastic energy before the release [J]	0.0033
Crack faces by origin	total: 19 138; interface_at_build: 4 999; torn: 8 249; knife_upper_bound: 5 890
Debris (evaluator)	2.887

Evaluator output (unmodified task evaluator, schema gar.evaluation.v1): success false, score 0.154.

Check	Value	Threshold	Unit	Passed	Note
one_piece	0	0.97	fraction	no	nothing has been removed from the frame yet
yield	0	0.09553	kg	no	nothing has been removed from the frame yet
no_rib_pieces_over_2mm	*none*	0	pieces	no	nothing has been removed from the frame yet
no_backbone_or_fin_support_fragments	*none*	0	pieces	no	nothing has been removed from the frame yet
workmanship_score	*none*	85	points	no	nothing has been removed from the frame yet
placed_in_tray	0	1	fraction	no	nothing has been removed from the frame yet
skin_down	*none*	45	deg	no	nothing has been removed from the frame yet
settled	*none*	0.001	m	no	nothing has been removed from the frame yet
knife_not_dropped	565.3	15	mm (handle offset in grasp)	no	ASSISTED run: the knife is on the phantom rig, not in the robot's hand (not a policy result)
no_blade_contact_with_holding_hand	0	0	ticks with contact	**yes**	
debris	2.887	1.206	g in pieces under 1 g (at most 10 pieces)	no	46 pieces
no_tissue_thrown	3.987	1	m/s fastest piece (14 nodes, at 63.54 s; the fastest single node 18.4 m/s, a bone rod node at 62.93 s); 50 nodes below the bench top	no	
time	70.11	180	s	**yes**	

8.4.4 A4 remove the belly bones (assisted)

The phantom rig removes the belly bones; the evaluator is unmodified.

Assistance: phantom knife rig; the left FR3 presses the fillet's tail through normal joint targets

Observable	Recorded value
Simulated time [s]	27.29
Wall time [s]	7651
Events	427
Assist-log entries (phantom rig or holds; a robot's own scripted hold, legal joint targets, is one)	3
Ledger: work put in [mJ]	1.021×10^4
Ledger: cutting dissipation [mJ]	224.1
Smallest volume ratio at the end	0.344

Evaluator output (unmodified task evaluator, schema gar.evaluation.v1): success false, score 0.333.

Check	Value	Threshold	Unit	Passed	Note
no_rib_pieces_over_2mm	36	0	pieces left in the fillet	no	
skin_hole_mm2	0	25	mm2 of skin gone from the fillet	no	nothing made yet: a quality check of a product passes only when the product exists
flesh_removed	0.001182	0.06	fraction of the fillet's mass	no	nothing made yet: a quality check of a product passes only when the product exists
time	25.79	120	s	**yes**	
debris	0.123	0.5203	g in pieces under 1 g (at most 10 pieces)	**yes**	3 pieces
no_tissue_thrown	1.105	1	m/s fastest piece (16 nodes, at 13.82 s; the fastest single node 7.87 m/s, a flesh node at 15.59 s); 0 nodes below the bench top	no	

8.4.5 A4 negative control: fillet not held

NEGATIVE CONTROL: the same run with nothing holding the fillet; the evaluator is unmodified. The control fails the same four checks as A4 itself (both score 0.333): the pair does not discriminate.

Assistance: phantom knife rig; nothing holds the fillet

Observable	Recorded value
Simulated time [s]	22.14
Wall time [s]	6063
Events	346
Assist-log entries (phantom rig or holds; a robot's own scripted hold, legal joint targets, is one)	2
Ledger: work put in [mJ]	1.029×10^4
Ledger: cutting dissipation [mJ]	123.4
Smallest volume ratio at the end	0.970

Evaluator output (unmodified task evaluator, schema gar.evaluation.v1): success false, score 0.333.

Check	Value	Threshold	Unit	Passed	Note
no_rib_pieces_over_2mm	36	0	pieces left in the fillet	no	
skin_hole_mm2	0	25	mm2 of skin gone from the fillet	no	nothing made yet: a quality check of a product passes only when the product exists
flesh_removed	0.0007122	0.06	fraction of the fillet's mass	no	nothing made yet: a quality check of a product passes only when the product exists
time	20.64	120	s	**yes**	
debris	0.07411	0.5203	g in pieces under 1 g (at most 10 pieces)	**yes**	2 pieces
no_tissue_thrown	2.414	1	m/s fastest piece (19 nodes, at 17.07 s; the fastest single node 10.5 m/s, a flesh node at 17.03 s); 0 nodes below the bench top	no	

8.4.6 B5 portions (assisted)

The phantom rig cuts the portions; the evaluator is unmodified.

Assistance: phantom knife rig; the left FR3 presses the side beside each cut through normal joint targets (creator-scripted)

Observable	Recorded value
Simulated time [s]	131
Wall time [s]	7.87×10^4
Events	834
Assist-log entries (phantom rig or holds; a robot's own scripted hold, legal joint targets, is one)	14
Ledger: work put in [mJ]	5455
Ledger: cutting dissipation [mJ]	3915
Smallest volume ratio at the end	0.717

Evaluator output (unmodified task evaluator, schema gar.evaluation.v1): success false, score 0.889.

Check	Value	Threshold	Unit	Passed	Note
full_portions	1	1	count	**yes**	
portions_150_7g	2.195	7	g worst deviation	**yes**	
cut_tilt_deg	3.286	10	deg from vertical (area-weighted)	**yes**	
no_tag_ends	0	0	pieces of 1-5 g	**yes**	
weighed	0	1	fraction of full portions	no	
time	129.5	180	s	**yes**	
debris	0.08062	4.763	g in pieces under 1 g (at most 10 pieces)	**yes**	2 pieces
no_tissue_thrown	0.4753	1	m/s fastest piece (7 nodes, at 61.46 s; the fastest single node 4.75 m/s, a flesh node at 61.35 s); 0 nodes below the bench top	**yes**	
products_resting	1	1	fraction of the removed pieces' mass on a support, off the knife	**yes**	

8.4.7 B5 negative control: cuts tilted 20 degrees

NEGATIVE CONTROL: the same cuts tilted 20 degrees; the evaluator is unmodified. The control fails full_ portions, portions_150_7g, cut_tilt_deg and no_tissue_thrown, which B5 passes (0.444 against 0.889; both fail weighed): the pair discriminates.

Assistance: phantom knife rig; the left FR3 presses the side beside each cut through normal joint targets (creator-scripted)

Observable	Recorded value
Simulated time [s]	131
Wall time [s]	8.009×10^4
Events	1 471
Assist-log entries (phantom rig or holds; a robot's own scripted hold, legal joint targets, is one)	14
Ledger: work put in [mJ]	6080
Ledger: cutting dissipation [mJ]	4225
Smallest volume ratio at the end	0.717

Evaluator output (unmodified task evaluator, schema gar.evaluation.v1): success false, score 0.444.

Check	Value	Threshold	Unit	Passed	Note
full_portions	0	1	count	no	
portions_150_7g	*none*	7	g worst deviation	no	
cut_tilt_deg	17.02	10	deg from vertical (area-weighted)	no	nothing made yet: a quality check of a product passes only when the product exists
no_tag_ends	0	0	pieces of 1-5 g	**yes**	
weighed	0	1	fraction of full portions	no	
time	129.5	180	s	**yes**	
debris	0.3135	4.763	g in pieces under 1 g (at most 10 pieces)	**yes**	5 pieces
no_tissue_thrown	3.487	1	m/s fastest piece (9 nodes, at 62.37 s; the fastest single node 5.53 m/s, a flesh node at 62.16 s); 18 nodes below the bench top	no	
products_resting	1	1	fraction of the removed pieces' mass on a support, off the knife	**yes**	

8.4.8 B2 trim the belly fat strip (assisted)

The phantom rig cuts the B2 trim line; the evaluator is unmodified.

Assistance: phantom knife rig; the left FR3 holds the side through normal joint targets

Observable	Recorded value
Simulated time [s]	38.72
Wall time [s]	2.268×10^4
Events	435
Assist-log entries (phantom rig or holds; a robot's own scripted hold, legal joint targets, is one)	6
Ledger: work put in [mJ]	7150
Ledger: cutting dissipation [mJ]	756.1
Smallest volume ratio at the end	0.997

Evaluator output (unmodified task evaluator, schema gar.evaluation.v1): success false, score 0.429.

Check	Value	Threshold	Unit	Passed	Note
trim_removed	0.07218	0.9	fraction of the reference trim	no	
loin_loss	0.0525	28.58	g (3 % of the side)	**yes**	
edge_raggedness_mm	2.405	2	mm RMS	no	
time	37.22	180	s	**yes**	
debris	0.7451	4.763	g in pieces under 1 g (at most 10 pieces)	no	15 pieces
no_tissue_thrown	2.472	1	m/s fastest piece (26 nodes, at 15.53 s; the fastest single node 7.54 m/s, a flesh node at 15.53 s); 0 nodes below the bench top	no	
products_resting	1	1	fraction of the removed pieces' mass on a support, off the knife	**yes**	

8.4.9 B2 negative control: the line 15 mm into the kept piece

NEGATIVE CONTROL: the same cut 15 mm into the loin; the evaluator is unmodified. Both fail trim_removed and the control scores lower (0.167 against 0.429), but its extra failure, loin_loss (1.8 g lost), fails only because nothing was trimmed (that check passes only when a product exists): the pair does not discriminate.

Assistance: phantom knife rig; left FR3 hold

Observable	Recorded value
Simulated time [s]	42.38
Wall time [s]	2.608×10^4
Events	395
Assist-log entries (phantom rig or holds; a robot's own scripted hold, legal joint targets, is one)	6
Ledger: work put in [mJ]	8431
Ledger: cutting dissipation [mJ]	804.9
Smallest volume ratio at the end	0.700

Evaluator output (unmodified task evaluator, schema gar.evaluation.v1): success false, score 0.167.

Check	Value	Threshold	Unit	Passed	Note
trim_removed	0	0.9	fraction of the reference trim	no	
loin_loss	1.787	28.58	g (3 % of the side)	no	nothing made yet: a quality check of a product passes only when the product exists
edge_raggedness_mm	2.394	2	mm RMS	no	
time	40.88	180	s	**yes**	
debris	1.787	4.763	g in pieces under 1 g (at most 10 pieces)	no	34 pieces
no_tissue_thrown	2.671	1	m/s fastest piece (51 nodes, at 37.81 s; the fastest single node 7.15 m/s, a flesh node at 37.77 s); 0 nodes below the bench top	no	

8.4.10 C1 remove the chain (assisted)

The phantom rig cuts the chain seam; the evaluator is unmodified.

Assistance: phantom knife rig; left FR3 hold

Observable	Recorded value
Simulated time [s]	48.74
Wall time [s]	2.358×10^4
Events	147
Assist-log entries (phantom rig or holds; a robot's own scripted hold, legal joint targets, is one)	5
Ledger: work put in [mJ]	1.436×10^4
Ledger: cutting dissipation [mJ]	885.4
Smallest volume ratio at the end	0.287

Evaluator output (unmodified task evaluator, schema gar.evaluation.v1): success false, score 0.333.

Check	Value	Threshold	Unit	Passed	Note
chain_separated	0	0.95	fraction of the chain's mass	no	
main_loss	0.0003045	0.03	fraction of the main muscle	no	nothing made yet: a quality check of a product passes only when the product exists
tail_split_mm	251.3	25	mm	no	
time	47.24	180	s	**yes**	
debris	0.143	2.671	g in pieces under 1 g (at most 10 pieces)	**yes**	3 pieces
no_tissue_thrown	1.922	1	m/s fastest piece (20 nodes, at 35.88 s; the fastest single node 11.3 m/s, a flesh node at 35.69 s); 0 nodes below the bench top	no	

8.4.11 C1 negative control: the line 12 mm into the kept piece

NEGATIVE CONTROL: the same cut 12 mm into the main muscle; the evaluator is unmodified. Both fail chain_separated, main_loss and tail_split_mm, and the control scores higher (0.500 against 0.333; C1 also throws tissue): the pair does not discriminate.

Assistance: phantom knife rig; left FR3 hold

Observable	Recorded value
Simulated time [s]	49.11
Wall time [s]	2.374×10^4
Events	231
Assist-log entries (phantom rig or holds; a robot's own scripted hold, legal joint targets, is one)	5
Ledger: work put in [mJ]	2.173×10^4
Ledger: cutting dissipation [mJ]	3231
Smallest volume ratio at the end	0.951

Evaluator output (unmodified task evaluator, schema gar.evaluation.v1): success false, score 0.500.

Check	Value	Threshold	Unit	Passed	Note
chain_separated	0	0.95	fraction of the chain's mass	no	
main_loss	0.001007	0.03	fraction of the main muscle	no	nothing made yet: a quality check of a product passes only when the product exists
tail_split_mm	268.2	25	mm	no	
time	47.61	180	s	**yes**	
debris	0.7113	2.671	g in pieces under 1 g (at most 10 pieces)	**yes**	10 pieces
no_tissue_thrown	0.738	1	m/s fastest piece (16 nodes, at 37.55 s; the fastest single node 9.66 m/s, a flesh node at 37.55 s); 0 nodes below the bench top	**yes**	

8.4.12 C3 medallions (assisted)

The phantom rig cuts the medallions; the evaluator is unmodified.

Assistance: phantom knife rig

Observable	Recorded value
Simulated time [s]	83.1
Wall time [s]	2.727×10^4
Events	193
Assist-log entries (phantom rig or holds; a robot's own scripted hold, legal joint targets, is one)	4
Ledger: work put in [mJ]	3.822×10^4
Ledger: cutting dissipation [mJ]	6434
Smallest volume ratio at the end	0.986

Evaluator output (unmodified task evaluator, schema gar.evaluation.v1): success false, score 0.333.

Check	Value	Threshold	Unit	Passed	Note
medallions	0	1	count	no	
thickness_25_5mm	*none*	5	mm worst	no	
thickness_spread_mm	*none*	5	mm	no	
cut_tilt_deg	51.06	10	deg from perpendicular	no	nothing made yet: a quality check of a product passes only when the product exists
no_tag_ends	1	0	pieces of 1-5 g	no	
time	81.6	180	s	**yes**	
debris	0.04166	2.671	g in pieces under 1 g (at most 10 pieces)	**yes**	1 pieces
no_tissue_thrown	1.843	1	m/s fastest piece (12 nodes, at 60.69 s; the fastest single node 9.17 m/s, a flesh node at 20.94 s); 0 nodes below the bench top	no	
products_resting	1	1	fraction of the removed pieces' mass on a support, off the knife	**yes**	

8.4.13 C3 negative control: cuts 10 mm off plan

NEGATIVE CONTROL: the same cuts 10 mm off plan; the evaluator is unmodified. Both fail medallions and the thickness and tilt checks, and the control scores higher (0.556 against 0.333; C3 also leaves tag ends and throws tissue): the pair does not discriminate.

Assistance: phantom knife rig

Observable	Recorded value
Simulated time [s]	83.05
Wall time [s]	2.42×10^4
Events	134
Assist-log entries (phantom rig or holds; a robot's own scripted hold, legal joint targets, is one)	4
Ledger: work put in [mJ]	3.391×10^4
Ledger: cutting dissipation [mJ]	5945
Smallest volume ratio at the end	0.944

Evaluator output (unmodified task evaluator, schema gar.evaluation.v1): success false, score 0.556.

Check	Value	Threshold	Unit	Passed	Note
medallions	0	1	count	no	
thickness_25_5mm	*none*	5	mm worst	no	
thickness_spread_mm	*none*	5	mm	no	
cut_tilt_deg	18.54	10	deg from perpendicular	no	nothing made yet: a quality check of a product passes only when the product exists
no_tag_ends	0	0	pieces of 1-5 g	**yes**	
time	81.55	180	s	**yes**	
debris	0.03409	2.671	g in pieces under 1 g (at most 10 pieces)	**yes**	1 pieces
no_tissue_thrown	0.5986	1	m/s fastest piece (32111 nodes, at 22.45 s; the fastest single node 9.74 m/s, a flesh node at 57.55 s); 0 nodes below the bench top	**yes**	
products_resting	1	1	fraction of the removed pieces' mass on a support, off the knife	**yes**	

8.4.14 B3 excise the dark spot (assisted)

Vertical side cuts around the melanin spot and an undercut above the skin; the evaluator is unmodified.

Assistance: phantom knife rig, and a creator-scripted left-FR3 hold through normal joint targets where the script uses one

Observable	Recorded value
Simulated time [s]	158.2
Wall time [s]	8.405×10^4
Events	300
Assist-log entries (phantom rig or holds; a robot's own scripted hold, legal joint targets, is one)	12
Ledger: work put in [mJ]	1219
Ledger: cutting dissipation [mJ]	468.7
Smallest volume ratio at the end	0.715
Undercut plane, world height [mm]	923.2
Undercut plane above the board (flesh and skin left under it) [mm]	9.49
Margin around the spot [mm]	1.5
Side cuts	8
Hold: press load target [N]	12

Evaluator output (unmodified task evaluator, schema gar.evaluation.v1): success false, score 0.333.

Check	Value	Threshold	Unit	Passed	Note
spot_removed	0	0.99	fraction of the spot's mass	no	
good_flesh_removed	0.03898	1.01	g (2 x the spot)	no	nothing made yet: a quality check of a product passes only when the product exists
skin_hole_mm2	0	0	mm2 of skin gone from the side	no	nothing made yet: a quality check of a product passes only when the product exists
time	156.7	120	s	no	
debris	0.03898	4.763	g in pieces under 1 g (at most 10 pieces)	**yes**	1 pieces
no_tissue_thrown	0.9095	1	m/s fastest piece (12 nodes, at 145.16 s; the fastest single node 7.33 m/s, a flesh node at 30.03 s); 0 nodes below the bench top	**yes**	

8.4.15 C2 denude the silver skin (assisted)

Shallow passes under the silver-skin band, each planned on the loin where it lies (a pass that would meet the holding hand, also when stopped up to 40 mm shorter, is skipped with the reason); the evaluator is unmodified.

Assistance: phantom knife rig, and a creator-scripted left-FR3 hold through normal joint targets where the script uses one

Observable	Recorded value
Simulated time [s]	106.1
Wall time [s]	4.201×10^4
Events	109
Assist-log entries (phantom rig or holds; a robot's own scripted hold, legal joint targets, is one)	5
Ledger: work put in [mJ]	2599
Ledger: cutting dissipation [mJ]	166.3
Smallest volume ratio at the end	0.944
Cut depth [mm]	2.2
Passes	4
Band stations (from, to)	0.1, 0.88
Hold: press load target [N]	10

Evaluator output (unmodified task evaluator, schema gar.evaluation.v1): success false, score 0.667.

Check	Value	Threshold	Unit	Passed	Note
silver_removed	0	0.9	fraction of silver area	no	
main_loss	0	0.05	fraction of the main muscle	no	nothing made yet: a quality check of a product passes only when the product exists
time	104.6	240	s	**yes**	
debris	0	2.671	g in pieces under 1 g (at most 10 pieces)	**yes**	0 pieces
no_tissue_thrown	0.5823	1	m/s fastest piece (31250 nodes, at 39.0 s; the fastest single node 24 m/s, a flesh node at 38.84 s); 0 nodes below the bench top	**yes**	
products_resting	1	1	fraction of the removed pieces' mass on a support, off the knife	**yes**	

8.5 Rest and weight (quality gate 4)

Worlds left alone after the settle, with no command: drift, micro-vibration, the load on the supports against the physical weight, and the energy ledger's residual.

8.5.1 30 s rest: trout, task A2, knife in hand

No command for 30 s, sampled every simulated second; UNASSISTED.

Observable	Recorded value
Seconds left alone	30
Tissue net displacement, max [μm]	5.47
Rod (thin bone) net displacement, max [μm]	1.41
Knife bodies net displacement, max [μm]	0.268
Rigid bones net displacement, max [μm]	2.43
Physical node mass [g]	242.8
Rigid bones [g]	45.7
Added stability inertia (no weight) [g]	31.2
Board load at the end [g]	288.5
Board load / (physical node mass + bones)	1.0000
Dissipation booked, viscous [mW]	0.0636
Dissipation booked, rod ties [mW]	0.000646
Dissipation booked, dampers [mW]	0.106
Dissipation booked, friction [mW]	0.0154
Energy change of the tissue [mW]	-4.04×10^{-5}
Work put in [mW]	0
Ledger residual [mW]	-0.186
Protective stops	0
Events	0
Wall seconds per simulated second	440

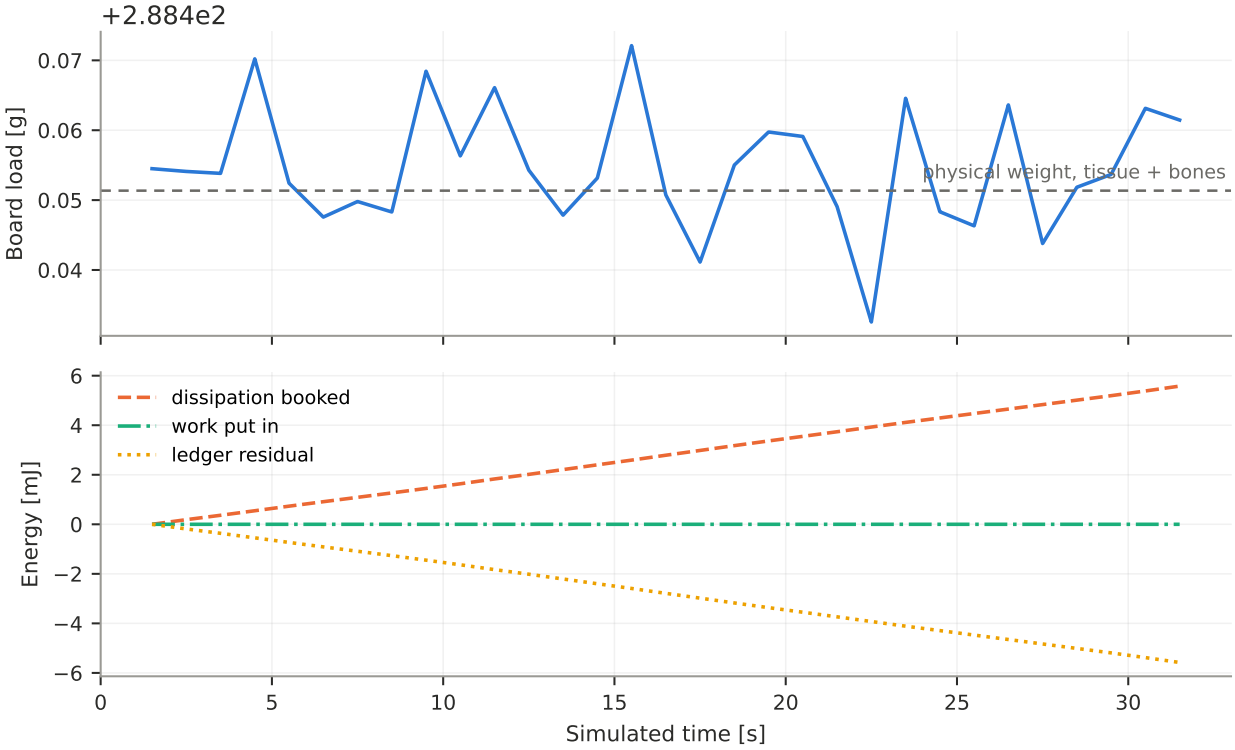


Figure 5: **30 s rest: trout, task A2, knife in hand.** Rest: the board's recorded load against the physical weight of tissue and bones (dashed), and the cumulative energy ledger (work put in, dissipation booked and residual) from the recorded samples.

8.5.2 30 s rest: trout, task A0, knife in the block

No command for 30 s, sampled every simulated second; UNASSISTED.

Observable	Recorded value
Seconds left alone	30
Tissue net displacement, max [μm]	5.18
Rod (thin bone) net displacement, max [μm]	1.29
Knife bodies net displacement, max [μm]	0
Rigid bones net displacement, max [μm]	2.18
Physical node mass [g]	242.8
Rigid bones [g]	45.7
Added stability inertia (no weight) [g]	31.2
Board load at the end [g]	288.4
Board load / (physical node mass + bones)	1.0000
Dissipation booked, viscous [mW]	0.0636
Dissipation booked, rod ties [mW]	0.000613
Dissipation booked, dampers [mW]	0.104
Dissipation booked, friction [mW]	0.0135
Energy change of the tissue [mW]	-3.91×10^{-5}
Work put in [mW]	0
Ledger residual [mW]	-0.182
Protective stops	0
Events	0
Wall seconds per simulated second	392

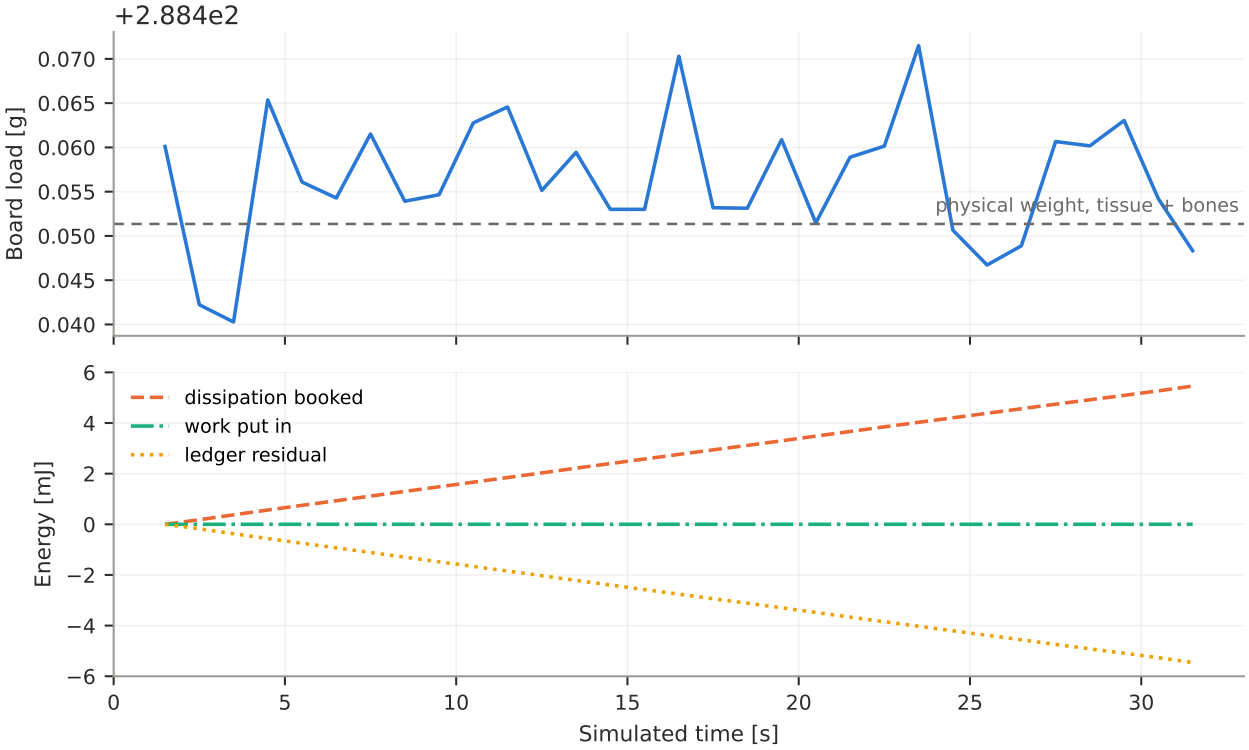


Figure 6: **30 s rest: trout, task A0, knife in the block.** Rest: the board's recorded load against the physical weight of tissue and bones (dashed), and the cumulative energy ledger (work put in, dissipation booked and residual) from the recorded samples.

8.5.3 30 s rest: fillet with rib strip, task A4

No command for 30 s, sampled every simulated second; UNASSISTED.

Observable	Recorded value
Seconds left alone	30
Tissue net displacement, max [μm]	0.253
Rod (thin bone) net displacement, max [μm]	0.135
Knife bodies net displacement, max [μm]	0.268
Rigid bones net displacement, max [μm]	*none*
Physical node mass [g]	104.6
Rigid bones [g]	0.0
Added stability inertia (no weight) [g]	13.1
Board load at the end [g]	104.6
Board load / (physical node mass + bones)	1.0000
Dissipation booked, viscous [mW]	0.0221
Dissipation booked, rod ties [mW]	1.6×10^{-5}
Dissipation booked, dampers [mW]	0.00977
Dissipation booked, friction [mW]	0.000334
Energy change of the tissue [mW]	-6.34×10^{-8}
Work put in [mW]	0
Ledger residual [mW]	-0.0322
Protective stops	0
Events	0
Wall seconds per simulated second	236

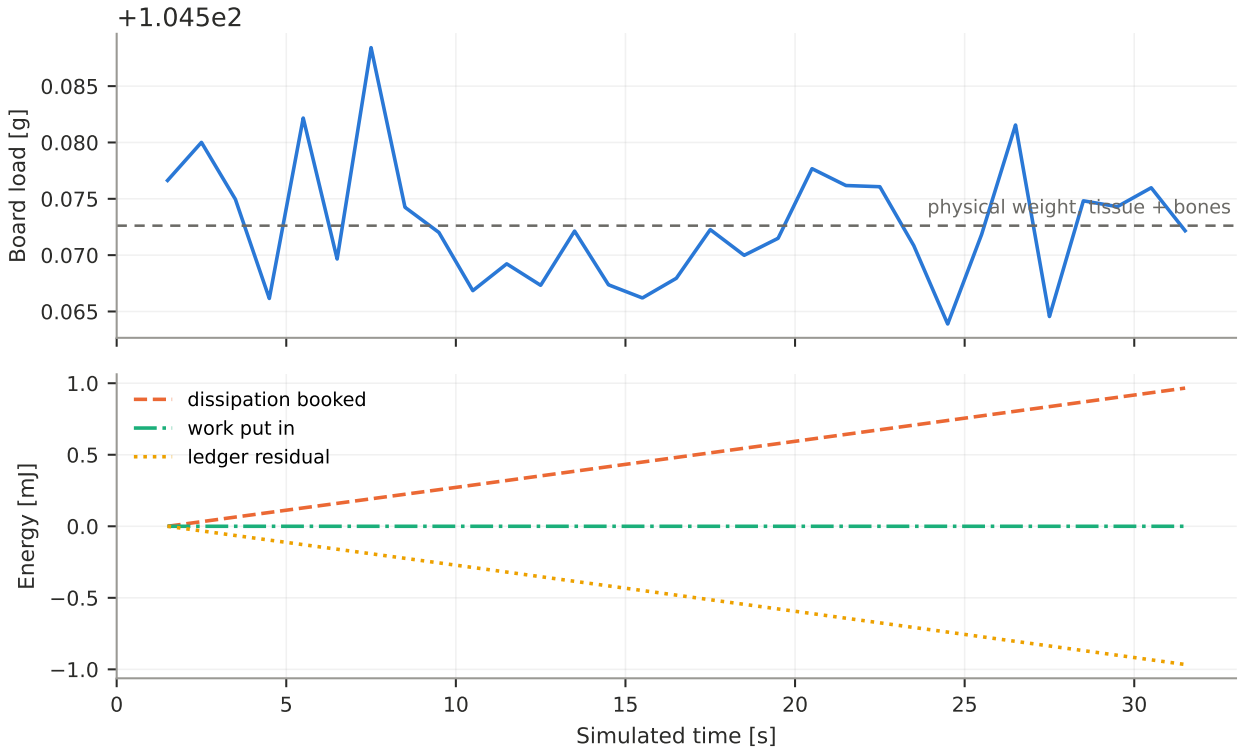


Figure 7: **30 s rest: fillet with rib strip, task A4.** Rest: the board's recorded load against the physical weight of tissue and bones (dashed), and the cumulative energy ledger (work put in, dissipation booked and residual) from the recorded samples.

8.5.4 30 s rest: salmon side, task B2

No command for 30 s, sampled every simulated second; UNASSISTED.

Observable	Recorded value
Seconds left alone	30
Tissue net displacement, max [μm]	8.99
Rod (thin bone) net displacement, max [μm]	2.17
Knife bodies net displacement, max [μm]	0.268
Rigid bones net displacement, max [μm]	0.149
Physical node mass [g]	954.2
Rigid bones [g]	4.5
Added stability inertia (no weight) [g]	34.2
Board load at the end [g]	958.6
Board load / (physical node mass + bones)	0.9999
Dissipation booked, viscous [mW]	0.00875
Dissipation booked, rod ties [mW]	9.17×10^{-5}
Dissipation booked, dampers [mW]	0.5
Dissipation booked, friction [mW]	0.0883
Energy change of the tissue [mW]	-2.17×10^{-5}
Work put in [mW]	0
Ledger residual [mW]	-0.597
Protective stops	0
Events	0
Wall seconds per simulated second	558

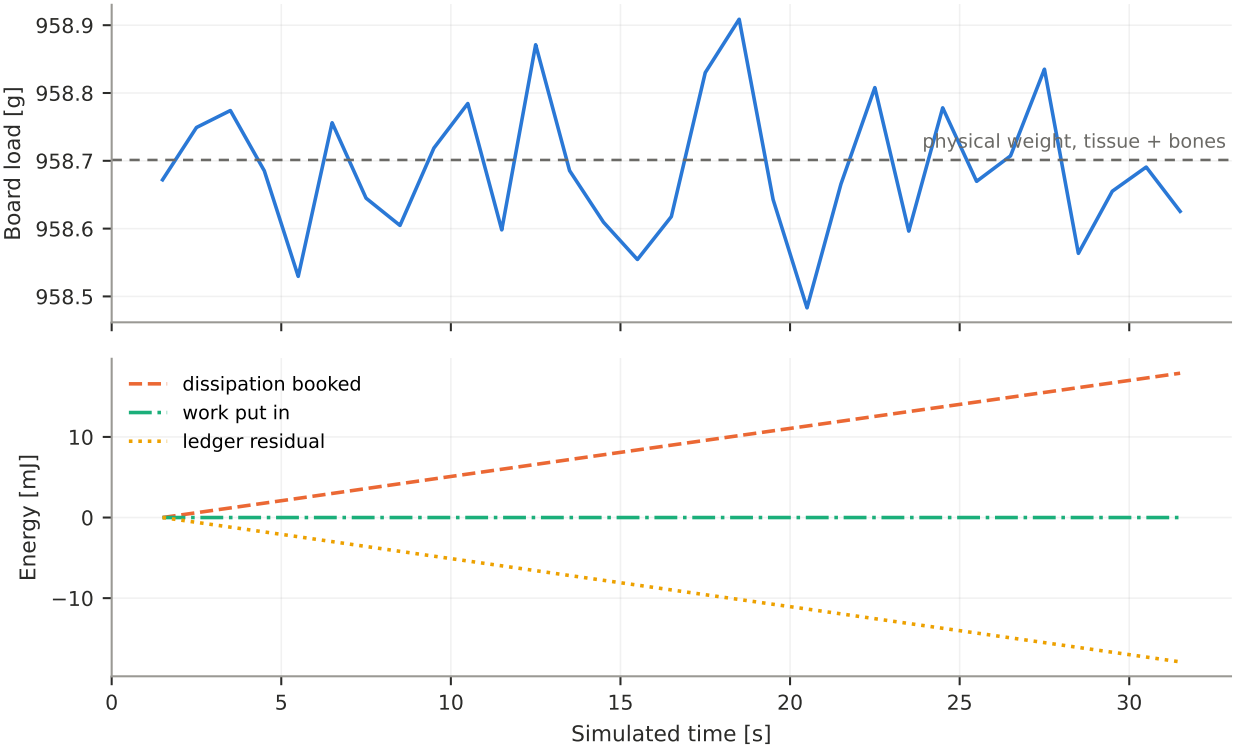


Figure 8: **30 s rest: salmon side, task B2.** Rest: the board's recorded load against the physical weight of tissue and bones (dashed), and the cumulative energy ledger (work put in, dissipation booked and residual) from the recorded samples.

8.5.5 30 s rest: pork tenderloin, task C1

No command for 30 s, sampled every simulated second; UNASSISTED. Its viscous term books a net -0.063 mW over the 30 s: energy a dissipative term should not put back; the cause is not found.

Observable	Recorded value
Seconds left alone	30
Tissue net displacement, max [μm]	0.15
Rod (thin bone) net displacement, max [μm]	*absent*
Knife bodies net displacement, max [μm]	0.268
Rigid bones net displacement, max [μm]	*none*
Physical node mass [g]	534.3
Rigid bones [g]	0.0
Added stability inertia (no weight) [g]	30.8
Board load at the end [g]	534.3
Board load / (physical node mass + bones)	1.0000
Dissipation booked, viscous [mW]	-0.063
Dissipation booked, rod ties [mW]	0
Dissipation booked, dampers [mW]	0.0513
Dissipation booked, friction [mW]	0.000168
Energy change of the tissue [mW]	1.36×10^{-7}
Work put in [mW]	0
Ledger residual [mW]	0.0116
Protective stops	0
Events	0
Wall seconds per simulated second	358

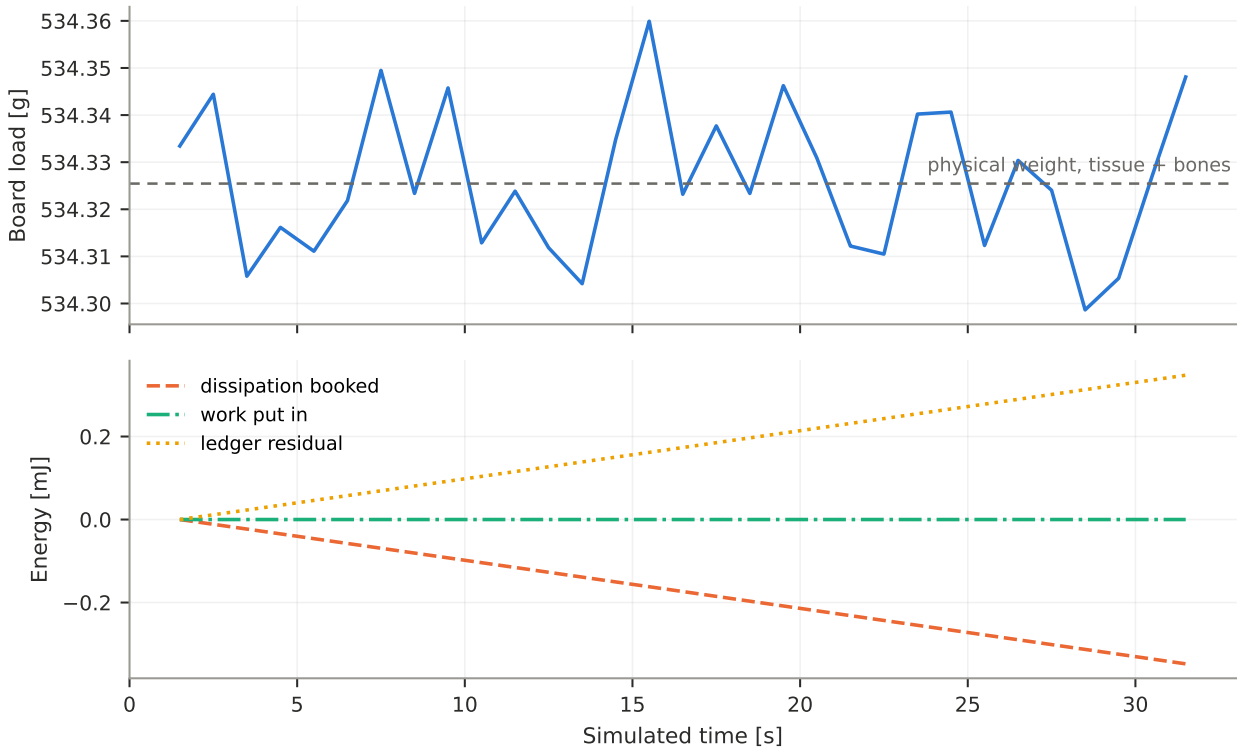


Figure 9: **30 s rest: pork tenderloin, task C1.** Rest: the board's recorded load against the physical weight of tissue and bones (dashed), and the cumulative energy ledger (work put in, dissipation booked and residual) from the recorded samples.

8.6 Legal abuse (quality gate 5)

Random legal commands, rams of the knife into the board and the backbone, the arms clashing and the knife dropped. Every episode must stay finite with tissue mass conserved.

8.6.1 Legal-abuse episodes (E15)

Random legal commands, knife rammed into the board and the backbone, arms clashing; 3 s each, recorded.

Assistance: none (legal robot commands only)

Each episode's final state and the worst sample over the episode. Lowest J is the smallest element volume ratio seen during the episode (negative means an element was inverted); Refused counts joint targets refused during protective stops.

Episode	Sim [s]	Finite	Mass err. [g]	Worst node [m/s]	Lowest J	Residual [J]	Work in [J]	Stops	Refused	Fault
random	3	true	0	0.0052	0.612	-0.000565	4.81×10^{-7}	4	25	*none*
ram_board	3	true	0	0.331	-0.310	0.00435	0.0792	2	217	*none*
ram_bone	3	true	0	0.889	0.061	0.00407	0.13	2	217	*none*
arm_clash	3	true	0	0.00566	0.612	-0.000572	3.05×10^{-9}	7	182	*none*

8.6.2 Knife dropped over the fish (E15)

The right hand opens with the knife above the board.

Assistance: none (legal robot commands only)

Observable	Recorded value
Finite	true
The knife touched tissue (a knife_to_tissue event)	false
At rest at the end	true
Knife ownership at the end ('held': MuJoCo integrates it, in a hand or resting on rigid geometry; 'free': the tissue solver)	held
Handle height at the end [m]	0.916
Handle speed at the end [m/s]	2.98×10^{-5}
Largest node speed at the end [m/s]	0.003
Fault	*none*

Evaluator output (unmodified task evaluator, schema gar.evaluation.v1): success false, score 0.308.

Check	Value	Threshold	Unit	Passed	Note
one_piece	0	0.97	fraction	no	nothing has been removed from the frame yet
yield	0	0.09553	kg	no	nothing has been removed from the frame yet
no_rib_pieces_over_2mm	*none*	0	pieces	no	nothing has been removed from the frame yet
no_backbone_or_fin_support_fragments	*none*	0	pieces	no	nothing has been removed from the frame yet
workmanship_score	*none*	85	points	no	nothing has been removed from the frame yet
placed_in_tray	0	1	fraction	no	nothing has been removed from the frame yet
skin_down	*none*	45	deg	no	nothing has been removed from the frame yet
settled	*none*	0.001	m	no	nothing has been removed from the frame yet
knife_not_dropped	390.8	15	mm (handle offset in grasp)	no	
no_blade_contact_with_holding_hand	0	0	ticks with contact	**yes**	
debris	0	1.206	g in pieces under 1 g (at most 10 pieces)	**yes**	0 pieces
no_tissue_thrown	4.014×10^{-6}	1	m/s fastest piece (19761 nodes, at 1.13 s; the fastest single node 0.00689 m/s, a flesh node at 1.14 s); 0 nodes below the bench top	**yes**	
time	2	180	s	**yes**	

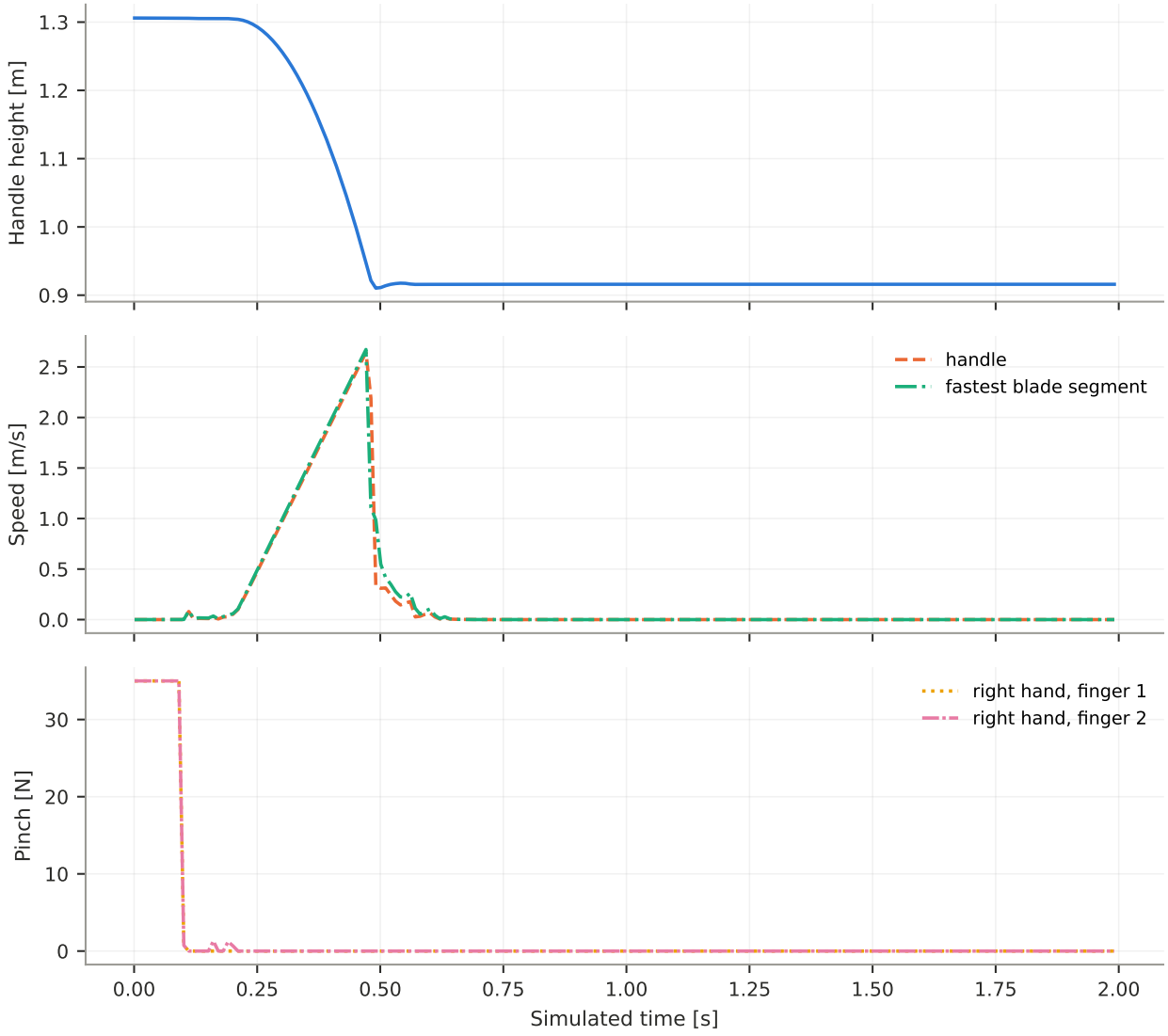


Figure 10: **Knife dropped over the fish (E15).** Knife drop: handle height, handle speed and the right hand's pinch on the handle, from the recorded samples; a dotted line would mark a knife_to_tissue event (the knife touching tissue), drawn only when one is recorded. **Not drawn:** event marker 'knife_to_tissue': no matching event recorded.

8.6.3 Knife dropped onto the fish (E15)

The A2 world; the right hand carries its held knife, edge-down, until the blade's middle is 60 mm above the trout's highest point over the median of its nodes, holds 0.3 s and opens, and the knife falls onto the fish. The drop height is measured when the hand opens. Gate 5 checks: finite, mass conserved, at most two ownership changes, rebound under 15 mm, no tissue node faster than the knife at impact, knife and tissue at rest within 1.5 s of the release. The run fails four of these (3 owner changes, a 15.2 mm rebound, a tissue node at 1.69 m/s against the knife's 1.00 m/s, the rest criteria met only 2.49 s after the release): gate 5 fails on it.

Assistance: none (legal robot commands only)

Observable	Recorded value
Finite	true
Landed on the fish	true
Hover reached	true
Blade middle above the fish's highest point at release [mm]	59.4
Blade middle to the nearest fish node at release [mm]	60.3
Blade speed at impact [m/s]	1
Mass error [g]	0
Knife ownership changes	3
Rebound [mm]	15.2
At rest after the release [s]	2.491
Fastest tissue node after the release [m/s]	1.69
Fault	*none*

8.6.4 Knife whip: fast wrist swings with the knife held

Joint 5 swung +-1.2 rad at 4.4 rad/s with the knife in the right hand.

Assistance: none

Observable	Recorded value
Swing amplitude [rad]	1.2
Peak wrist speed [rad/s]	4.4
Handle slide in the grip, max [mm]	3.19
Peak pinch [N]	226.6
Peak blade speed [m/s]	1.21
No-slip iterations	50
Protective stops	1
Fault	*none*

8.7 Mechanics against references (quality gate 6)

Each mechanism alone against a physical expectation: the coupon cut against Atkins' law, the grip, interface fracture energy, flank friction, the arm's motion generator, and the head-off energy ledger.

8.7.1 Coupon press cut across the sheets (E2)

An 80 x 40 x 20 mm fish-muscle block on the board, the knife pushed down at 20 mm/s.

Assistance: phantom kinematic knife rig

Observable	Recorded value
Tissue step, nominal [s]	2×10^{-5}
Mesh edge [mm]	3.5
Slice ratio	0
Sharpness index	0.22
Sheet orientation	across
Press plateau, push [N]	24.81
Atkins prediction, push [N]	24
Plateau / prediction	1.034
Plateau, slice [N]	0.0144
Atkins prediction, slice [N]	0
Peak [N]	29.06
Fragments	2
Mass before [g]	66.6186
Mass after [g]	66.6186
Ledger residual / work	-0.00254
Cut faces	538
Wall seconds per simulated second	73

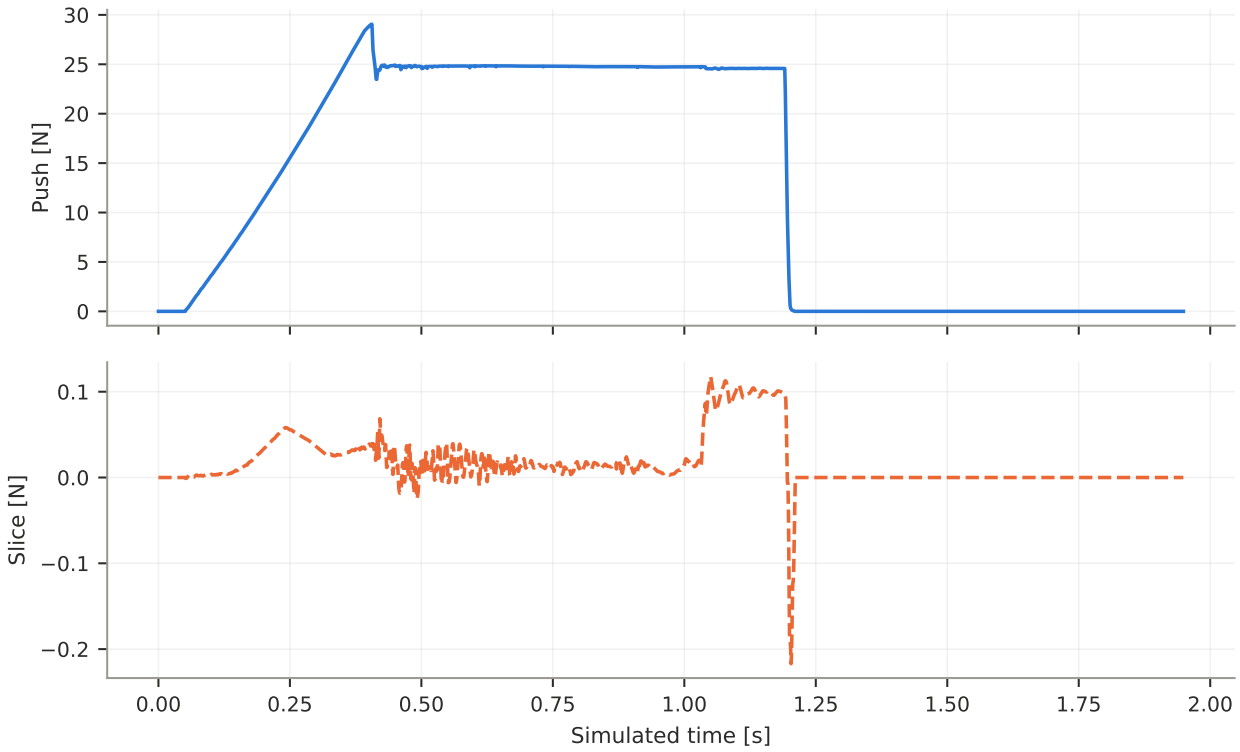


Figure 11: **Coupon press cut across the sheets (E2).** Coupon cut: vertical (push) and horizontal (slice) load on the knife from the recorded trace.

8.7.2 Coupon press cut, 2.5 mm mesh (E2)

An 80 x 40 x 20 mm fish-muscle block on the board, the knife pushed down at 20 mm/s.

Assistance: phantom kinematic knife rig

Observable	Recorded value
Tissue step, nominal [s]	2×10^{-5}
Mesh edge [mm]	2.5
Slice ratio	0
Sharpness index	0.22
Sheet orientation	across
Press plateau, push [N]	25.03
Atkins prediction, push [N]	24
Plateau / prediction	1.043
Plateau, slice [N]	0.0331
Atkins prediction, slice [N]	0
Peak [N]	29.09
Fragments	1
Mass before [g]	67.2
Mass after [g]	67.2
Ledger residual / work	-0.00505
Cut faces	1 195
Wall seconds per simulated second	124

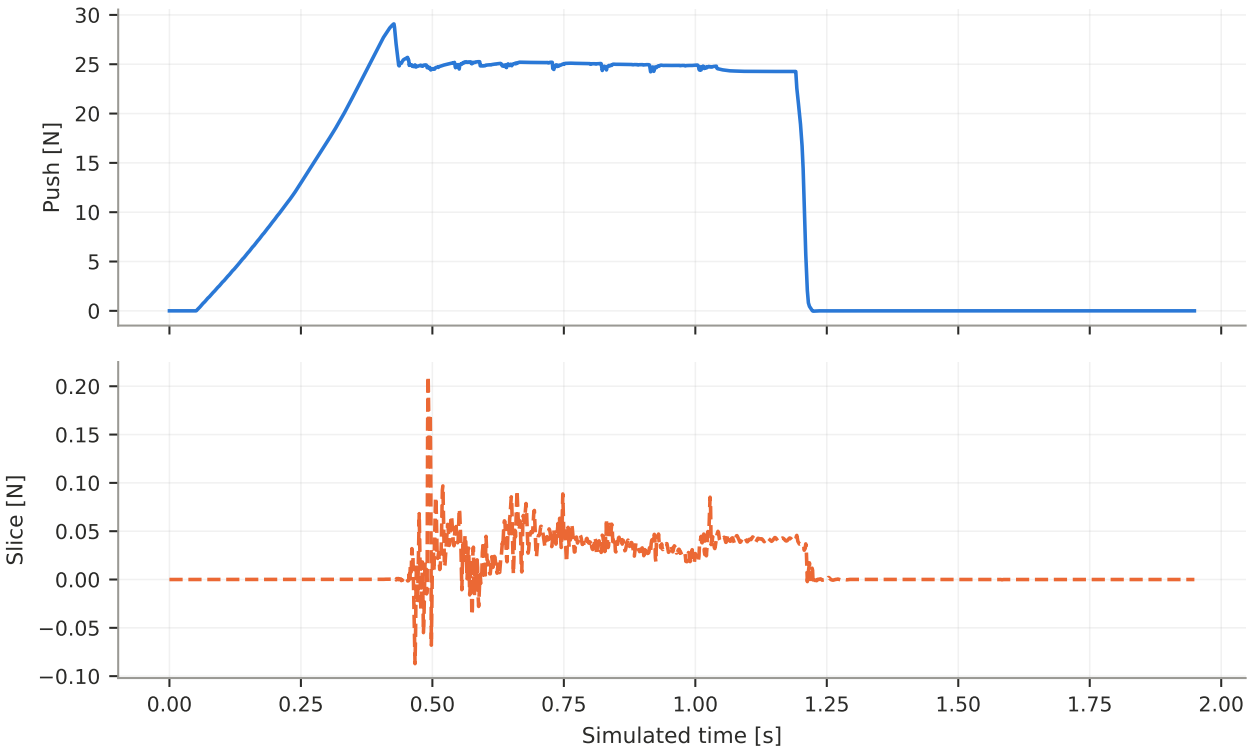


Figure 12: **Coupon press cut, 2.5 mm mesh (E2).** Coupon cut: vertical (push) and horizontal (slice) load on the knife from the recorded trace.

8.7.3 Coupon press cut, tissue step 1e-5 s (E2)

An 80 x 40 x 20 mm fish-muscle block on the board, the knife pushed down at 20 mm/s.

Assistance: phantom kinematic knife rig

Observable	Recorded value
Tissue step, nominal [s]	1×10^{-5}
Mesh edge [mm]	3.5
Slice ratio	0
Sharpness index	0.22
Sheet orientation	across
Press plateau, push [N]	24.81
Atkins prediction, push [N]	24
Plateau / prediction	1.034
Plateau, slice [N]	0.0157
Atkins prediction, slice [N]	0
Peak [N]	29.08
Fragments	2
Mass before [g]	66.6186
Mass after [g]	66.6186
Ledger residual / work	-0.000616
Cut faces	538
Wall seconds per simulated second	101

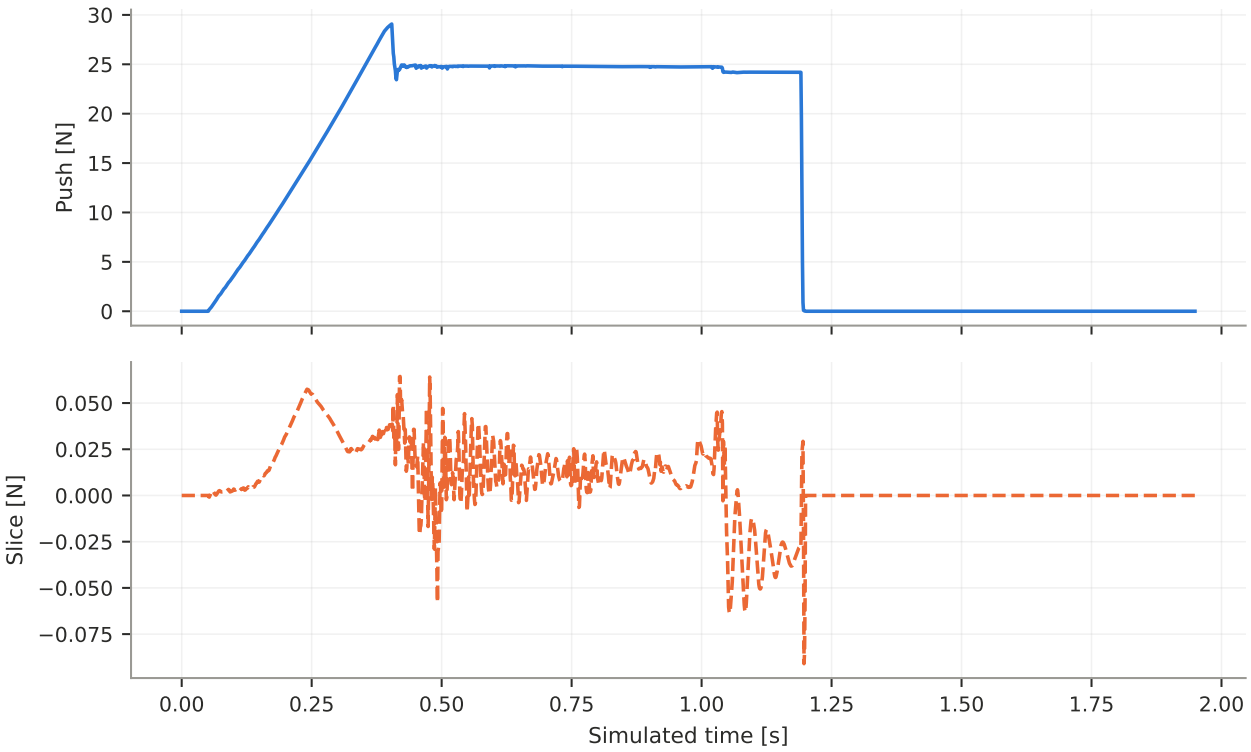


Figure 13: **Coupon press cut, tissue step 1e-5 s (E2).** Coupon cut: vertical (push) and horizontal (slice) load on the knife from the recorded trace.

8.7.4 Coupon slice cut, slice ratio 1 (E2)

An 80 x 40 x 20 mm fish-muscle block on the board, the knife pushed down at 20 mm/s.

Assistance: phantom kinematic knife rig

Observable	Recorded value
Tissue step, nominal [s]	2×10^{-5}
Mesh edge [mm]	3.5
Slice ratio	1
Sharpness index	0.22
Sheet orientation	across
Press plateau, push [N]	14.57
Atkins prediction, push [N]	12
Plateau / prediction	1.214
Plateau, slice [N]	-10.67
Atkins prediction, slice [N]	12
Peak [N]	19.29
Fragments	1
Mass before [g]	66.6186
Mass after [g]	66.6186
Ledger residual / work	-0.00171
Cut faces	533
Wall seconds per simulated second	77

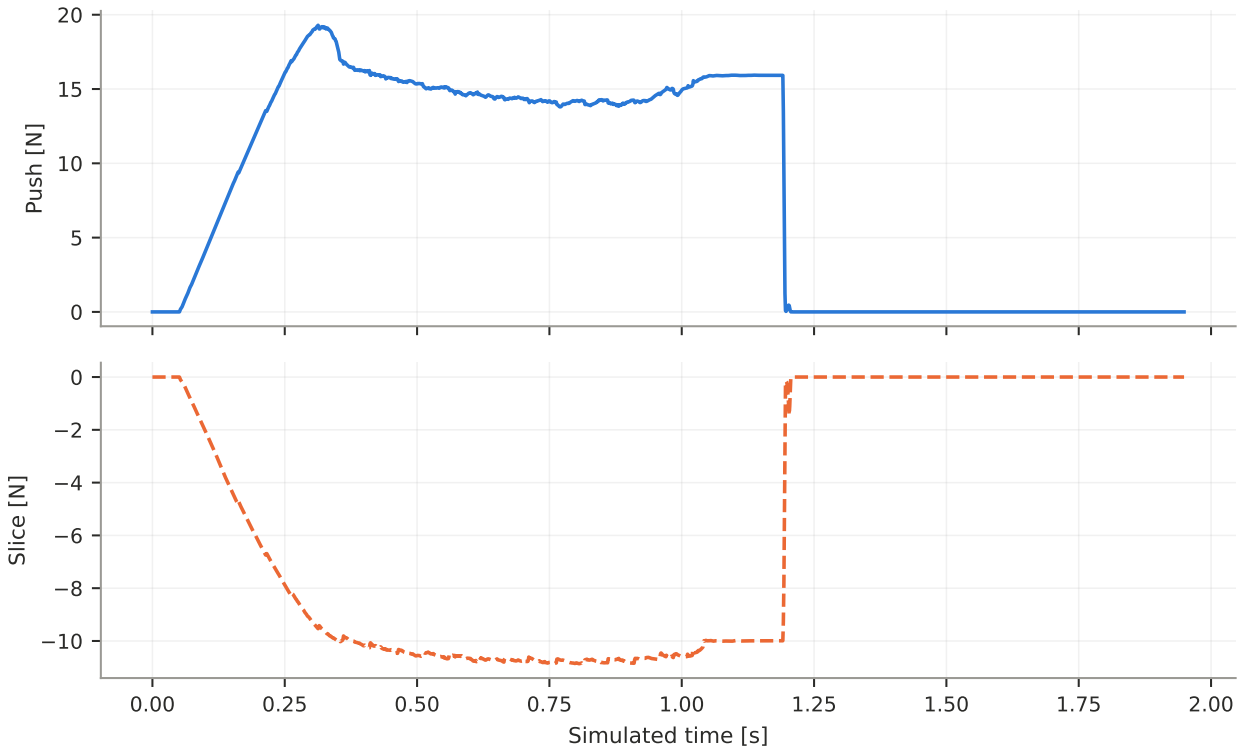


Figure 14: **Coupon slice cut, slice ratio 1 (E2).** Coupon cut: vertical (push) and horizontal (slice) load on the knife from the recorded trace.

8.7.5 Skin-on coupon: the press cut through a 0.5 mm skin on the board (SKIN_R 2 kJ/m^2)

An ASSISTED kinematic fillet blade presses at 20 mm/s through the 80 x 40 x 20 mm fish-muscle block (the world's coupon material, tearing on) to 2 mm into the board, with the world's skin membrane (0.5 mm, SKIN_E, the cutting toughness given) on the block's bottom face, the face on the board (the bottom skin layer is where B5's cuts stay joined). The edge must cut the skin pressed onto the board; a pure press needs R l of the skin over the 80 mm cut.

Observable	Recorded value
Skin cutting toughness SKIN_R [J/m^2]	2000
Skin thickness [mm]	0.5
Fragments	2
Fragment masses [g]	33.58, 33.04
Press plateau in the flesh, 6-14 mm up [N]	24.96
Largest force in the bottom 3 mm (the skin) [N]	174.4
Atkins' R l of the skin, pure press [N]	160
Crack faces	747
Severed samples	2 544
Carried on the lifted blade [g]	0

8.7.6 Skin-on coupon, SKIN_R 1 kJ/m^2 (sensitivity)

An ASSISTED kinematic fillet blade presses at 20 mm/s through the 80 x 40 x 20 mm fish-muscle block (the world's coupon material, tearing on) to 2 mm into the board, with the world's skin membrane (0.5 mm, SKIN_E, the cutting toughness given) on the block's bottom face, the face on the board (the bottom skin layer is where B5's cuts stay joined). The edge must cut the skin pressed onto the board; a pure press needs R l of the skin over the 80 mm cut.

Observable	Recorded value
Skin cutting toughness SKIN_R [J/m^2]	1000
Skin thickness [mm]	0.5
Fragments	2
Fragment masses [g]	33.58, 33.04
Press plateau in the flesh, 6-14 mm up [N]	24.96
Largest force in the bottom 3 mm (the skin) [N]	98.76
Atkins' R l of the skin, pure press [N]	80
Crack faces	747
Severed samples	1 847
Carried on the lifted blade [g]	0

8.7.7 Skin-on coupon, SKIN_R 3 kJ/m^2 (sensitivity)

An ASSISTED kinematic fillet blade presses at 20 mm/s through the 80 x 40 x 20 mm fish-muscle block (the world's coupon material, tearing on) to 2 mm into the board, with the world's skin membrane (0.5 mm, SKIN_E, the cutting toughness given) on the block's bottom face, the face on the board (the bottom skin layer is where B5's cuts stay joined). The edge must cut the skin pressed onto the board; a pure press needs R l of the skin over the 80 mm cut.

Observable	Recorded value
Skin cutting toughness SKIN_R [J/m^2]	3000
Skin thickness [mm]	0.5
Fragments	2
Fragment masses [g]	33.58, 33.04
Press plateau in the flesh, 6-14 mm up [N]	24.96
Largest force in the bottom 3 mm (the skin) [N]	244.7
Atkins' R l of the skin, pure press [N]	240
Crack faces	785
Severed samples	4 847
Carried on the lifted blade [g]	0

8.7.8 The same coupon without skin

An ASSISTED kinematic fillet blade presses at 20 mm/s through the 80 x 40 x 20 mm fish-muscle block (the world's coupon material, tearing on) to 2 mm into the board. Reference for the skin-on runs.

Observable	Recorded value
Skin cutting toughness SKIN_R [J/m^2]	*none*
Skin thickness [mm]	*none*
Fragments	2
Fragment masses [g]	33.58, 33.04
Press plateau in the flesh, 6-14 mm up [N]	25.14
Largest force in the bottom 3 mm (the skin) [N]	24.76
Atkins' R l of the skin, pure press [N]	*none*
Crack faces	757
Severed samples	1 968
Carried on the lifted blade [g]	0

8.7.9 Sensitivity of the coupon cut to the open values (one perturbed at a time)

ASSISTED kinematic fillet blade through the world's coupon (80 x 40 x 20 mm fish muscle with its tearing law and sheets) to 2 mm into the board; MU_BLADE_TISSUE 0/0.5/1.0, FISH_SIGMA_T 25/100 kPa, SHEET_SIGMA 5/20 kPa, SHEET_G 1/10 J/m2 replaced one at a time in the probe's process only, on a straight press (20 mm/s), a drifting tilted press (10 mm, 7 deg) and a sawing descent (10 mm/s, +-20 mm at 1.5 Hz). A coupon, not a task: how the cut's outcome responds to the open values.

Observable	Recorded value
straight press: divides with	nominal, MU_BLADE_TISSUE=0, MU_BLADE_TISSUE=0.5, MU_BLADE_TISSUE=1, FISH_SIGMA_T=25000, FISH_SIGMA_T=100000, SHEET_SIGMA=20000, SHEET_G=1, SHEET_G=10
straight press: stays whole with	SHEET_SIGMA=5000
straight press: press plateau range [N]	25.14, 25.14
straight press: most carried on the lifted blade [g]	0
drifting tilted press: divides with	nominal, MU_BLADE_TISSUE=0, MU_BLADE_TISSUE=0.5, MU_BLADE_TISSUE=1, FISH_SIGMA_T=25000, FISH_SIGMA_T=100000, SHEET_SIGMA=5000, SHEET_SIGMA=20000, SHEET_G=1, SHEET_G=10
drifting tilted press: stays whole with	*none*
drifting tilted press: press plateau range [N]	24.78, 26.7
drifting tilted press: most carried on the lifted blade [g]	2.85
sawing descent: divides with	nominal, MU_BLADE_TISSUE=0, MU_BLADE_TISSUE=0.5, MU_BLADE_TISSUE=1, FISH_SIGMA_T=25000, FISH_SIGMA_T=100000, SHEET_SIGMA=5000, SHEET_SIGMA=20000, SHEET_G=1, SHEET_G=10
sawing descent: stays whole with	*none*
sawing descent: press plateau range [N]	22.12, 22.71
sawing descent: most carried on the lifted blade [g]	33.04
Runs	30
That process carried this world's identity	true

8.7.10 Knife held 10 s, dual FR3 (E11)

10 s with the knife in the right hand's inserts at the default grip, no command.

Assistance: none

Observable	Recorded value
Contact impedance ratio	10
No-slip iterations	50
Seconds held	10
Handle creep [mm/s]	2.18×10^{-5}
Axis drift [deg]	-1.5×10^{-5}
Pinch, finger 1 at the end [N]	35
Pinch, finger 2 at the end [N]	35
Protective stops	0
Wall time [s]	345.3

8.7.11 Knife held 10 s, dual Panda (E11)

The dual-Panda setup: the knife in the right hand's inserts at the default grip, no command.

Assistance: none

Observable	Recorded value
Contact impedance ratio	10
No-slip iterations	50
Seconds held	10
Handle creep [mm/s]	0.000286
Axis drift [deg]	0.00806
Pinch, finger 1 at the end [N]	35
Pinch, finger 2 at the end [N]	35
Protective stops	0
Wall time [s]	338.5

8.7.12 Interface fracture energy paid by the blade

1 mJ of interface debt placed on a blade segment in the air, held still, then moved.

Assistance: phantom knife rig (knife in the air)

Observable	Recorded value
Debt placed on the blade [mJ]	1
Debt left after the still hold [mJ]	1
Interface energy booked at the end [mJ]	1
Debt left at the end [mJ]	3.26×10^{-7}
Work put in at the end [mJ]	1.061
Ledger residual at the end [mJ]	0.0152
Residual / work put in	0.0144
Verdict: unpaid while still	true
Verdict: paid	true
Verdict: booked	true
Verdict: residual small	true

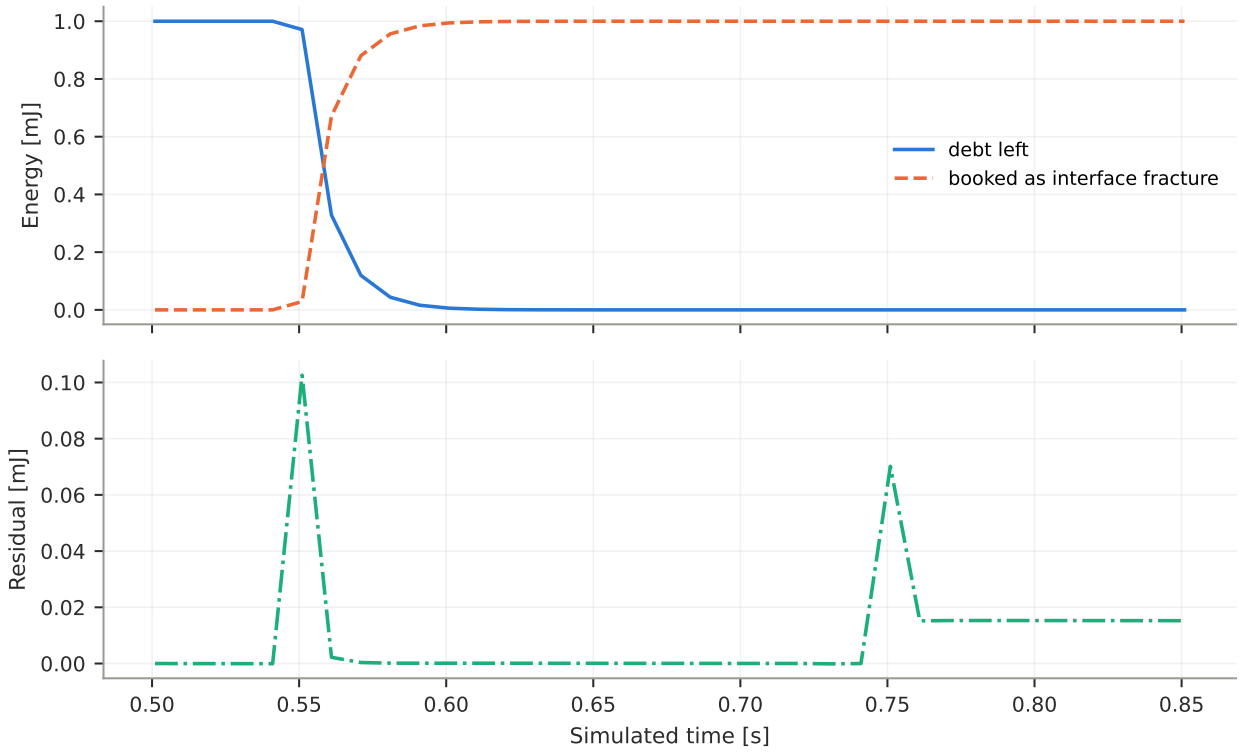


Figure 15: **Interface fracture energy paid by the blade.** Interface fracture energy placed on the blade: the debt left, the energy booked to the interface term and the ledger residual, from the recorded samples.

8.7.13 Flank friction while sawing

The blade pressed flat 1 mm into a coupon and sawn, its edge outside the block, meant to isolate friction in the ledger. With tearing on, the sawn flat tears the top layer: the 151 faces counted are tears, made in 58 tear events, the ledger misses the work by about 10 %, and the probe does not isolate friction. Open.

Assistance: phantom knife rig

Observable	Recorded value
Press into the coupon [mm]	1
Saw amplitude [mm]	20
Saw frequency [Hz]	1.5
Seconds	3
Friction dissipation [mJ]	73.69
Viscous dissipation [mJ]	11.94
Damper dissipation [mJ]	1.052
Interface fracture booked [mJ]	7.202
Topology (cut and torn faces) booked [mJ]	-2.601
Rod ties booked [mJ]	0.1628
Work put in [mJ]	83.94
Energy change [mJ]	0.644
Ledger residual [mJ]	-8.15
Residual / work [%]	-9.71
Faces cut	151

8.7.14 Arm motion generator: far joint steps

Far joint targets on five joints; the rate limiter must keep every joint under its speed limit without tripping the velocity reflex.

Assistance: none

Observable	Recorded value
Protective stops	0
1: peak speed / limit	0.972
2: peak speed / limit	0.627
4: peak speed / limit	0.746
6: peak speed / limit	0.457
7: peak speed / limit	0.562
1: overshoot [mrad]	0
2: overshoot [mrad]	32
4: overshoot [mrad]	10.5
6: overshoot [mrad]	0
7: overshoot [mrad]	0.329

8.7.15 Coupon slice cut, slice ratio 1, frictionless blade (E2)

The slice cut with blade-tissue friction off, isolating the cutting law from friction.

Assistance: phantom kinematic knife rig

Observable	Recorded value
Slice ratio	1
Blade friction	0
Mesh edge [mm]	3.5
Press plateau, push [N]	14.57
Atkins prediction, push [N]	12
Plateau / prediction	1.214
Fragments	2
Ledger residual / work	-0.0017

8.7.16 Finite-difference and conservation audit of the force laws

Every force kernel against the derivative of its booked energy, sums of forces and torques, and the integrators' float32 resolution. Two of the 287 checks fail: a rod node advanced at 2 mm/s travels 1.69 % off (threshold 1 %; float32 rounding in the rod projection), and the three-node bending hinge's forces carry a net torque of 1.0 % of the sum of |r||f| (threshold 1e-5); that check runs tissue.k_rod_bend, a kernel the world never launches (Tissue.n_hinges is only ever 0: the bones' rods use bodies.k_rods_pbd, whose checks pass). The run is required to complete, not to pass every check.

Assistance: none

Observable	Recorded value
Cases	71
Checks	287
Checks failed	2
Wall time [s]	533

8.7.17 MuJoCo box-box contact with a detection gap

MuJoCo 3.11's box-box collider with and without a contact gap, the reason the world runs with none.

Assistance: none

Observable	Recorded value
MuJoCo	3.11.0
Gap changes the constraint rows	false
Bogus deep contacts	1

8.7.18 Phantom rig force cap: the knife driven 10 mm into the board

The force-limited rig servo driven 10 mm into the board.

Assistance: phantom knife rig

Observable	Recorded value
Commanded into the board [mm]	10
Servo force, max [N]	149.3
Force cap [N]	150
Holding lag [mm]	6.58
Blade below the board top [mm]	1.15
Ledger residual / work	0.000215
Finite	true

8.7.19 Tearing in tension, sheets across

A fish-muscle bar with a 4 mm notch, pulled apart across the sheets by spring grips.

Assistance: none

Observable	Recorded value
Governing strength [Pa]	1×10^4
Peak nominal stress [Pa]	8248
Peak / strength	0.825
Broke at [s]	0.65
Torn faces	383

8.7.20 Tearing in tension, sheets along

A fish-muscle bar with a 4 mm notch, pulled apart along the sheets by spring grips.

Assistance: none

Observable	Recorded value
Governing strength [Pa]	5×10^4
Peak nominal stress [Pa]	3.497×10^4
Peak / strength	0.699
Broke at [s]	3.32
Torn faces	784

8.7.21 Flat-punch puncture of a fish-muscle slab (E1)

A flat punch of the 12.5 mm cylinder's area pressed into a slab at 1 mm/s; Torgersen et al. 2014: 6.6-20.9 N, yield at 6-8 mm.

Assistance: none

Observable	Recorded value
Peak force [N]	16.83
Depth at peak [mm]	15
Torn faces	367

8.7.22 Grip creep in the recorded runs

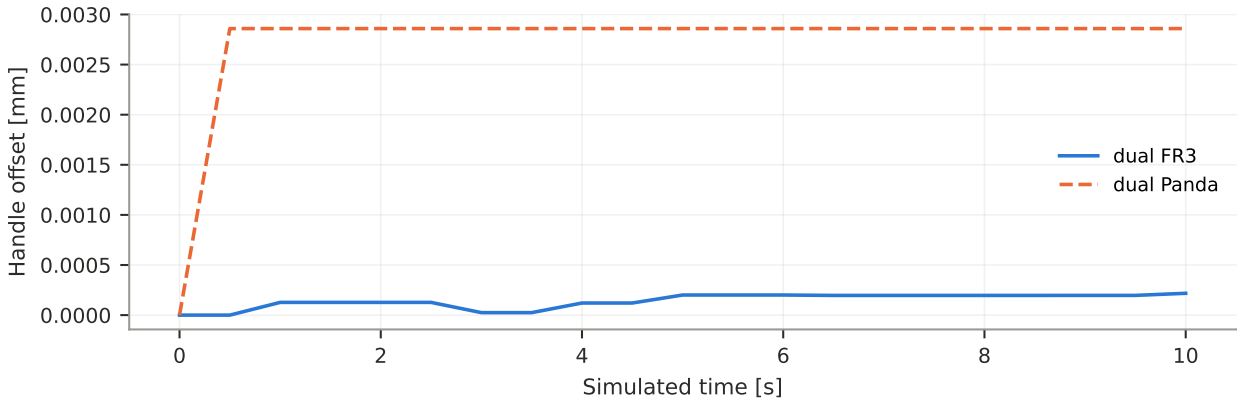


Figure 16: Offset of the knife handle in the grip over 10 s with no command, from the recorded samples of the recorded runs.

8.8 Runtime (quality gate 7)

Wall-clock seconds per simulated second through the public interface with an RGB observation, against the limit of 200. Startup is reported separately; host load is recorded.

8.8.1 Operating cost through the public interface (hold) (E16)

Six 0.5 s actions of the holding arm, each followed by an RGB observation, in process through the public gar API (gar.create, act_policy; not the HTTP server). Every action waits up to 15 min for a quiet host (1-min load under 12, the GPU under 30 % busy with other work, no other process rendering; sampled before and after the action) and is marked quiet or not; the steady figures leave out the first action (the warm-up); gate 7 judges the quiet ones.

Observable	Recorded value
Robot	dual-fr3
Task	fillet-one-side
Camera	overhead
Action length [s]	0.5
Startup: create + reset [s]	137.7
Steady wall/sim, mean	59.9
Steady wall/sim, max	60.1
Physics only wall/sim, mean	58.53
Quiet-host actions (gate 7)	2
Quiet-host wall/sim, mean	60.02
Quiet-host wall/sim, max	60.1
Mean of the steady actions under 200 (a cut stage's cutting actions, a hold's actions after the first; gate 7 judges the largest quiet one)	true
Clock agrees	true
Every RGB frame decoded	true
Tets	73 273

Every cycle as recorded (action then RGB observation).

Wall [s]	Sim [s]	Wall / sim	Physics wall / sim	Load before (1 min)	GPU before [%]	Load after	GPU after [%]	Render work-ers	Cut faces	Quiet host
30.17	0.5	60.33	58.87	4.04	2.8	3.88	1	1	4 999	true
30.02	0.5	60.04	58.66	3.81	1	4.28	40.6	1	4 999	false
29.97	0.5	59.94	58.53	4.28	1	3.77	4	1	4 999	true
30.05	0.5	60.1	58.74	3.77	1	3.72	17.6	1	4 999	true
29.87	0.5	59.74	58.36	3.72	1	3.54	39	1	4 999	false
29.85	0.5	59.7	58.37	3.54	1	3.6	40.6	1	4 999	false

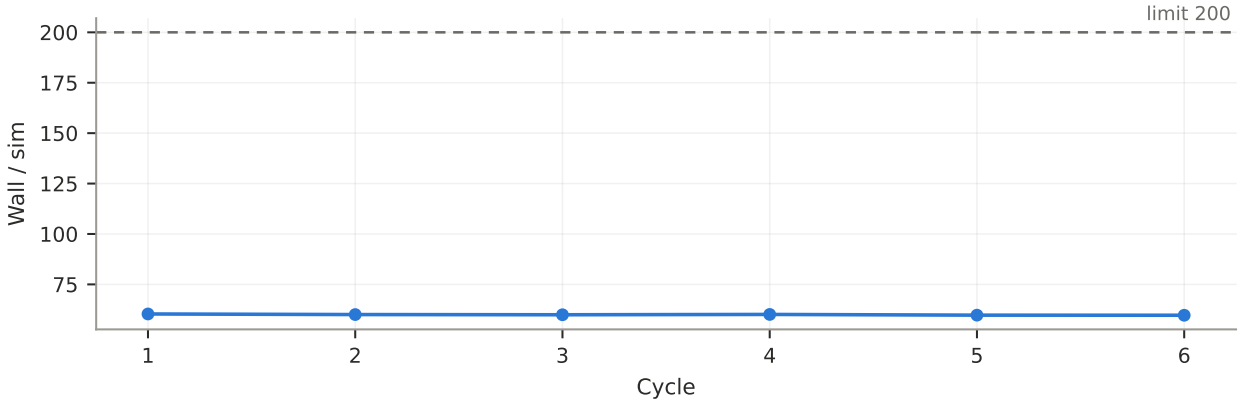


Figure 17: **Operating cost through the public interface (hold) (E16).** Wall seconds per simulated second for every recorded cycle of the public-interface loop, against the limit of 200.

8.8.2 Operating cost through the public interface while cutting (trout) (E16)

The held knife brought to 4 mm above the trout (two approach actions), then pressed 6 mm per 0.5 s action and drawn along the blade, each action followed by an RGB observation, in process through the public gar API (contact collision profile). Cutting actions are those that made crack faces (gate 7 requires them); then ten 0.1 s strokes after the slices (the regime after a large tear). Every action waits up to 15 min for a quiet host (1-min load under 12, the GPU under 30 % busy with other work, no other process rendering; sampled before and after the action) and is marked quiet or not; gate 7 judges the quiet ones.

Observable	Recorded value
Robot	dual-fr3
Task	fillet-one-side
Cameras	overhead
Action length [s]	0.5
Startup: create + reset [s]	135.9
Actions that cut	13
Steady wall/sim, mean	118.7
Steady wall/sim, max	215
Non-cutting actions, mean wall/sim	82.07
Physics only wall/sim, mean	113.8
Quiet-host actions (gate 7)	10
Quiet-host wall/sim, mean	118.2
Quiet-host wall/sim, max	215
Mean of the steady actions under 200 (a cut stage's cutting actions, a hold's actions after the first; gate 7 judges the largest quiet one)	true
Clock agrees	true
Every RGB frame decoded	true
Tets	73 273

Every cycle as recorded (action then RGB observation).

Action	Wall [s]	Sim [s]	Wall / sim	Physics wall / sim	Load before (1 min)	GPU before [%]	Load after	GPU after [%]	Render work-ers	Cut faces	New crack faces	Topology [s]	Quiet host
approach_ above	59.53	1	59.53	58.49	2.42	20.8	2.92	1	1	4 999	0	2.75	true
approach_ surface	30.54	0.5	61.08	58.97	2.92	1	24.7	21	1	4 999	0	1.37	false
press_0	31.25	0.5	62.5	60.97	9.6	1	6.77	10.4	1	4 999	0	1.62	true
press_1	34.18	0.5	68.37	66.72	6.31	1	4.8	40.6	1	5 007	8	3.82	false
press_2	34.36	0.5	68.72	67.22	4.58	1	3.58	22.2	1	5 012	5	3.92	true
press_3	36.64	0.5	73.28	71.79	3.58	1	2.95	16.8	1	5 025	13	6.07	true
press_4	43.5	0.5	87	85.46	2.79	1	2.94	1	1	5 131	106	12.2	true
slice_0	54.42	0.5	108.8	107.4	2.94	1	3.02	1	1	5 347	216	22.4	true
slice_1	81.01	0.5	162	160.2	3.17	1	4.43	1	1	6 228	881	48	true
after_0	21.5	0.1	215	207.2	4.23	1	3.88	1	1	6 255	27	14.1	true
after_1	16.34	0.1	163.4	155.9	3.88	1	3.57	31.8	1	6 259	4	9.1	false
after_2	15.26	0.1	152.6	144.8	3.57	1	3.48	22.6	1	6 268	9	7.86	true
after_3	10.41	0.1	104.1	96.95	3.48	1	3.31	1	1	6 268	0	3.51	true
after_4	10.73	0.1	107.3	99.41	3.31	1	3.4	1.2	1	6 276	8	3.69	true
after_5	12.96	0.1	129.6	121.9	3.4	1	3.24	40.6	1	6 279	3	5.72	false
after_6	8.954	0.1	89.54	81.61	3.22	1	3.19	1	1	6 279	0	2	true
after_7	11.57	0.1	115.7	108.3	3.19	1	3.15	29.8	1	6 279	0	4.5	true
after_8	8.959	0.1	89.59	81.8	3.15	1	3.2	1	1	6 280	1	1.99	true
after_9	11.74	0.1	117.4	109.8	3.2	1	3.23	1	1	6 283	3	4.63	true

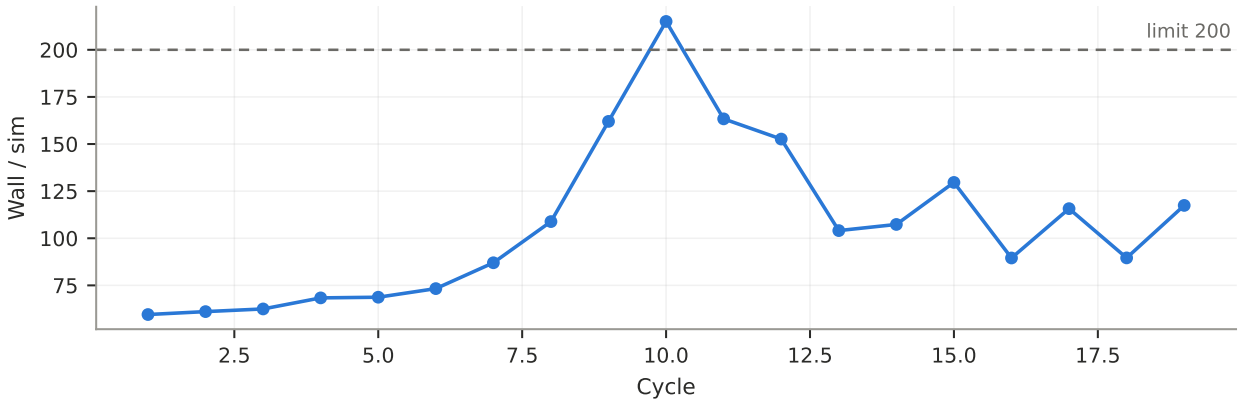


Figure 18: **Operating cost through the public interface while cutting (trout) (E16).** Wall seconds per simulated second for every recorded cycle of the public-interface loop, against the limit of 200.

8.8.3 Operating cost while cutting (trout), second run (E16)

The held knife brought to 4 mm above the trout (two approach actions), then pressed 6 mm per 0.5 s action and drawn along the blade, each action followed by an RGB observation, in process through the public gar API (contact collision profile). Cutting actions are those that made crack faces (gate 7 requires them); then ten 0.1 s strokes after the slices (the regime after a large tear). Every action waits up to 15 min for a quiet host (1-min load under 12, the GPU under 30 % busy with other work, no other process rendering; sampled before and after the action) and is marked quiet or not; gate 7 judges the quiet ones.

Observable	Recorded value
Robot	dual-fr3
Task	fillet-one-side
Cameras	overhead
Action length [s]	0.5
Startup: create + reset [s]	136.2
Actions that cut	15
Steady wall/sim, mean	128.3
Steady wall/sim, max	239.6
Non-cutting actions, mean wall/sim	71.47
Physics only wall/sim, mean	123.1
Quiet-host actions (gate 7)	11
Quiet-host wall/sim, mean	133.9
Quiet-host wall/sim, max	239.6
Mean of the steady actions under 200 (a cut stage's cutting actions, a hold's actions after the first; gate 7 judges the largest quiet one)	true
Clock agrees	true
Every RGB frame decoded	true
Tets	73 273

Every cycle as recorded (action then RGB observation).

Action	Wall [s]	Sim [s]	Wall / sim	Physics wall / sim	Load before (1 min)	GPU before [%]	Load after	GPU after [%]	Render work-ers	Cut faces	New crack faces	Topology [s]	Quiet host
approach_ above	59.51	1	59.51	58.45	9.55	1	16.5	40.6	1	4 999	0	2.77	false
approach_ surface	30.55	0.5	61.1	59.18	10.2	1	7.35	40.6	1	4 999	0	1.38	false
press_0	31.31	0.5	62.63	61.13	7.35	1	5.59	10.2	1	4 999	0	1.64	true
press_1	34.14	0.5	68.29	66.66	5.3	1	4.36	1	1	5 007	8	3.85	true
press_2	34.06	0.5	68.11	66.6	4.36	1	3.89	17	1	5 012	5	3.81	true
press_3	37.01	0.5	74.01	72.52	3.82	1	3.58	40.6	1	5 025	13	6.17	false
press_4	44.71	0.5	89.43	87.87	3.58	1	3.27	40.6	1	5 111	86	13.3	false
slice_0	54.02	0.5	108	106.6	3.27	1	3.62	1	1	5 314	203	21.7	true
slice_1	80.71	0.5	161.4	159.6	3.49	1	3.73	1	1	6 219	905	47.6	true
after_0	23.96	0.1	239.6	231.8	3.67	1	3.96	8.6	1	6 253	34	16.5	true
after_1	15.83	0.1	158.3	151	3.96	1	3.71	40.6	1	6 257	4	8.5	false
after_2	16.14	0.1	161.4	153.6	3.71	1	3.57	1	1	6 269	12	8.73	true
after_3	13.98	0.1	139.8	132.5	3.57	1	3.49	1	1	6 277	8	6.76	true
after_4	12.21	0.1	122.1	114.4	3.49	1	3.31	1	1	6 278	1	4.96	true
after_5	13.76	0.1	137.6	129.9	3.31	1	3.3	1	1	6 281	3	6.58	true
after_6	13.54	0.1	135.4	127.6	3.3	1	3.31	1	1	6 282	1	6.22	true
after_7	13.06	0.1	130.6	122.9	3.2	1	3.24	1	1	6 283	1	5.71	true
after_8	13.08	0.1	130.8	122.7	3.24	1	3.2	40.6	1	6 286	3	5.77	false
after_9	10.26	0.1	102.6	95.19	3.1	1	3.24	1	1	6 286	0	3.28	true

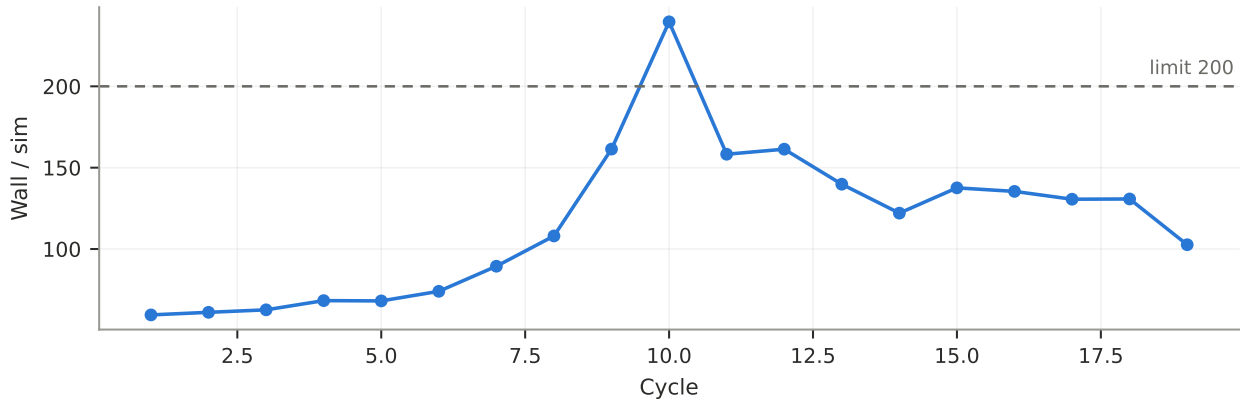


Figure 19: **Operating cost while cutting (trout), second run (E16).** Wall seconds per simulated second for every recorded cycle of the public-interface loop (second trout run), against the limit of 200.

8.8.4 Operating cost through the public interface while cutting (salmon side, 93,851 tets) (E16)

The held knife brought to 4 mm above the salmon side (task portion) (two approach actions), then pressed 6 mm per 0.5 s action and drawn along the blade, each action followed by an RGB observation, in process through the public gar API (contact collision profile). Cutting actions are those that made crack faces (gate 7 requires them); then ten 0.1 s strokes after the slices (the regime after a large tear). Every action waits up to 15 min for a quiet host (1-min load under 12, the GPU under 30 % busy with other work, no other process rendering; sampled before and after the action) and is marked quiet or not; gate 7 judges the quiet ones.

Observable	Recorded value
Robot	dual-fr3
Task	portion
Cameras	overhead
Action length [s]	0.5
Startup: create + reset [s]	152.9
Actions that cut	11
Steady wall/sim, mean	115.4
Steady wall/sim, max	156.9
Non-cutting actions, mean wall/sim	96.17
Physics only wall/sim, mean	111.9
Quiet-host actions (gate 7)	7
Quiet-host wall/sim, mean	118.8
Quiet-host wall/sim, max	156.9
Mean of the steady actions under 200 (a cut stage's cutting actions, a hold's actions after the first; gate 7 judges the largest quiet one)	true
Clock agrees	true
Every RGB frame decoded	true
Tets	93 851

Every cycle as recorded (action then RGB observation).

Action	Wall [s]	Sim [s]	Wall / sim	Physics wall / sim	Load before (1 min)	GPU before [%]	Load after	GPU after [%]	Render work-ers	Cut faces	New crack faces	Topology [s]	Quiet host
approach_above	80.76	1	80.76	79.81	3.47	12.4	11.8	47.8	1	293	0	2.47	false
approach_surface	41.22	0.5	82.44	80.51	11.2	1	7.64	6.2	1	293	0	1.46	true
press_0	40.71	0.5	81.42	80.15	7.19	1	5.1	1	1	293	0	1.56	true
press_1	42.52	0.5	85.03	83.61	4.85	1	4.4	38.6	1	294	1	3.29	false
press_2	50.84	0.5	101.7	100.4	4.4	1	4.19	1	1	320	26	11.2	true
press_3	56.81	0.5	113.6	112.3	4.19	1	4.09	17.2	1	366	46	16.9	true
press_4	62.76	0.5	125.5	124.2	4.09	1	4.14	16.8	1	433	67	22.5	true
slice_0	64.97	0.5	129.9	128.4	4.13	1	3.82	40.6	1	502	69	24.6	false
slice_1	78.43	0.5	156.9	155.4	3.67	1	4.03	18	1	870	368	37.6	true
after_0	11.16	0.1	111.6	105.5	4.03	1	4.55	35.8	1	871	1	2.76	false
after_1	11.06	0.1	110.6	104.2	4.55	1	4.34	3.8	1	881	10	2.61	true
after_2	10.44	0.1	104.4	98.54	4.34	1	4.12	1	1	881	0	2.14	true
after_3	11.08	0.1	110.8	104.7	4.12	1	3.87	1	1	885	4	2.75	true
after_4	11.08	0.1	110.8	105	3.87	1	3.61	40.6	1	886	1	2.69	false
after_5	11.29	0.1	112.9	107	3.48	1	3.48	16	1	887	1	2.9	true
after_6	10.45	0.1	104.5	98.75	3.28	1	3.16	1	1	887	0	2.11	true
after_7	10.75	0.1	107.5	101.7	2.98	1	2.83	9.6	1	887	0	2.44	true
after_8	10.35	0.1	103.5	97.71	2.68	1	2.58	1	1	887	0	2.05	true
after_9	10.48	0.1	104.8	99.07	2.45	1	2.38	40.6	1	887	0	2.26	false

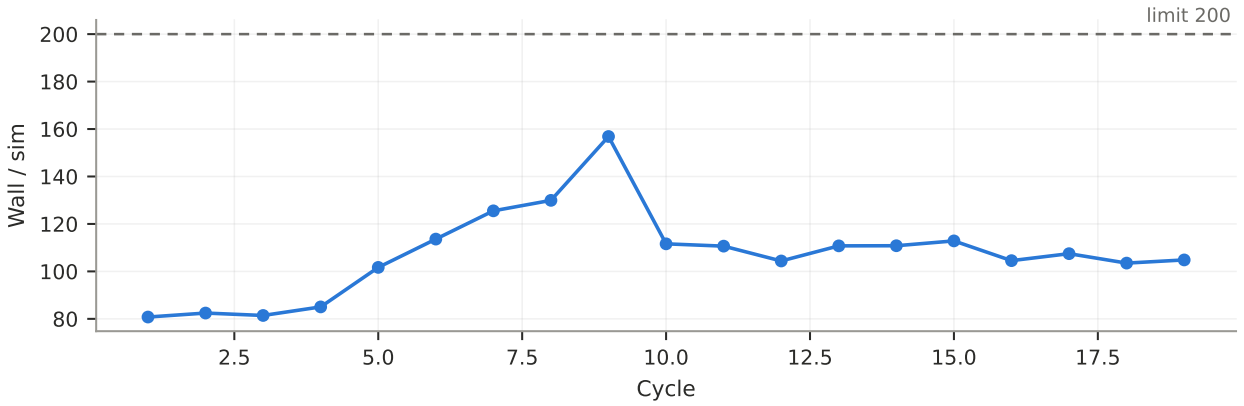


Figure 20: **Operating cost through the public interface while cutting (salmon side, 93,851 tets) (E16).** Wall seconds per simulated second for every recorded cycle of the public-interface loop, against the limit of 200.

8.8.5 Operating cost through the public interface while cutting (tenderloin, 149,862 tets) (E16)

The held knife brought to 4 mm above the pork tenderloin (task medallions) (two approach actions), then pressed 12 mm per 0.5 s action (with 6 mm, five presses did not cut the pork) and drawn along the blade, each action followed by an RGB observation, in process through the public gar API (contact collision profile). Cutting actions are those that made crack faces (gate 7 requires them); then ten 0.1 s strokes after the slices (the regime after a large tear). Every action waits up to 15 min for a quiet host (1-min load under 12, the GPU under 30 % busy with other work, no other process rendering; sampled before and after the action) and is marked quiet or not; gate 7 judges the quiet ones.

Observable	Recorded value
Robot	dual-fr3
Task	medallions
Cameras	overhead
Action length [s]	0.5
Startup: create + reset [s]	177.4
Actions that cut	6
Steady wall/sim, mean	90.58
Steady wall/sim, max	129.3
Non-cutting actions, mean wall/sim	64.51
Physics only wall/sim, mean	89.04
Quiet-host actions (gate 7)	2
Quiet-host wall/sim, mean	72.01
Quiet-host wall/sim, max	80.45
Mean of the steady actions under 200 (a cut stage's cutting actions, a hold's actions after the first; gate 7 judges the largest quiet one)	true
Clock agrees	true
Every RGB frame decoded	true
Tets	149 862

Every cycle as recorded (action then RGB observation).

Action	Wall [s]	Sim [s]	Wall / sim	Physics wall / sim	Load before (1 min)	GPU before [%]	Load after	GPU after [%]	Render work-ers	Cut faces	New crack faces	Topology [s]	Quiet host
approach_above	58.78	1	58.78	57.9	10.6	1	6.24	37	1	1 603	0	2.53	false
approach_surface	29.84	0.5	59.67	57.95	6.24	1	5.4	40.6	1	1 603	0	1.25	false
press_0	30.46	0.5	60.92	59.35	5.4	1	4.47	17	1	1 603	0	1.65	true
press_1	31.78	0.5	63.56	62.17	4.47	1	3.82	26.6	1	1 604	1	2.65	true
press_2	32.75	0.5	65.51	64.15	3.82	1	3.64	40.6	1	1 606	2	3.64	false
press_3	64.63	0.5	129.3	127.8	3.51	1	4	40.6	1	2 023	417	33.6	false
press_4	59.44	0.5	118.9	116.9	3.84	1	3.58	51.2	1	2 256	233	28.7	false
slice_0	42.89	0.5	85.79	84.27	3.58	1	3.72	40.6	1	2 277	21	13.2	false
slice_1	40.23	0.5	80.45	78.88	3.58	1	3.6	1	1	2 384	107	10.4	true
after_0	6.608	0.1	66.08	59.82	3.55	1	3.39	1	1	2 384	0	0.25	true
after_1	6.62	0.1	66.2	60.06	3.39	1	3.33	1	1	2 384	0	0.252	true
after_2	6.595	0.1	65.95	59.96	3.33	1	3.2	15.4	1	2 384	0	0.248	true
after_3	6.599	0.1	65.99	59.97	3.2	1	3.1	40.6	1	2 384	0	0.253	false
after_4	6.598	0.1	65.98	59.92	3.09	1	3.08	40.6	1	2 384	0	0.248	false
after_5	6.585	0.1	65.85	59.79	3.08	1	3.63	40.6	1	2 384	0	0.253	false
after_6	6.602	0.1	66.02	59.96	3.5	1	3.58	40.6	1	2 384	0	0.248	false
after_7	6.582	0.1	65.82	59.77	3.58	1	3.49	24.2	1	2 384	0	0.249	true
after_8	6.577	0.1	65.77	59.78	3.49	1	3.42	17.2	1	2 384	0	0.248	true
after_9	6.558	0.1	65.58	59.61	3.42	1	3.35	17.4	1	2 384	0	0.246	true

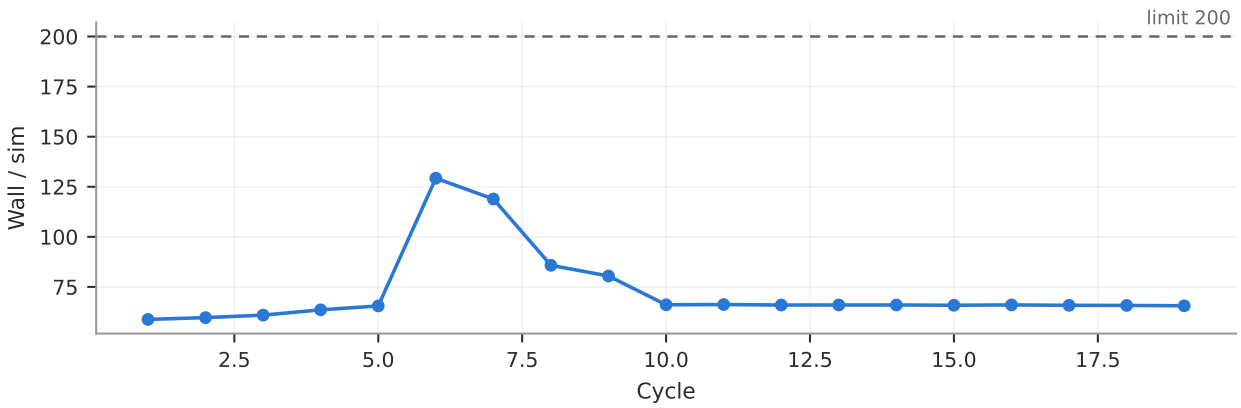


Figure 21: **Operating cost through the public interface while cutting (tenderloin, 149,862 tets) (E16).** Wall seconds per simulated second for every recorded cycle of the public-interface loop, against the limit of 200.

8.8.6 Operating cost with all four policy cameras (trout) (E16)

The trout cut of the cutting runtime stage with four policy cameras (overhead, supervisor, both wrists): 0.5 s actions, then ten 0.1 s strokes after the slices, each action followed by four RGB observations (8 samples per pixel, denoised: the observations' default). Cutting actions are those that made crack faces; every action waits up to 15 min for a quiet host and gate 7 judges the quiet ones.

Observable	Recorded value
Robot	dual-fr3
Task	fillet-one-side
Cameras	overhead, supervisor, left_wrist_rgb, right_wrist_rgb
Action length [s]	0.5
Startup: create + reset [s]	138.9
Actions that cut	13
Steady wall/sim, mean	127.2
Steady wall/sim, max	224.1
Non-cutting actions, mean wall/sim	98.31
Physics only wall/sim, mean	112.9
Quiet-host actions (gate 7)	8
Quiet-host wall/sim, mean	132.2
Quiet-host wall/sim, max	224.1
Mean of the steady actions under 200 (a cut stage's cutting actions, a hold's actions after the first; gate 7 judges the largest quiet one)	true
Clock agrees	true
Every RGB frame decoded	true
Tets	73 273

Every cycle as recorded (action then RGB observation).

Action	Wall [s]	Sim [s]	Wall / sim	Physics wall / sim	Load before (1 min)	GPU before [%]	Load after	GPU after [%]	Render work-ers	Cut faces	New crack faces	Topology [s]	Quiet host
approach_above	61.08	1	61.08	58.46	2.65	1.4	3.08	40.6	1	4 999	0	2.73	false
approach_surface	31.87	0.5	63.74	58.86	3.15	1	3.12	20.8	1	4 999	0	1.35	true
press_0	32.73	0.5	65.46	61	3.12	1	3.24	20.8	1	4 999	0	1.62	true
press_1	35.61	0.5	71.21	66.56	3.14	1	3.21	20.8	1	5 007	8	3.77	true
press_2	35.63	0.5	71.26	66.78	3.2	1	3.37	20.8	1	5 012	5	3.8	true
press_3	38.04	0.5	76.07	71.56	3.26	1	3.18	40.6	1	5 025	13	6.06	false
press_4	48.48	0.5	96.97	92.44	3.18	1	3.16	2.6	1	5 167	142	15.3	true
slice_0	55.87	0.5	111.7	107.3	3.07	1	3.36	40.6	1	5 444	277	22.3	false
slice_1	85.17	0.5	170.3	165.5	3.25	1	3.12	15	1	6 310	866	50.3	true
after_0	22.41	0.1	224.1	201.3	3.11	1	3.2	20.8	1	6 325	15	13.5	true
after_1	14.24	0.1	142.4	120.1	3.2	1	3.16	40.6	1	6 331	6	5.62	false
after_2	16.31	0.1	163.1	140.4	3.06	1	3.3	1	1	6 343	12	7.52	true
after_3	12.88	0.1	128.8	106.8	3.12	1	3.31	40.6	1	6 344	1	4.32	false
after_4	13.7	0.1	137	114.3	3.31	1	2.95	40.6	1	6 348	4	4.99	false
after_5	12.81	0.1	128.1	105.4	2.87	1	5.48	4.8	1	6 357	9	4.18	true
after_6	13.22	0.1	132.2	109.9	5.48	1	4.64	11.4	1	6 359	2	4.7	true
after_7	12.92	0.1	129.2	106.8	4.35	1	3.82	35	1	6 359	0	4.3	false
after_8	13.63	0.1	136.3	113.4	3.82	1	3.65	33.8	1	6 359	0	4.98	false
after_9	13.41	0.1	134.1	111.7	3.52	1	3.56	40.6	1	6 359	0	4.71	false

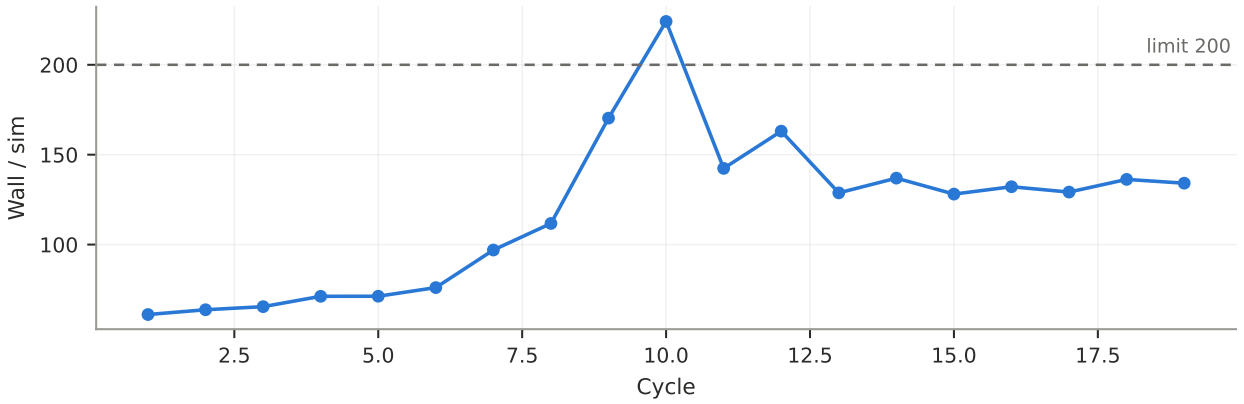


Figure 22: **Operating cost with all four policy cameras (trout) (E16).** Wall seconds per simulated second for every recorded cycle of the public-interface loop, against the limit of 200.

8.9 Interfaces (quality gate 8)

The GAR interface check: command line, HTTP with both keys, refusal of privileged state to the policy, checkpoint round trip, dashboard screenshots.

8.9.1 GAR interface check

CLI, HTTP with the owner and the policy key, refusal of privileged state to the policy, checkpoint round trip, dashboard screenshots.

Observable	Recorded value
Checks	cli_all_exit_0: true; dashboard_200: true; policy_action_200: true; policy_checkpoint_forbidden: true; policy_state_forbidden: true; owner_save_restore_200: true; dashboard_frames_in_browser: true; owner_geometry_200: true
Stopped cleanly	true

8.9.2 GAR filleting test suite

The filleting interface test suite: counts of its cases' outcomes.

Observable	Recorded value
All passed	true
Exit status	0
Passed	28
Failed	0
Errors	0
Skipped	0
Wall time [s]	895.7

9 Rendered evidence

The badge burned into a frame by the recorder (ASSISTED, UNASSISTED, SCRIPTED robot) is part of the evidence and is not edited.

9.1 Films of the recorded runs

Each run that records its state is filmed from its recording: every recorded frame is rendered by the path tracer from the recorded physics state, encoded, decoded back and counted, and review frames are written from the decoded video. Each figure below is a film's review sheet.

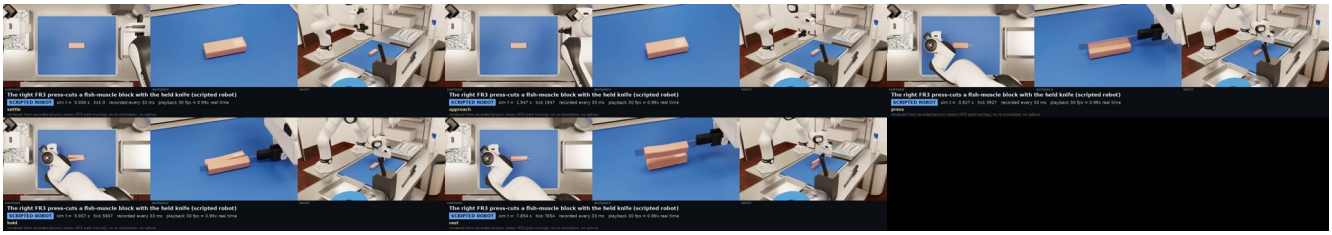


Figure 23: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): The right FR3 press-cuts a fish-muscle block with the held knife (scripted robot). 239 frames rendered, 239 decoded back from the MP4; 30 fps = 0.99x real time; ticks 0–7854; cameras overhead, workpiece, bench.

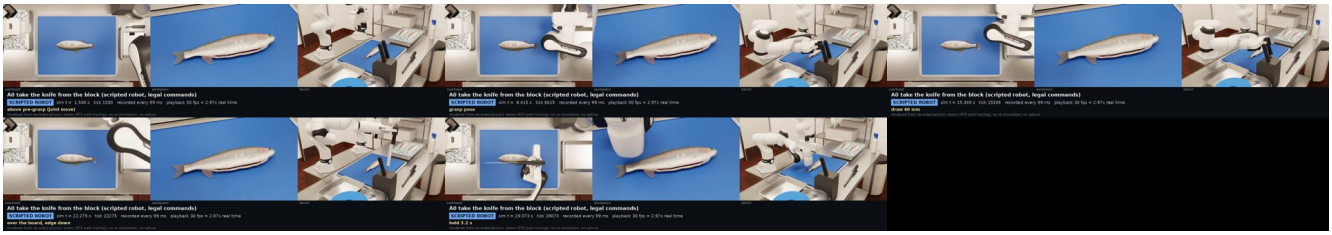


Figure 24: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): A0 take the knife from the block (scripted robot, legal commands). 280 frames rendered, 280 decoded back from the MP4; 30 fps = 2.97x real time; ticks 1500–29073; cameras overhead, workpiece, bench.

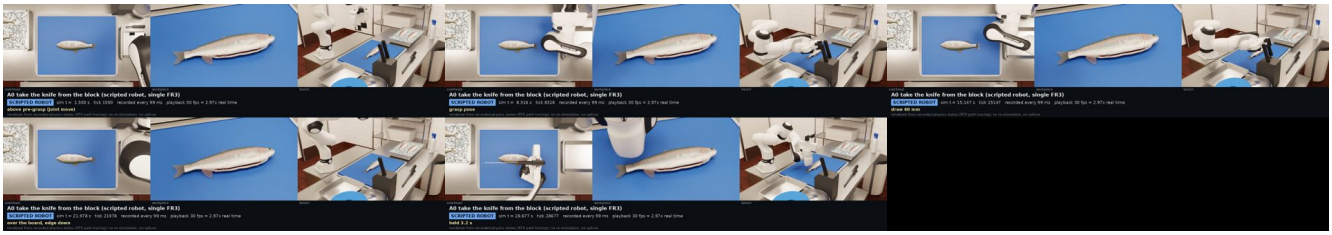


Figure 25: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): A0 take the knife from the block (scripted robot, single FR3). 276 frames rendered, 276 decoded back from the MP4; 30 fps = 2.97x real time; ticks 1500–28677; cameras overhead, workpiece, bench.

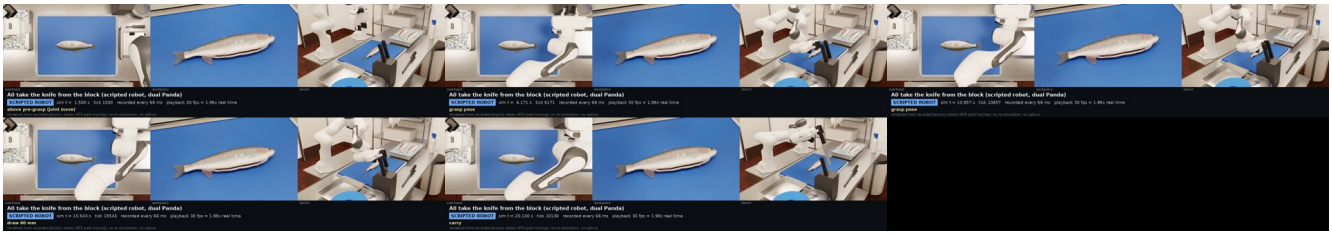


Figure 26: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): A0 take the knife from the block (scripted robot, dual Panda). 284 frames rendered, 284 decoded back from the MP4; 30 fps = 1.98x real time; ticks 1500–20130; cameras overhead, workpiece, bench.

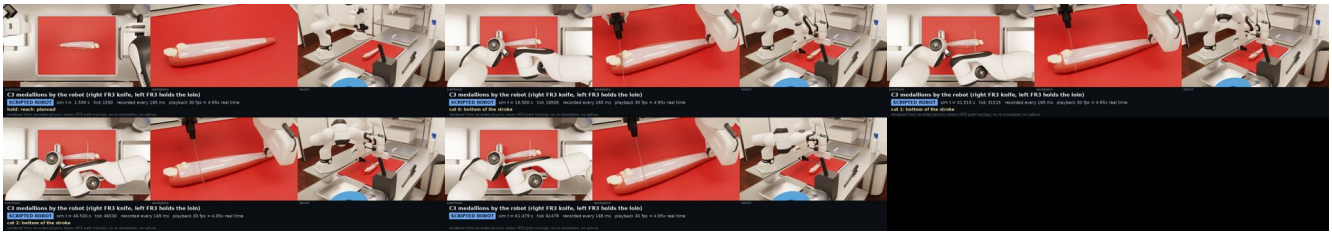


Figure 27: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): C3 medallions by the robot (right FR3 knife, left FR3 holds the loin). 365 frames rendered, 365 decoded back from the MP4; 30 fps = 4.95x real time; ticks 1500–61479; cameras overhead, workpiece, bench.

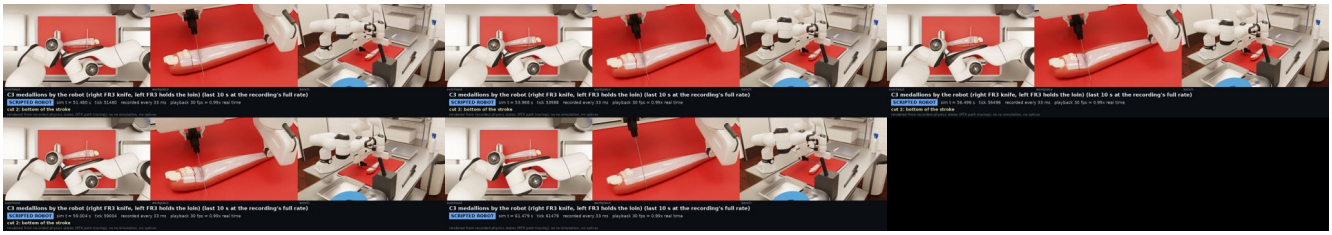


Figure 28: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): C3 medallions by the robot (right FR3 knife, left FR3 holds the loin) (last 10 s at the recording's full rate). 304 frames rendered, 304 decoded back from the MP4; 30 fps = 0.99x real time; ticks 51480–61479; cameras overhead, workpiece, bench.



Figure 29: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): A1 head-off by the robot (right FR3 knife, left FR3 holds the tail). 148 frames rendered, 148 decoded back from the MP4; 30 fps = 0.99x real time; ticks 1500–6336; cameras overhead, workpiece, bench.

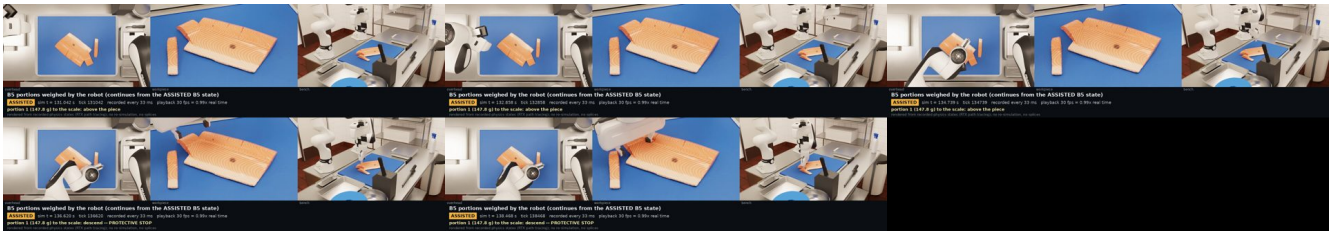


Figure 30: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): B5 portions weighed by the robot (continues from the ASSISTED B5 state). 227 frames rendered, 227 decoded back from the MP4; 30 fps = 0.99x real time; ticks 131042–138468; cameras overhead, workpiece, bench.

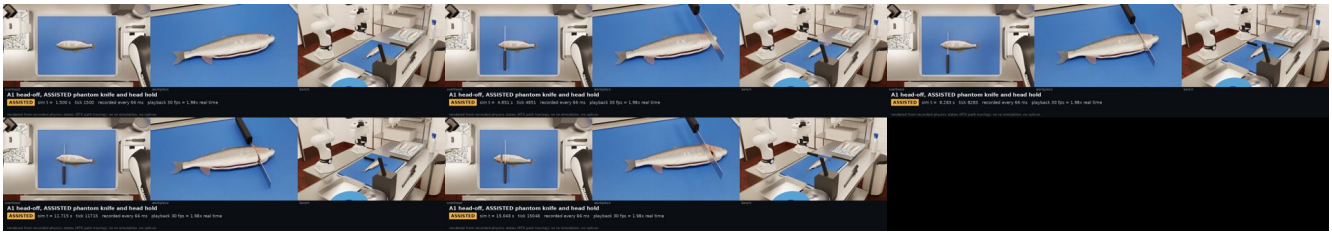


Figure 31: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): A1 head-off, ASSISTED phantom knife and head hold. 207 frames rendered, 207 decoded back from the MP4; 30 fps = 1.98x real time; ticks 1500–15048; cameras overhead, workpiece, bench.

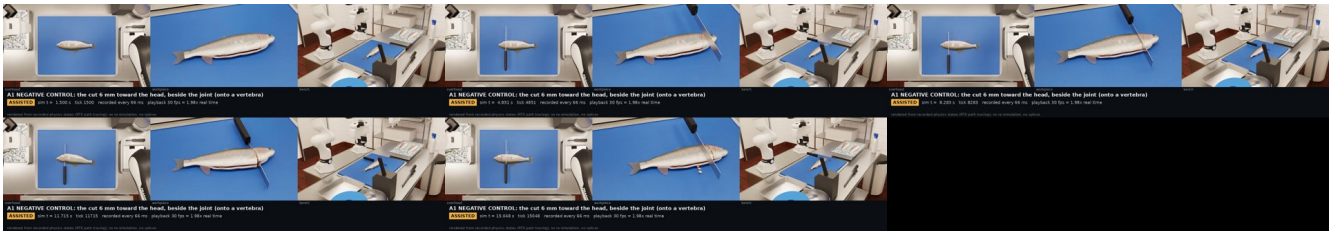


Figure 32: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): A1 NEGATIVE CONTROL: the cut 6 mm toward the head, beside the joint (onto a vertebra). 207 frames rendered, 207 decoded back from the MP4; 30 fps = 1.98x real time; ticks 1500–15048; cameras overhead, workpiece, bench.

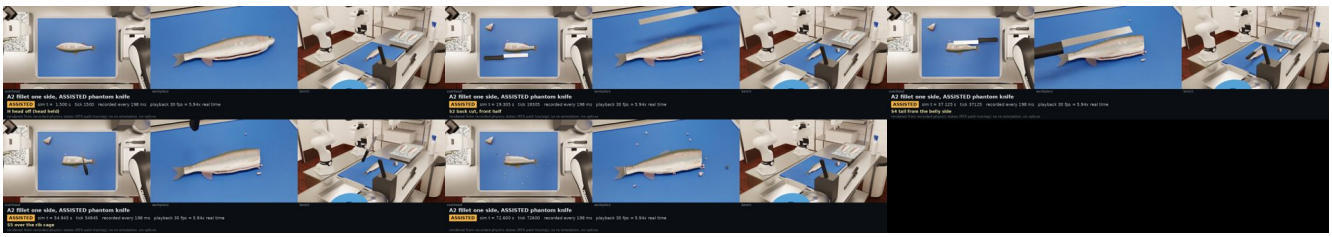


Figure 33: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): A2 fillet one side, ASSISTED phantom knife. 361 frames rendered, 361 decoded back from the MP4; 30 fps = 5.94x real time; ticks 1500–72600; cameras overhead, workpiece, bench.

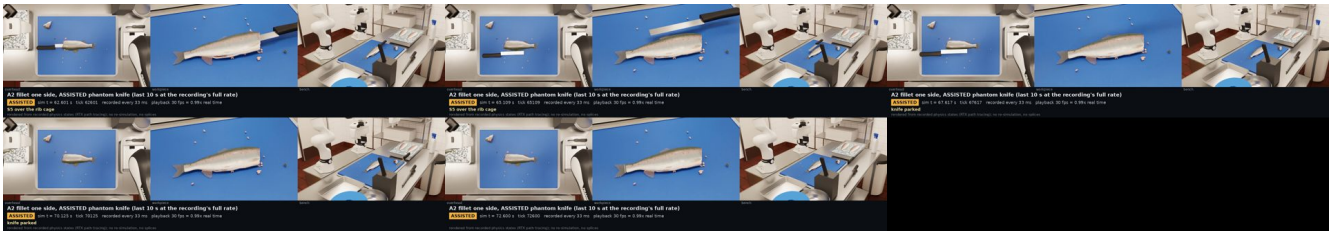


Figure 34: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): A2 fillet one side, ASSISTED phantom knife (last 10 s at the recording's full rate). 304 frames rendered, 304 decoded back from the MP4; 30 fps = 0.99x real time; ticks 62601–72600; cameras overhead, workpiece, bench.

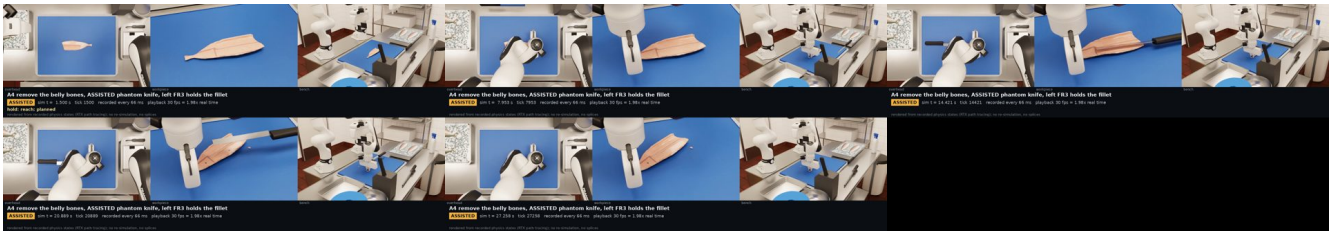


Figure 35: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): A4 remove the belly bones, ASSISTED phantom knife, left FR3 holds the fillet. 392 frames rendered, 392 decoded back from the MP4; 30 fps = 1.98x real time; ticks 1500–27258; cameras overhead, workpiece, bench.

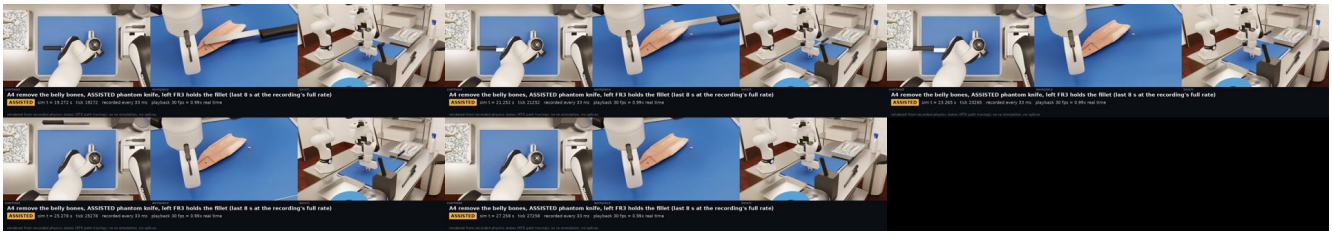


Figure 36: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): A4 remove the belly bones, ASSISTED phantom knife, left FR3 holds the fillet (last 8 s at the recording's full rate). 243 frames rendered, 243 decoded back from the MP4; 30 fps = 0.99x real time; ticks 19272–27258; cameras overhead, workpiece, bench.

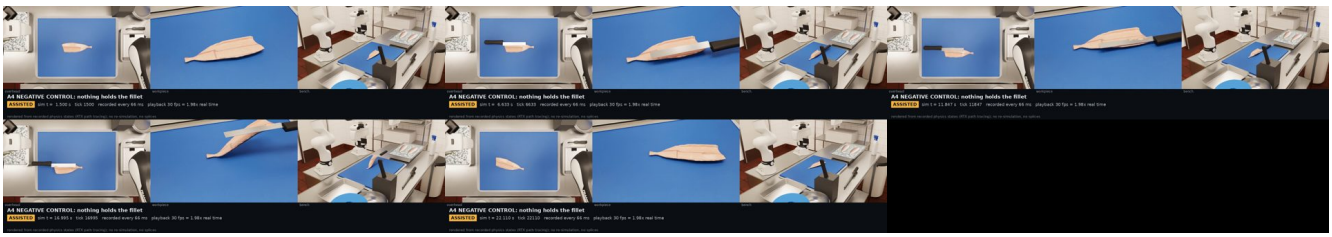


Figure 37: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): A4 NEGATIVE CONTROL: nothing holds the fillet. 314 frames rendered, 314 decoded back from the MP4; 30 fps = 1.98x real time; ticks 1500–22110; cameras overhead, workpiece, bench.

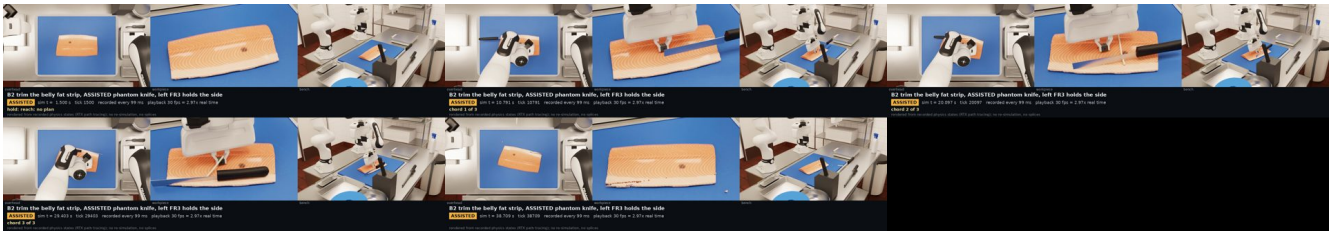


Figure 38: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): B2 trim the belly fat strip, ASSISTED phantom knife, left FR3 holds the side. 377 frames rendered, 377 decoded back from the MP4; 30 fps = 2.97x real time; ticks 1500–38709; cameras overhead, workpiece, bench.

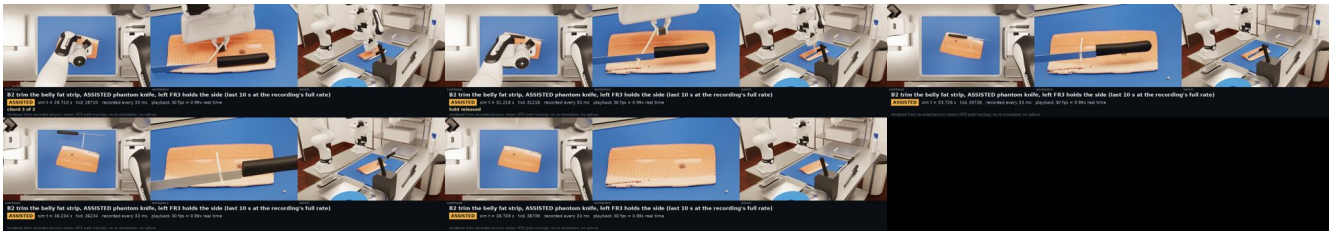


Figure 39: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): B2 trim the belly fat strip, ASSISTED phantom knife, left FR3 holds the side (last 10 s at the recording's full rate). 304 frames rendered, 304 decoded back from the MP4; 30 fps = 0.99x real time; ticks 28710–38709; cameras overhead, workpiece, bench.

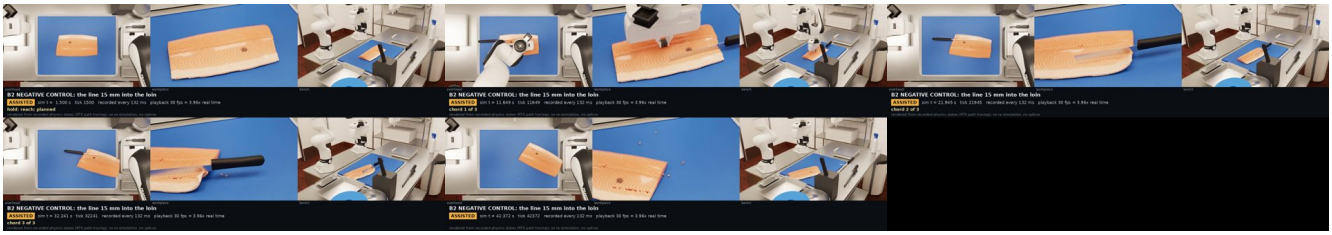


Figure 40: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): B2 NEGATIVE CONTROL: the line 15 mm into the loin. 311 frames rendered, 311 decoded back from the MP4; 30 fps = 3.96x real time; ticks 1500–42372; cameras overhead, workpiece, bench.

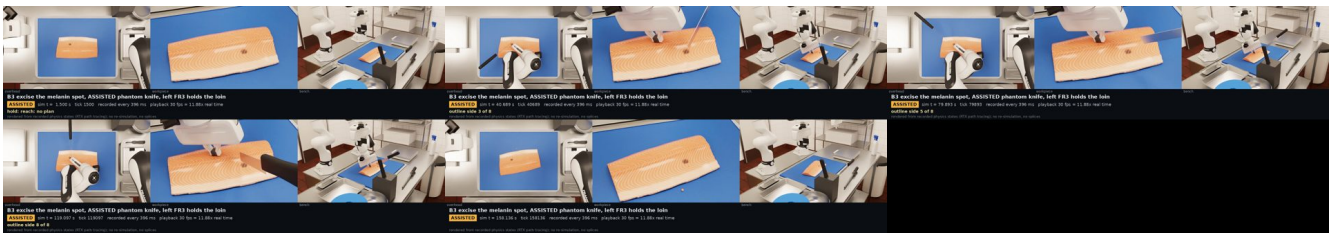


Figure 41: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): B3 excise the melanin spot, ASSISTED phantom knife, left FR3 holds the loin. 397 frames rendered, 397 decoded back from the MP4; 30 fps = 11.88x real time; ticks 1500–158136; cameras overhead, workpiece, bench.

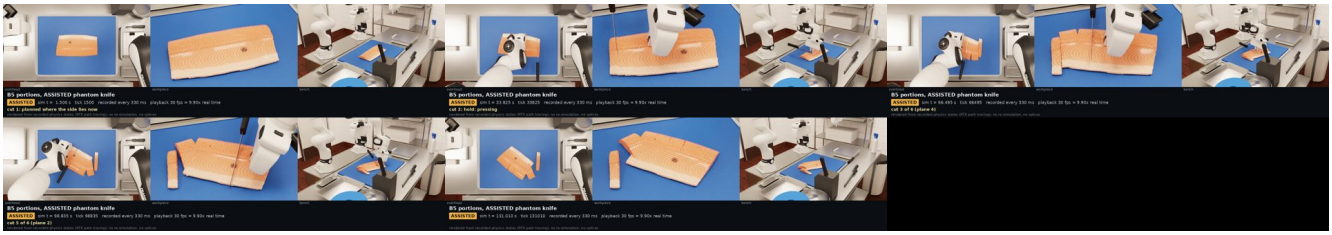


Figure 42: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): B5 portions, ASSISTED phantom knife. 394 frames rendered, 394 decoded back from the MP4; 30 fps = 9.90x real time; ticks 1500–131010; cameras overhead, workpiece, bench.

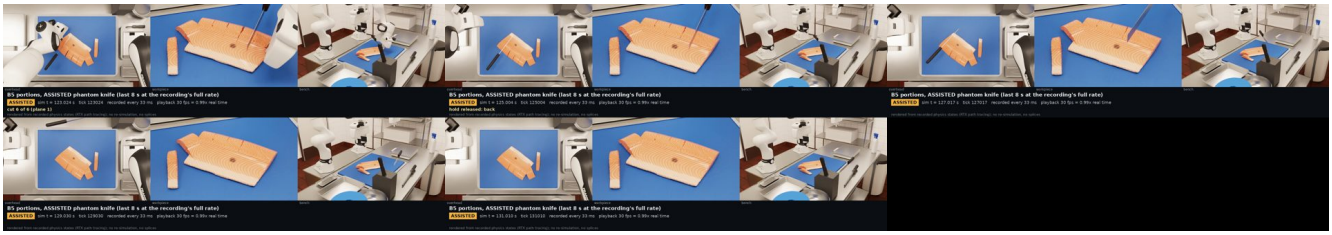


Figure 43: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): B5 portions, ASSISTED phantom knife (last 8 s at the recording's full rate). 243 frames rendered, 243 decoded back from the MP4; 30 fps = 0.99x real time; ticks 123024–131010; cameras overhead, workpiece, bench.



Figure 44: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): B5 NEGATIVE CONTROL: cuts tilted 20 deg. 394 frames rendered, 394 decoded back from the MP4; 30 fps = 9.90x real time; ticks 1500–131010; cameras overhead, workpiece, bench.

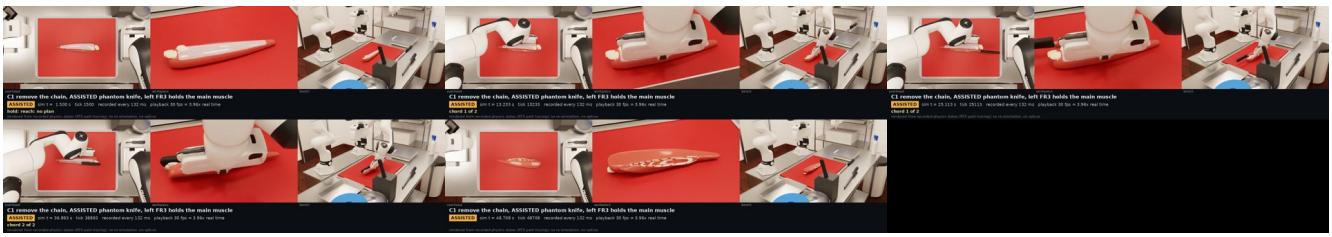


Figure 45: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): C1 remove the chain, ASSISTED phantom knife, left FR3 holds the main muscle. 359 frames rendered, 359 decoded back from the MP4; 30 fps = 3.96x real time; ticks 1500–48708; cameras overhead, workpiece, bench.



Figure 46: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): C1 remove the chain, ASSISTED phantom knife, left FR3 holds the main muscle (last 8 s at the recording's full rate). 243 frames rendered, 243 decoded back from the MP4; 30 fps = 0.99x real time; ticks 40722–48708; cameras overhead, workpiece, bench.

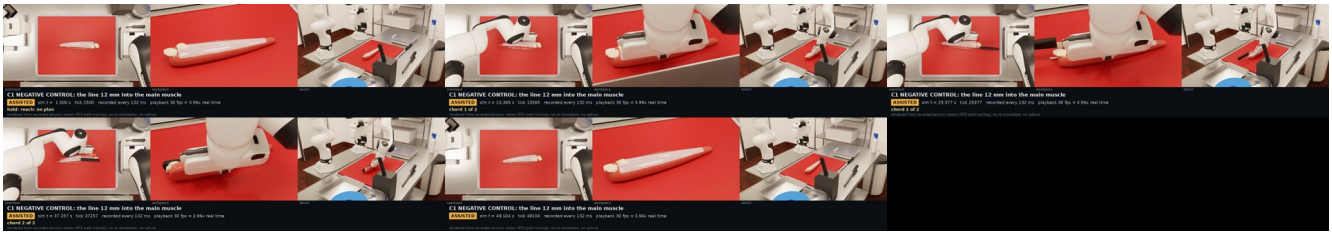


Figure 47: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): C1 NEGATIVE CONTROL: the line 12 mm into the main muscle. 362 frames rendered, 362 decoded back from the MP4; 30 fps = 3.96x real time; ticks 1500–49104; cameras overhead, workpiece, bench.

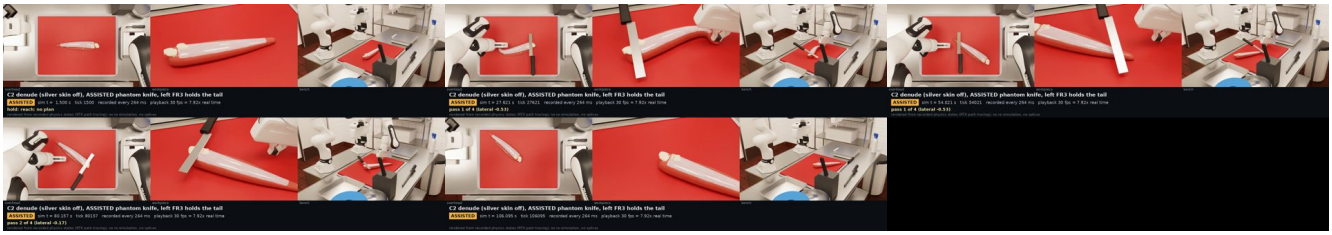


Figure 48: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): C2 denude (silver skin off), ASSISTED phantom knife, left FR3 holds the tail. 398 frames rendered, 398 decoded back from the MP4; 30 fps = 7.92x real time; ticks 1500–106095; cameras overhead, workpiece, bench.



Figure 49: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): C2 denude (silver skin off), ASSISTED phantom knife, left FR3 holds the tail (last 8 s at the recording's full rate). 243 frames rendered, 243 decoded back from the MP4; 30 fps = 0.99x real time; ticks 98109–106095; cameras overhead, workpiece, bench.

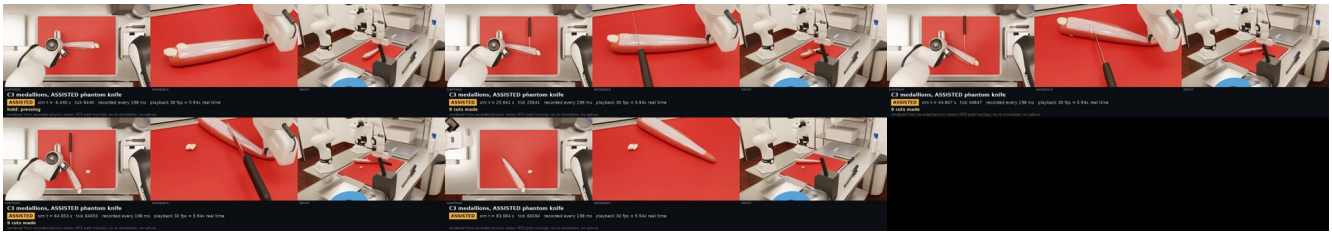


Figure 50: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): C3 medallions, ASSISTED phantom knife. 389 frames rendered, 389 decoded back from the MP4; 30 fps = 5.94x real time; ticks 6440–83094; cameras overhead, workpiece, bench.

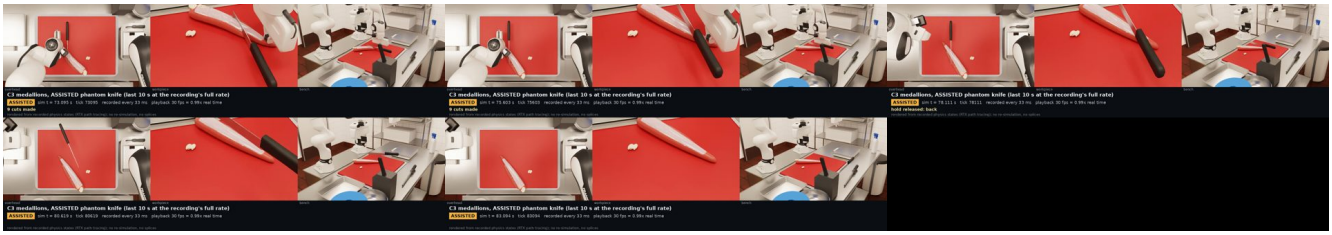


Figure 51: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): C3 medallions, ASSISTED phantom knife (last 10 s at the recording's full rate). 304 frames rendered, 304 decoded back from the MP4; 30 fps = 0.99x real time; ticks 73095–83094; cameras overhead, workpiece, bench.



Figure 52: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): C3 NEGATIVE CONTROL: cuts 10 mm off plan. 388 frames rendered, 388 decoded back from the MP4; 30 fps = 5.94x real time; ticks 6440–83028; cameras overhead, workpiece, bench.

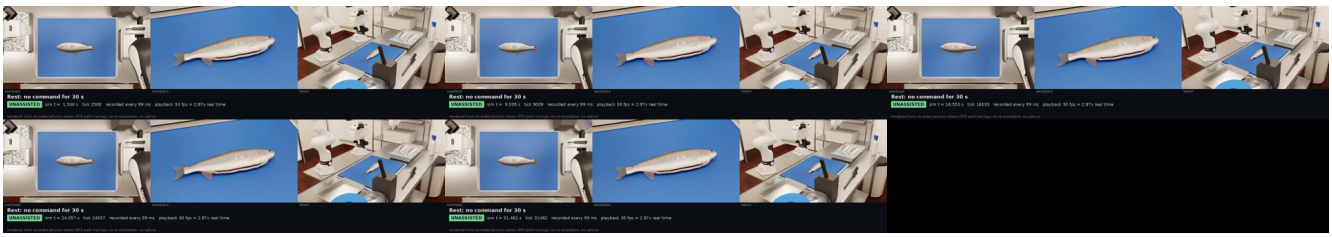


Figure 53: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): Rest: no command for 30 s. 304 frames rendered, 304 decoded back from the MP4; 30 fps = 2.97x real time; ticks 1500–31482; cameras overhead, workpiece, bench.

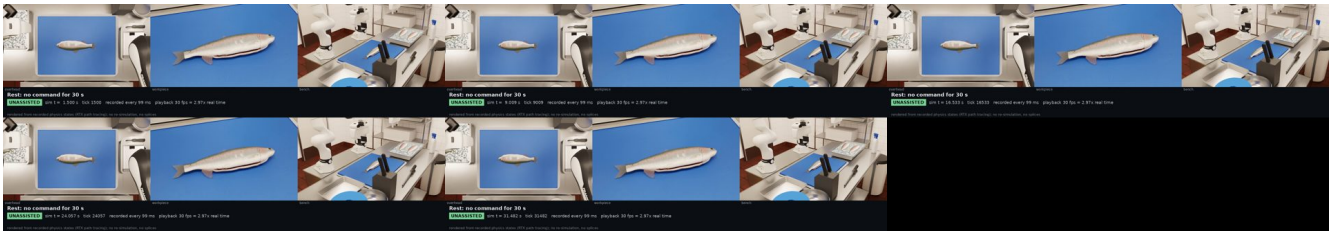


Figure 54: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): Rest: no command for 30 s. 304 frames rendered, 304 decoded back from the MP4; 30 fps = 2.97x real time; ticks 1500–31482; cameras overhead, workpiece, bench.

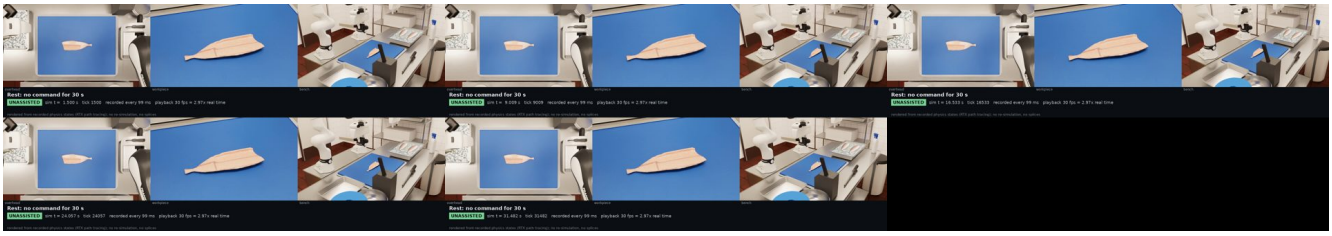


Figure 55: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): Rest: no command for 30 s. 304 frames rendered, 304 decoded back from the MP4; 30 fps = 2.97x real time; ticks 1500–31482; cameras overhead, workpiece, bench.

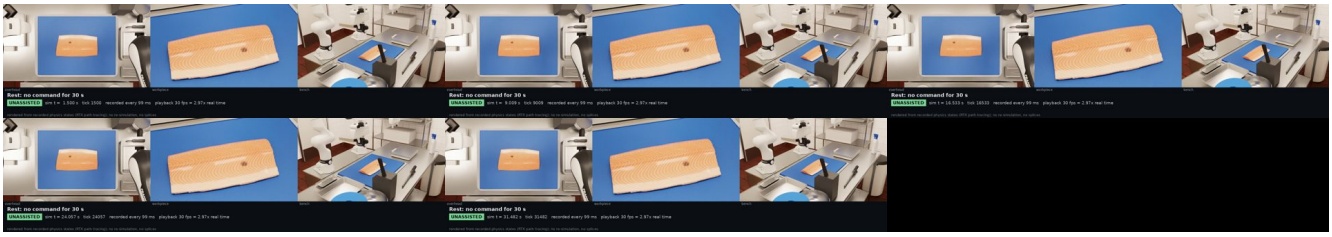


Figure 56: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): Rest: no command for 30 s. 304 frames rendered, 304 decoded back from the MP4; 30 fps = 2.97x real time; ticks 1500–31482; cameras overhead, workpiece, bench.

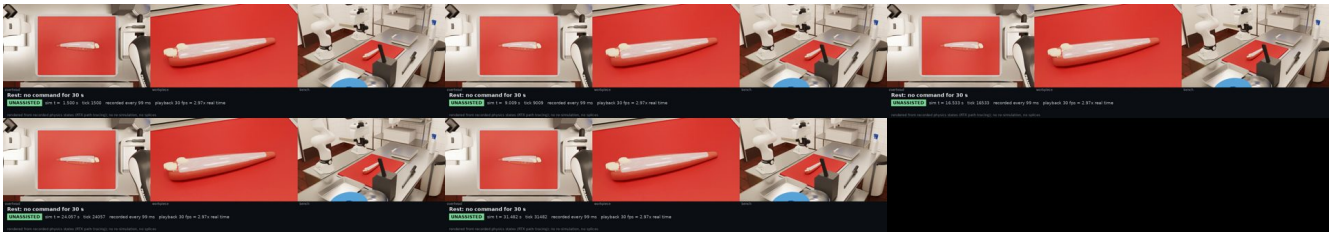


Figure 57: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): Rest: no command for 30 s. 304 frames rendered, 304 decoded back from the MP4; 30 fps = 2.97x real time; ticks 1500–31482; cameras overhead, workpiece, bench.

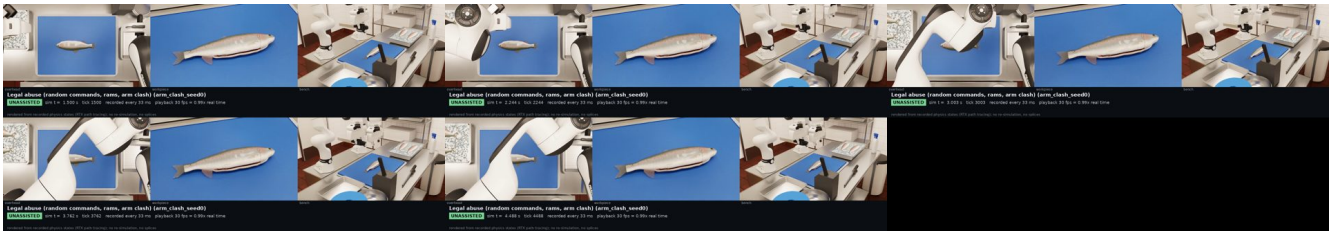


Figure 58: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): Legal abuse (random commands, rams, arm clash) (arm_clash_seed0). 92 frames rendered, 92 decoded back from the MP4; 30 fps = 0.99x real time; ticks 1500–4488; cameras overhead, workpiece, bench.

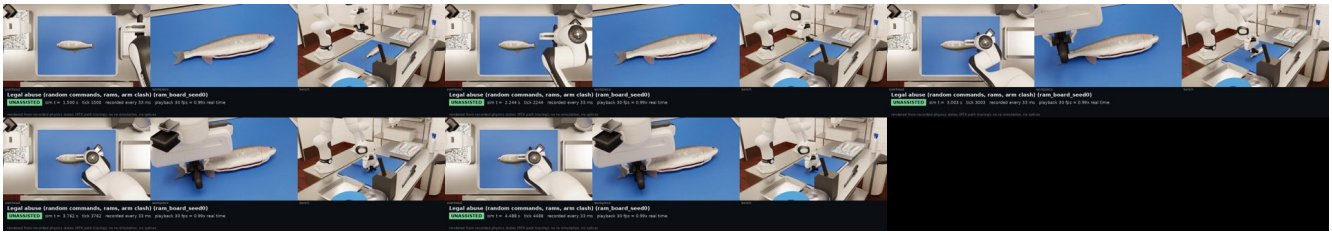


Figure 59: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): Legal abuse (random commands, rams, arm clash) (ram_board_seed0). 92 frames rendered, 92 decoded back from the MP4; 30 fps = 0.99x real time; ticks 1500–4488; cameras overhead, workpiece, bench.

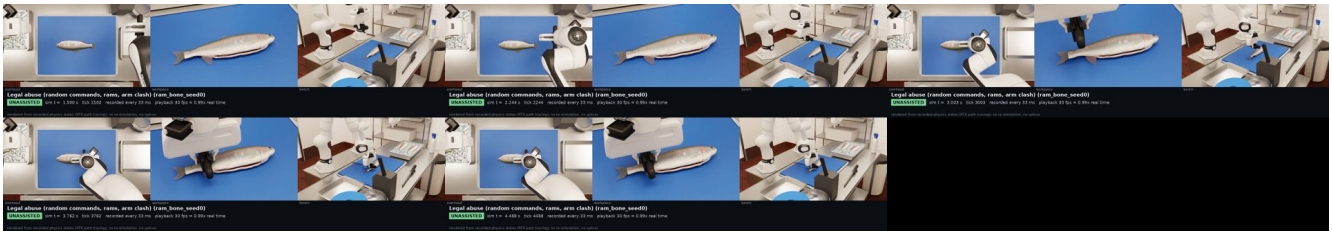


Figure 60: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): Legal abuse (random commands, rams, arm clash) (ram_bone_seed0). 92 frames rendered, 92 decoded back from the MP4; 30 fps = 0.99x real time; ticks 1500–4488; cameras overhead, workpiece, bench.

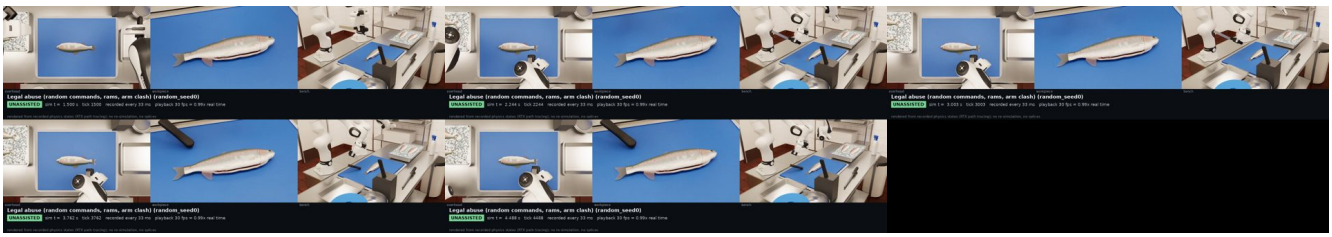


Figure 61: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): Legal abuse (random commands, rams, arm clash) (random_seed0). 92 frames rendered, 92 decoded back from the MP4; 30 fps = 0.99x real time; ticks 1500–4488; cameras overhead, workpiece, bench.

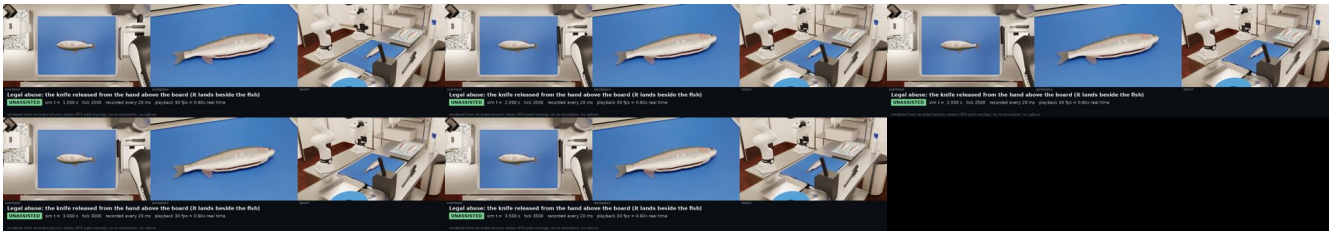


Figure 62: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): Legal abuse: the knife released from the hand above the board (it lands beside the fish). 101 frames rendered, 101 decoded back from the MP4; 30 fps = 0.60x real time; ticks 1500–3500; cameras overhead, workpiece, bench.

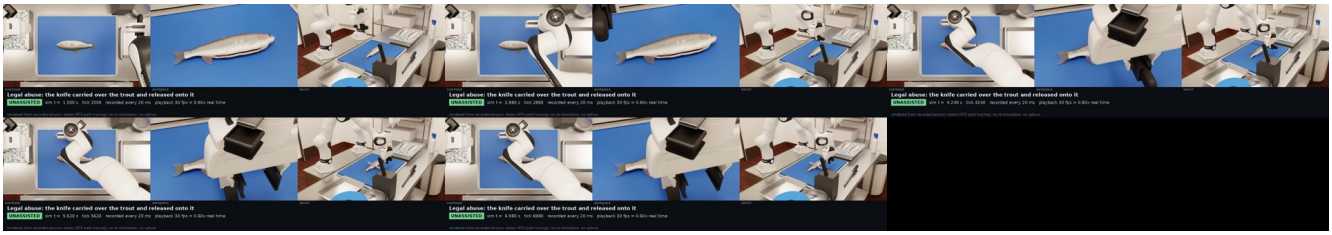


Figure 63: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): Legal abuse: the knife carried over the trout and released onto it. 275 frames rendered, 275 decoded back from the MP4; 30 fps = 0.60x real time; ticks 1500–6980; cameras overhead, workpiece, bench.

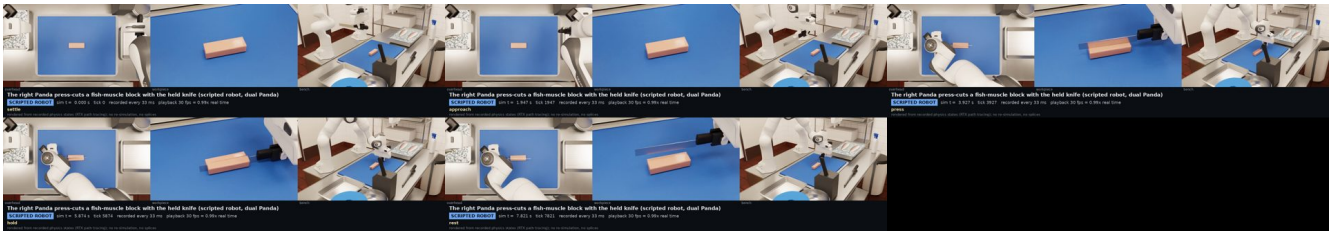


Figure 64: Review sheet (decoded frames at 0, 25, 50, 75 and 100 %): The right Panda press-cuts a fish-muscle block with the held knife (scripted robot, dual Panda). 238 frames rendered, 238 decoded back from the MP4; 30 fps = 0.99x real time; ticks 0–7821; cameras overhead, workpiece, bench.

10 Measured laws of this solver stack

The behaviours below were measured on Newton 1.5 with MuJoCo at a 1 ms tick, coupled to this environment's explicit GPU tissue solver. Each is a law of the tools, not of fish, and each sets a design rule of the world described above. Items marked *standing* are engine limits that remain in the world and are disclosed in Section 11; the others state the rule the world follows and what happens without it.

10.1 Coupling between the two solvers

1. **The explicit coupling budget.** Every coupling that crosses the boundary between MuJoCo's 1 ms tick and the tissue solver is an explicit spring, and it is stable only when $k\,\Delta t_r^2/m \ll 1$. Each failure measured had $k\,\Delta t_r^2/m \geq 1$: a light blade segment against flank contacts, backbone segments bonded to many flesh nodes, rib roots on a segment, and a released handle against the blade chain's stiff root joint. Light, stiffly coupled bodies therefore live in the substepped tissue solver, and MuJoCo keeps the arms and the knife as one rigid body. *Standing design rule.*

2. **A filter on an explicit reaction is a Maxwell element.** At high frequency a low-pass on a reaction acts as a damper $c = k\tau$ with its own explicit limit $c\,\Delta t/m < 2$; on the blade it makes the oscillation worse.
3. **One-tick-lagged contact on a light finger is unstable.** Fingers that meet a free knife only through the tissue solver, one tick late, build a large force and drive MuJoCo non-finite during a legal grasp. Fingers therefore meet the knife in MuJoCo, where a free knife's copy is kinematic. *Standing*: a finger cannot push a free knife in MuJoCo.
4. **Two copies of one knife fling a closing hand.** If any unpinched knife belongs to the tissue solver, a hand closing on the knife in the block meets both copies at once. The knife has one owner at a time (Section 5.2).
5. **A kinematic body touching static geometry resets MuJoCo.** An immovable body pressed into an immovable body has no solution; MuJoCo answers with unbounded accelerations and resets its state. Outside the held mode the knife's MuJoCo geometry collides with the robots only, and the world faults on MuJoCo's instability warnings.
6. **A hand-back must check overlap with the static world.** A rigid blade handed back to MuJoCo inside the block's slot starts inside the slot walls; the hand-back gate measures signed distances to the robot and to the static geometry.

10.2 Tissue solver

7. **Isosurface stuffing removes the mesher's slivers**, and with them the spurious dissipation of inverted slivers under a pressed edge.
8. **A discontinuous contact rule is an energy source**: a force that jumps when a node changes side injects energy at every event.
9. **Thin stiff layers need plane stress, real gradients and their own damping time**; each of the three errors alone diverges the trout's skin within a fraction of a millisecond.
10. **Kick tests expose non-conservative forces** that a static test hides.
11. **Stability mass is inertia, never weight.** *Standing*: the scaled nodes answer fast loads more slowly than real tissue.
12. **The cohesive law's dissipation is the spent fraction, not the damage variable** (Section 4.2).
13. **A split that costs nothing is a free cut**: geometric release of bonds, ties and joints must be paid for by the blade (Section 4.3). *Standing*: the release itself is geometric.
14. **The flank friction probe does not isolate friction with tearing on.** The blade sawn flat on the block, its edge outside it, tears the top layer (58 tears, 151 faces), and the ledger misses $-9.7\,\%$ of the work (Section 8). *Open.*
15. **Rest dissipation without a source**: light rod and surface nodes vibrate at kilohertz frequencies with micrometre amplitude at rest; the rods' part is float32 rounding (see below), the rest stands. *Standing.*
16. **Squeezed fragments can be ejected** by explicit penalty contact between the blade and the board. *Standing*; the floor catches them.
17. **A new sheet next to an old one makes crumbs**, which stay attached across the new sheet.

10.3 Robot and control

18. **An instant speed step overshoots the velocity reflex**; an acceleration-limited generator does not (joint step probe, Section 8).
19. **The hand's force mode must latch only when blocked.**
20. **The grip holds with the qualified MuJoCo contact settings** and creeps without them (grip probes, Section 8).

21. **Streamed Cartesian motion trails its path.** A hand following IK targets through the rate-limited generator trails the streamed pose; approaching the knife handle too fast, the trailing hand meets the handle with an insert lip and trips a protective stop. Every streamed move logs its largest trailing distance, and the approach to the handle runs at 12 mm/s.

10.4 Energy, precision and contact

22. **The joint torque must be the derivative of the joint energy.** A small-angle form errs by 1.3 % at 2° and 26 % at 30°; the world uses the exact form.
23. **Float32 world coordinates quantise slow motion** at bench height: positions are advanced with compensated summation and a rod node's velocity is carried as its own displacement.
24. **A rib's rest curvature turns with the flesh or the hinge it belongs to.** As a world-frame vector, a rigidly rotated rib bends itself up to 30 % of its breaking moment.
25. **Releasing a link while it is held open must not count its stored energy twice.**
26. **Dampers must not resist rigid rotation.** The membrane, bond and tie dashpots damp only deformation.
27. **The rod projection's loss of kinetic energy is booked** in the ledger.
28. **The knife creeps in the grip under fast wrist motion.** With the stiffest stable insert contact time constant (2 ms) the creep per swing is 1.2 mm, against 2.8 mm with contacts as soft as rubber pads. *Standing*: the residual creep under extreme wrist motion.
29. **The slice push exceeds Atkins' frictionless prediction by a constant** of about 1.5–2.6 N (2.57 N in the recorded run), the bevel wedging the tissue apart; the total force is within 6 %. *Standing*; the slice gate stays failed.
30. **A crumb rule that rejoins a crumb to both sides bridges the cut**; a crumb with no neighbour on its own side joins one side.
31. **MuJoCo's box-box collider reports a deep false contact when a contact gap is set.** During a legal wrist swing the handle's grip box and an insert lip, 0.25 mm apart, give two contacts 38.8 mm deep in MuJoCo 3.11: the solver pushes with 5 kN and the collision reflex reads 254 N m. Two bare boxes at the same poses reproduce it with a 2 mm gap and the lip listed first. A gap only adds inactive contacts in MuJoCo, so every shape has none: the active contacts are unchanged, the generated contacts per tick drop from about 480 to 8, and the swing stops once, physically, when the blade tip strikes the arm's own forearm.
32. **A centroid-segment crossing test cannot see a cut sheet that runs through the centroids.** On the 2.5 mm coupon the cut plane is a lattice plane; the recorded sheet weaves ±0.3 mm across a row of centroids, and faces between two tets of that row are never crossed. Faces with a tet on the sheet are decided by the tets' sides of it. In the recorded run the 2.5 mm press keeps 3 faces of the bottom layer whole 1.0–1.7 mm up and stays one piece: the separation of the last layer under the edge is marginal (Section 8).
33. **Rigid bodies need compensated summation too.** With plain summation a servoed handle pressing 36 N reports 1 mm/s for 0.4 s while its height stays within 0.1 µm, and its damper books work against a body that does not move (a 6 % hole in the ledger). With compensated summation of rigid positions and orientations the press closes to −0.8 %.
34. **A friction dashpot stiffer than its body's explicit limit chatters.** Many samples of a 3 g blade segment sticking on the board at once exceed what an explicit step can carry; the sticking velocity flips every substep and friction books energy no force supplied: 18.4 J against 0.05 J of work in one second with the rig's blade pressed into the board. Each sample's dashpot is capped at half its body's mass per sample per substep; the same probe books 1.1 mJ (residual −0.1 %; +0.02 % in the recorded run) with the rig's servo held at its 150 N cap.
35. **An assisted rig that follows a pose with unbounded force can break the solver.** Driven 6 mm into the held head (the A1 control), a pose-following rig ends in non-finite MuJoCo loads. The rig is a force-limited servo on the handle (150 N, 15 N m, authored); with it and the friction cap the control runs to its end.

36. **A projection that books the kinetic energy it removes books a static load as dissipation.** Booked that way, the thin bones' inextensibility projection counts the velocity increment of a steady load, cancelled within the same substep, as dissipated ($F^2\,\Delta t^2/2m$ per substep): 12.9 W over pin bones loaded by their bonds beside the A1 control's cut, with nothing moving, and a held root's work as negative dissipation. It books the explicit forces' and the root's work along the actual displacement less the kinetic energy gained; the same stretch closes to −0.5 % of the work with the energy stored outside the tets counted.

37. **A bone whose bond to the flesh has failed must still move with the flesh around it.** Without the flesh's drag, a pin's first free node swings round its root at up to 10.7 m/s inside the flesh. The drag stays while the bone is within about one element of its material point: 2.9 m/s at most from the same state.

38. **A velocity dashpot cannot hold static friction.** Capped for explicit stability, a friction dashpot on the rigid samples gives the rig's pressed blade an effective μ of 0.01 against 0.25 at 20 mm/s, and a cap that ignores rotation leaves bones and blade segments 2.5–6.5 times over the explicit limit. Stick-slip anchors capped by each sample's effective mass at the contact point reproduce $\mu = 0.25$ with the ledger closed to −0.011 %.

39. **A stiff damper sized on a stability inertia must not reach a lighter axis.** A blade hinge bent 28 deg flat-wise turns part of an edge-wise damper sized on an inflated inertia onto the next segment's twist axis; the twist grows to 4800 rad/s in 23 ms and the A2 run goes non-finite. Sized on the twist inertia, the damper keeps every segment below 1.9 rad/s from the same state.

40. **An explicit step bound must count the damping.** A bound that holds $K\Delta t^2 \le 2.25\,m$ and leaves the Kelvin–Voigt damping out lets stretched collar flesh in A1's control reach $c\,\Delta t = 2.04$, and the run goes non-finite. The bound holds $K\Delta t^2 + 2C\Delta t \le 2.25\,m$ with sound bounds of both.

41. **Flesh tears where it is pulled apart.** The sheet and tensile laws with fixed-plane cracks and extrinsic cohesive bonds (Section 4.4) break a notched bar at a net-section stress 1.24 and 1.05 times the governing strength (0.82 and 0.70 on the gross section, across and along the sheets) with the ledger closed to 1.1 %; a per-face test opens no face in a bar stretched 4 times.

42. **A numerical damping must not stiffen with stretch.** On the Green strain rate the damping's tangent grows as the stretch squared; beside A1's cut the step's demand reaches 3212 substeps, and run at the cap the damper's overshoot books 0.81 J of false dissipation in 0.1 s. Normalised by $3/\max(3, I_C)$, the cut closes to a few mJ per second against 113–307 mJ of work.

43. **A body's own springs belong in its stored energy, not in its work.** A rod projection that books its springs' start-of-step work, which exceeds their energy change by $\frac{1}{2}dx^T H dx$ per step, books hundreds of times a flailing rib's kinetic energy as loss every 0.1 s.

44. **A start state must be at rest, and the flesh's own damping does not bring a trout there in 1.5 s.** The numerical Kelvin–Voigt term is a damping ratio of about $\omega\tau/2 \approx 0.004$ at a 13 Hz mode: after a plain 1.5 s hands-off settle the trout's last vertebra still swings 258 μm peak-to-peak (e-fold ≈ 2.7 s), and a 2 s rest moves tissue nodes 421 μm. The settle therefore relaxes its first 1.2 s (velocities scaled by $e^{-80\Delta t}$ per tick, the energy removed booked); left alone for 3 s from that start the tissue moves 0.19 μm (median), 5.0 μm at most, the bones 0.55 μm.

45. **MuJoCo's no-slip post-solver stops at its tolerance.** At the default 1e-6 it stops after about nine iterations whatever the cap (10, 50 and 200 give bit-identical holds), and the held knife turns in the grip under its own weight at 0.0012 deg/s, its tip 0.14 mm in 30 s. At 1e-10 it takes about 31 iterations and the turn is a thousand times smaller for under 1 ms more per tick: in the recorded rests the knife moves 0.2 μm in 30 s.

46. **A projection must not re-evaluate its constraints on rounded positions.** Read from float32 positions at bench height (one spacing is 6e-8 m), the rods' projection, bending and ties correct each position's rounding as if it were deformation, which becomes velocity (6 mm/s per spacing per substep): the rod nodes jitter at 4.6 mm/s rms and the rest drains 1.98 mW. Evaluated on true position differences the rods jitter at 0.1 mm/s and the drain is 0.05–0.84 mW across the presets. A free rod's internal oscillation still grows from rest (momentum exact); in the fish the ties damp it.

47. **The tearing law must not act in the knife edge's process zone.** The edge law loads single material points, so the stress beside the edge is the mesh's; tearing there cuts faces whose cohesive bonds survive the

blade off its slab and hold cut halves together (a coupon press cut stays one piece through 17 such bonds). Skipping the elements within 1.5 mean edge lengths of the edge, the same cuts divide; the robot press cut divides at three speeds.

48. **A separated half can still ride up on the knife.** After a press cut through an unheld block that slid 12 mm under the blade's lateral load, one half rises about 50 mm with the withdrawn knife, carried by it; the mechanism is not isolated (the clamp through friction below is released where the kerf is not slanted).

49. **In air a MuJoCo-owned knife's blade follows MuJoCo's knife at its rest shape.** Its flexible segments meet only tissue while MuJoCo's rigid copy meets the board; left dynamic in air, a dropped knife's blade swings through the board at 2–4.5 m/s indefinitely.

50. **A runtime measurement labelled 'cut' must be shown to cut.** A knife started 0.33 m above the fish never reaches it, and its 99.7x measures a knife in air; the cutting runtime run approaches the workpiece and must make crack faces.

51. **A tet the blade passes through clamps it through friction.** Exempting such tets' far-side nodes from the blade releases the clamp on a kinematic blade but bends a flexible one (0.35 J created in the robot's press cut); keeping the contact and removing its friction carries nothing on the coupon and brings the press plateau to 24.8 N against $R\ell = 24$ N. A4's ledger stays open ($-87.5\,\%$; $-0.1\,\%$ with the exemption, so A4's energy comes from those nodes' normal push). A piece whose own whole elements lie on both sides of a slanted kerf still rides up.

52. **A blade that switches owner must hand over its weight.** Otherwise the held blade's switch to dynamic puts its 0.16 J of potential energy into the world unbooked, and the switch back books it as damping.

53. **The flesh needs structural damping.** Without it, a knife dropped on the trout bounces 67 mm and the fish moves 1.7 s later. With one Maxwell branch (loss tangent up to 0.09) and the knife's owner held for 100 ms off the tissue, the 60 mm drop changes owner 3 times, rebounds 15.2 mm, throws a tissue node at 1.69 m/s against the knife's 1.00 m/s and meets the rest criteria only 2.49 s after the release. *Standing.*

54. **In C2 the rig sits at its cap with the knife in free air**, 53–70 mm from any tissue node; the source is not found. *Standing.*

55. **The fastest "tissue" is a hanging sliver, not a thrown piece.** The nodes a speed check catches at 2–8 m/s are crack-surface copies that own a single sliver tet ringing over sub-millimetre amplitudes after a tear or split; the check reads pieces (fragment centroids), and ringing single-element crumbs still trip it in several runs. *Standing.*

56. **A rig that aims at the board's surface stops short of it.** Under the tissue's 15–25 N the force-limited servo holds B5's edge 0.9–2.4 mm up, outside the 0.6 mm window in which the edge cuts the last layer: the rigs aim 1.5 mm into the board, and B5 cuts a portion. A wider window for a stalled edge breaks ordinary presses into crumbs.

57. **A penalty scaled to a node's mass throws it at metres per second** when the node is found inside a shape; no contact pushes a node away faster than 1 m/s.

58. **A product's quality checks pass only when the product exists**; otherwise a do-nothing run passes them.

59. **A4's rib layer is representable by its carriers** (3.5 % of the fillet, budget 6 %), not by a flat stroke (8.8 %).

60. **Four policy cameras cost the path tracer's samples**, not the cameras: 20 s per observation at 32 samples, 2.2 s at 8.

61. **A blade landing and sawing on the board books energy no force supplied** through the rigid samples' damping (A1: $-242\,\%$, a second run $-73\,\%$); a still press and slow drags close. *Standing.*

62. **The cut's outcome depends on two open values within their ranges:** the sheet strength and the blade-tissue friction (coupon sensitivity). *Standing.*

63. **The blade pierces the salmon side's collarbone and carries it off** (B2), unseen by every check. *Standing.*

11 Limitations

Passing component probes establishes how this model behaves; it does not establish agreement with a real fish, a real knife or a real cook. This version is a knife-work benchmark with a physically cut workpiece, not a calibrated digital twin of any specimen, and it contains no autonomous policy. The items below are what the world does not do, or does with a known error. They are unfinished work, not accepted properties. Read them before relying on the environment.

1. **No robot solution, one assisted product, no placement.** No autonomous robot solution of any cutting task exists or is claimed, and no robot run makes a task product. One assisted run does: B5 cuts a 147.8 g portion within the specification, but four of its six cuts stay joined and the portion is not weighed, so the task is not complete. The robot's weighing reaches the portion and stops: it lies flush against its neighbours and the fingers come down on them; sliding a portion clear is not built. A1's head does not separate (its cut goes through a vertebral joint in bulk flesh); no run lays a fillet skin-down in the tray.
2. **Not calibrated to a specimen.** Fish flesh stiffness and toughness are derived from cod and tuna data, not measured on trout; the FR3's joint acceleration limit is recalled, not re-read; the knife inserts are authored; and task thresholds without a published source are labelled authored in the evaluator.
3. **Open parameters.** 20 of the 83 ledger entries have no digital basis (*open* as their primary class; 21 carry *open* somewhere in their label, Table 2), among them every interface's fracture energy and strength, blade-tissue friction and chilled fat stiffness. They are placeholders; their basis states a proposed sweep range. On the coupon, one at a time, the bulk tensile strength and the sheets' fracture energy change nothing in its range, while the sheet strength at 5 kPa leaves a press cut whole and a blade friction of 1.0 lets a sawn half ride out: the cut's outcome is not robust to those two; no sweep on a whole task has been run. 12 entries are read by no world module (`FISH_HEAD_E`, `FISH_HEAD_R`, `SKIN_FLESH_G`, `SKIN_FLESH_SIGMA`, `BONE_CRUSH`, `BONE_SHEAR`, `PIN_PULL_SALMON`, `PIN_BREAK_SALMON`, `PIN_LENGTH_SALMON`, `TAUO_BLADE_TISSUE`, `FRANKA_HAND`, `CODEX_BONE`): each is recorded for reference only and marked so in the ledger with the reason (for example the rigid head's toughness; SKIN_FLESH_G and SKIN_FLESH_SIGMA since the salmon's belly seam was removed: the skin shares the flesh's surface nodes). Several numbers are written inline in the code without a provenance label (Table 3).
4. **Mesh resolution.** Cuts follow element faces, so a cut face is ragged at a fraction of the mesh edge (coupon 3.5 mm, trout 3.5 mm, trout_fillet_rib_strip 3.5 mm, salmon 5 mm, tenderloin 3.5 mm) and the thinnest removable layer is one element. Skinning (B4) is refused for this reason, and C2's muscle-loss budget is tight at this resolution.
5. **Transient inversion under ramming.** Legal rams of the knife into the board or onto the backbone crush elements past inversion for a moment; they recover, and the evaluator counts them as bruises, but the state is not physical while it lasts and the board ram leaves energy unexplained in the ledger (legal-abuse results). There is no crush failure model for tissue beyond the reduced edge resistance of Section 3.5.
6. **Stability mass.** Explicit stability adds inertia, not weight, to stiff or tiny nodes (mass budget in Section 8); those nodes respond more slowly to fast loads than real tissue would.
7. **Interface release is geometric.** A blade releases a bond or tie when its slab passes it and then pays the fracture energy; the bond does not first load the edge.
8. **Robot contact.** The robots touch tissue and a free knife through boxes on the fingers, pads, inserts and palm only; the arms' other links touch the tissue and a free knife only in MuJoCo, one-sided. A free knife's MuJoCo copy is kinematic, so a finger cannot push it there.
9. **Explicit-contact ejection.** A fragment squeezed between the blade and the board can be thrown off faster than the knife moves; the floor catches it, but the ejection is an artifact of explicit penalty contact. No contact pushes a node away faster than 1 m/s, yet in the recorded runs two of A2's crumbs are accelerated to about 1.4 m/s on the board and fly off it to the floor, and B5's tilted control leaves two small pieces below the bench.
10. **The scale** reads only the tissue's load on its platform, not a hand pressing on it.

11. **Unsupported tasks and start states.** B1 (no physical tweezers) and B4 (resolution) are refused; the B and C tasks start from the whole workpiece; the belly membrane (B2) and the IMPS suspension check (C3) are not modelled.

12. **Silver skin.** The tenderloin's silver skin is a membrane whose modulus, thickness and bond to the muscle are open parameters, and it has no failure law of its own: it parts only where the elements under it are cut (Section 4). At the 3.5 mm mesh a removed silver skin carries at least the tetrahedra under it; the silver-layer probe measures that mass against C2's 5 % budget.

13. **Viscoelasticity and structural damping.** One Maxwell branch (loss tangent up to 0.09 near 10 Hz, fit to fish muscle's $G''/G' \approx 0.1$) damps the flesh's low modes; muscle's slow relaxation over seconds and the frequency spread of its loss are not modelled, and the branch's two values are derived from a loss tangent, not from a relaxation test of raw fish or pork. The start state is brought to rest by relaxing the first 1.2 s of the 1.5 s hands-off settle (the energy removed is booked).

14. **Rendering.** The head's skin surface and the fins are render-only surfaces moved with the physics; the fins are drawn translucent with rays, the salmon's fat lines and the cut faces' myomere rings are drawn per pixel from the anatomy's own phase field (only when the anatomy rebuilt from its seed reproduces the recorded colours), and the silver skin is a glossy see-through sheet. Still: fin outlines are polygonal, the gill-cover margin is painted on collar tissue the trout's rigid head does not include (it stays on the body after A2's head-off), ribs under the cavity lining are not visible, and cut faces' rings are faceted near their centres.

15. **Tearing** is tension-only and mesh-dependent (no regularisation); it does not act beside the knife's edge. In the recorded runs tears are 2–26 % of the pork runs' crack faces made after the build (the interfaces split at build left out), 29–58 % of the salmon runs', 58 % of A2's and 73 % of A4's.

16. **Debris, work carried on the knife and thrown tissue.** Cuts leave sub-gram fragments, and the withdrawn knife lifts work it is embedded in: the held tenderloin in C3 (up to 103 mm), part of the side in B2 and C1, a half of the robot's press-cut block at 20 and 30 mm/s; every recording is measured for it (gates 2 and 3). The thrown-tissue check reads the fastest piece; ringing single-element crumbs still trip it in several runs.

17. **Energy created where the blade meets bone or the board.** The demonstrations' ledgers: A1 −242 % (its rigid samples' damping while the flexible blade lands and saws on the board; a second run of the same code −73 %), A2 −17 %, B5 and B3 −13 %, A4 −5.6 %; C2 −6.1 %; the other pork runs, B2 and the robot's press cuts close within 4 %.

18. **The blade pierces a bone.** In B2 and its control the blade pierces the salmon side's collarbone and carries it 325 to 348 mm, up to 149 mm above the board; no check tracks rigid bones on the knife.

19. **No knife point.** The physics blade is 25 mm high to a flat end face; the tip techniques of filleting (pushing the tip through into the gut cavity, working with the first two inches) cannot be performed.

20. **No humanoid.** The G1 is refused without a G1 run: there is no Dex3 knife grip, no tissue colliders for its links and no standing controller under cutting loads; ten of eleven tasks accept only the dual FR3.

21. **Rest micro-vibration.** From the relaxed start the resting workpieces' nodes move by micrometres in 30 s; the tissue nodes jitter at 0.2–0.4 mm/s rms and the resting world books −0.01–0.59 mW of dissipation (rest results; negative in the tenderloin, where the Maxwell branch's first-order booking books too little loss); every assisted run's residual has to be read against this floor.

22. **Runtime** through the public API on a quiet host: holding still about 60 wall seconds per simulated second; cutting at most 80 (tenderloin; 129 in an action the host was not quiet for) and 157 (salmon); on the trout every action is under the limit of 200 except the first 0.1 s stroke after a slice of about 900 crack faces, 215, 240 and 224 times real time in three runs (the host-side splitting of those faces most of it); four policy cameras cost 2.2 s per observation. Beside other jobs the long assisted runs took 300–900 times real time (Section 8).

12 Reproduction

12.1 Environment

Newton 1.5.0, Warp 1.16.0 and MuJoCo 3.11.0 on the CPU for the rigid world, the tissue solver on the GPU, and an Isaac Sim 5.1 render worker for films and RGB observations (`ISAAC_RENDER_PYTHON`).

12.2 Running a task

The environment is `filleting`. A task is resolved from its preset, task and seed and served over HTTP:

```
gar doctor filleting
gar tasks resolve filleting --selector-json '{"preset": "trout", "task": "fillet-one-side",
"seed": 0}' --output filleting.task.json
gar run --task-file filleting.task.json --port 8080 --policy-key
gar verify filleting
gar stop filleting
```

`gar verify` returns the task evaluator's record (Section 6). The robot setup is chosen with `--robot` (`dual-fr3` by default; Section 5).

12.3 Module map

Modules	Responsibility
`world.py`, `scene.py`, `layout.py`	Rigid world, the knife's ownership, grip, rate limiter and protective stops; the physical layout.
`tissue.py`, `bodies.py`, `cohesive.py`, `tearing.py`	Tissue elements, contact, the edge law, rigid bones, joints, rods, bonds, ties, skin, tearing, the energy ledger.
`topology.py`, `mesher.py`	Crack recording and insertion; isosurface-stuffing mesher.
`workpiece.py`, `assembly.py`, `anatomy/`	Workpiece generation and assembly; the rigid-tissue coupling tick.
`knife.py`, `params.py`	Blade geometry; the parameter ledger.
`task.py`, `variants.py`, `env.py`, `robot_setup.py`	Tasks, the evaluator, task selection, robot setups and the environment interface.
`render.py`, `render_worker.py`, `frames.py`, `dressing.py`	Render-only path from recorded states: observations and room dressing.

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Source labels. Each source was read in full, read as an abstract or quoted table only, seen as a search summary only, or recalled from memory. The labels in the parameter ledger (Table 2) state which kind of source each value rests on; a source listed here is not a claim that every value attributed to it was measured on the object pair simulated.